A geometric model for instantaneous phase and frequency of multivariate signals

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Abstract

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1 Introduction

The notion of phase is historically central to many problems in signal processing, ranging from fundamental analysis (delay estimation, correlation or spectral analysis) to filtering, together with its daily usage in numerous applications (radar, acoustic imaging, remote sensing, etc.). For non-stationary signals, instantaneous phase and amplitude are of major interest to track their time evolution and at the roots of time-frequency analysis with the well known *analytic signal* [1, 2].

The definition of an instantaneous phase/frequency (together with instantaneous amplitude) for non-stationary multivariate signals was proposed in a series of papers [3–6]. In particular, in the bivariate case [3], authors introduced the concept of *modulated elliptical signal* that was later generalized to arbitrary dimensions. Extending instantaneous attributes to multivariate signals allows to provide *local descriptors* for their analysis. These attributes are equivalent to local descriptors of curves in Euclidean spaces of dimension equal to or greater than two. In particular, the bivariate [3, 4] ($n = 2$) and trivariate [5] ($n = 3$) cases have proven to be of great interest in many application fields ranging from oceanography [3, 4] to seismology [7] or gravitational astronomy [8], to name only a few.

Recall that for the univariate case, providing that the complex-valued signal $\mathsf{x}(\mathsf{t})$ is an analytic signal (satisfying the Bedrosian conditions [2]), then it can be expressed in an AM-FM (Amplitude Modulation-Phase Modulation) form the following way:

$$\mathsf{x}(\mathsf{t}) = \mathsf{a}_{\mathsf{x}}(\mathsf{t}) e^{\boldsymbol{j}\phi_{\mathsf{x}}(\mathsf{t})} \tag{1}$$

with $\forall \mathsf{t}$, $a_{\mathsf{x}}(t) \in \mathbb{R}_+$ is its *instantaneous amplitude* and $\phi_{\mathsf{x}}(\mathsf{t}) \in [0, 2\pi[$ is its *instantaneous phase*, or equivalently with $e^{\boldsymbol{j}\phi_{\mathsf{x}}(\mathsf{t})} \in \mathsf{U}(1)$, where $\mathsf{U}(1)$ is the one-dimensional unitary group, which is isomorphic to $\mathcal{S}^1$, the real sphere of dimension 1. JF:2e partie de phrase bizarre

As exposed in [6], when generalizing this signal expression from the univariate case to the multivariate case, it seems natural to seek for a real positive quantity for the instantaneous amplitude (an element of $\mathbb{R}_+$) and for a unit complex factor (an element of $\mathcal{S}^1$) from which one can obtain the instantaneous phase. This means for a multivariate complex-valued signal $\mathbf{x}(\mathsf{t}) \in \mathbb{C}^n$ that its expression, at each time instant t should be like:

$$\boldsymbol{x}(t) = a_{\boldsymbol{x}}(t) e^{\boldsymbol{j}\varphi_{\boldsymbol{x}}(t)} \boldsymbol{p}(t) \tag{2}$$

where $\mathsf{a}_{\mathbf{x}}(\mathsf{t}) \in \mathbb{R}_+$ is the *instantaneous amplitude*, $\phi_{\mathbf{x}}(t)$ is the *instantaneous phase* (or equivalently with $e^{\boldsymbol{j}\phi_{\mathbf{x}}(t)} \in \mathsf{U}(1)$) and $\mathbf{p}(\mathsf{t}) \in \mathcal{S}^{2n-2}$ is a normalized n-dimensional complex-valued vector containing the remaining parameters of $\mathbf{x}(\mathsf{t})$. The phase factor $e^{\boldsymbol{j}\phi_{\mathbf{x}}(t)}$ contains the phase term whared by the n components of $\mathbf{x}(\mathsf{t})$.

In the univariate case, only the instantaneous amplitude and phase are necessary to parametrize a signal with values in $\mathbb{C}$, and it is only for $n \geq 2$ that extra parameters appear, contained in $\mathbf{p}(\mathsf{t})$. As an example, in the bivariate case $\mathbf{p}(\mathsf{t})$ is an element of $\mathcal{S}^2$, the unit sphere in $\mathbb{R}^3$. Thus it can be parametrized by two angles $\theta \in [0, 2\pi]$ and $\xi \in [-\pi/2, \pi/2]$ which are related to the (instantaneous) parameters of the ellipse that the bivariate signal draws locally (see [4, 9]). Examples of multivariate signals of the form (2) are given in [6]: a *multivariate pure oscillation* $\mathbf{x}(\mathsf{t}) = \mathsf{a}_{\mathbf{x}}(\mathsf{t}) e^{\omega_0 \mathsf{t}} \mathbf{x_0}$ and the *multivariate phase signal* $\mathbf{x}(\mathsf{t}) = \mathsf{a}_{\mathbf{x}}(\mathsf{t}) e^{\boldsymbol{j}\phi_{\mathbf{x}}(t)} \mathbf{x_0}$, where $\mathbf{p}(\mathsf{t}) = \mathbf{x}_0$ for both and with x_0 a constant. In the sequel, we will be interested in multivariate signals for which $\mathbf{p}(\mathsf{t})$ has time fluctuations and it will be demonstrated the effect of those fluctuations on the instantaneous phase $\phi_{\mathbf{x}}(\mathsf{t})$.

In Lilly and Olhede [6], the instantaneous parameters are defined via a variational approach, where the multivariate signal is expressed via a *local modulation expansion.* The first two coefficients of the expansion, forming the *deviation vectors*, can be related to intrinsic analytic velocity and acceleration and are central to the concept of multivariate oscillation. In particular, the instantaneous phase is demonstrated to be the minimizer of the first order deviation vector. In this work, we adopt the multivariate signal model given in (2) which is based on [6]. However, we highlight an effect that was not foreseen by authors in [6]: the instantaneous phase may incorporate an extra term that is only due to the time fluctuations of $\mathbf{p}(\mathsf{t})$ in the model (2). This term is a actually *geometric phase*, a concept well known in physics [10] first introduced in quantum mechanics [11] before being identified in many classical and quantum systems [12].

This papers thus introduces a concept/effect that was ignored along the way in signal processing when considering multivariate non-stationary modulated signals. The contributions presented hereafter are the following:

- Evidence of the existence of a geometric phase for multivariate complex non-stationary signals
- Geometric description (based on fiber bundle theory) of the concept of instantaneous phase
- New/Corrected definition for instantaneous phase and frequency for multivariate signals
- Original recursive estimator for the geometric phase of a multivariate signal
- Illustrative examples (with emphasis on bivariate case) showcasing the apparition of geometric phase and how it affects the instantaneous frequency estimation.

The structure of the paper is as follows: Section 2 illustrates how the instantaneous phase and frequency concepts have to be carefully handled in the multivariate case and how they may incorporate a geometric component. Section 3 provides a mathematical description of the geometric concept phase that can affect multivariate complex signals using the language of fiber bundles. In section 5, the practical issue of estimating the geometric phase of a multivariate signal is addressed, while section 6 introduces several illustrative examples where geometric phases occur, for bivariate and multivariate signals.

1.1 Notations

JF:introduire tableau de notations. Introduire $L^2(\mathbb{C}^d)$ + footnote associée.

notation pour les dérivées. vecteurs etc.

discuter produit scalaire dans $\mathbb{C}^d$, $\boldsymbol{x}^\dagger \mathbf{y}$. Insister sur le caractere instantané des produits scalaires / normes

JF:we often refer to section 3 here; maybe its worth saying upfront that technical details related to geo diff are given in Section 3.

2 The instantaneous phase of a multivariate signal

The concept of phase is well known in signal and image processing. As recalled in the Introduction it is a fundamental notion central in many applications. To discuss the concept for multivariate signals, it is interesting to first examine the case of scalar signals and recall some basic ideas which allows the understanding of phases. As well known since the 19th century, using the complex representation of signals permits an easy manipulation of phases, as angles between vectors. This particular point puts also forward the central idea of the relativity of phases: The phase has a meaning only when there is a reference.

We elaborate in this section on these two basic ideas, first recalling known facts in the scalar case, and then on developping the notion of instantaneous phase for multivariate signals and on examining the consequence of the multivariate character on the notion.

2.1 Univariate case

Consider a univariate complex signal $x(t) \in L^2(\mathbb{C})$ indexed by time[1]. It can obviously be written in a unique manner as $x(t) = a_x(t)\exp(\boldsymbol{j}\varphi_x(t))$. The instantaneous phase then writes

$$\phi_{x,t_0}(t) = \arg\left(x^*(t_0)x(t)\right) = \varphi_x(t) - \varphi_x(t_0) \tag{3}$$

where t_0 is anrbitrary time chosen as a reference.

Is well known [2] that under the AM-FM (Amplitude Modulated, Frequency Modulated) hypothesis, $a_x(t)$ and $\varphi_x(t)$ are called the *instantaneous amplitude* and *instantaneous phase* respectively. They are assumed to be smooth functions of t in the sequel (at least differentiable). The condition for the complex-valued signal $x(t)$ to be interpretable as a AM-FM signal is the Bedrosian condition [2] that reads $|\dot{\varphi}_x(t)| \gg |\dot{a}(t)|/|a(t)|$, where the dot notation stands for time derivation. From (3), the instantaneous frequency usually defined as $\nu_x(t) = (2\pi)^{-1}\dot{\varphi}_x(t)$ is recovered since

$$\frac{1}{2\pi}\dot{\phi}_{x,t_0}(t) = \frac{1}{2\pi}\dot{\varphi}_x(t) = \nu_x(t)$$

Thus in the univariate case, the definition proposed here of the instantaneous phase relative to t_0 is in accordance with the usual definition found in the literature.

Furthermore, for such univariate AM-FM signal $x(t)$ as defined in (1), the variations of its instantaneous phase $\phi_x(t)$ during a time interval $[t_0, t]$ fulfills the following equality:

$$\int_{t_0}^{t} \frac{\operatorname{Im}\left(x^*(s)\dot{x}(s)\right)}{|x(s)|^2}\, ds = \varphi_x(t) - \varphi_x(t_0) = \phi_{x,t_0}(t) \tag{4}$$

since $\operatorname{Im}\left(x^*(s)\dot{x}(s)\right)/||x(s)||^2 = \dot{\varphi}_x(s) \in \mathbb{R}$ is the *instantaneous frequency* (up to $(2\pi)^{-1}$), denoted $\omega_x(t)$ in [4, 6]. Equation (4) can be directly checked using the expression of $\boldsymbol{x}(t)$ in (1) and by simple derivation, noting that $x^*(t)\dot{x}(t) = a_x(t)\dot{a}_x(t) + \boldsymbol{j}a_x^2(t)\dot{\varphi}_x(t)$, and $||x(t)||^2 = a_x^2(t)$. Literally, expression (4) means that, for a univariate AM-FM signal, the phase difference between two time instants is obtained by integrating the quantity $\operatorname{Im}\left(x^*(s)\dot{x}(s)\right)/||x(s)||^2$. This quantity also as the interpretation of a moment of the instantaneous energy density. Indeed, it can be shown

$$\langle \nu_x \rangle = \int \nu |X(\nu)|^2 d\nu = \int \frac{\operatorname{Im}\left(x^*(t)\dot{x}(t)\right)}{2\pi|x(t)|^2}|x(t)|^2 dt = \int \nu_x(t)|x(t)|^2 dt$$

where $X(nu)$ stands for the Fourier transform of $x(t)$. In quantume mechanics, this reflects the fact that knowing perfectly the speed v of a particle (formal equivalence between frequency ν for signals and momentum mv for a particle) forbids the perfect knowledge of the position r of the particle (formal equivalence between t for a signal and r for a particle). For scalar signals, function $\nu_x(t)$ has therefore the meaning of an instantaneous frequency.

[1] In the sequel, notation $L^2(\mathbb{E})$ stands for the vector space of finite energy signals indexed by $\mathbb{R}$ (typically time) and taking values in $\mathbb{E}$.

Surprisingly, or may be counterintuitively, the basic and well-known results for univariate signals does not generalize to multivariate signals. This is a central result for what follows.

2.2 In-phase multivariate signals

Consider two complex-valued multivariate signals $\boldsymbol{x}(t)$ and $\boldsymbol{y}(t)$. One can ask whether these two signals are "in-phase " or "out-of-phase" at a time instant t, or to quantify/measure the phase shift between them. This question is easily handled in the univariate case, but becomes less evident in the multivariate case In the bivariate case which is used to describe polarization, this question was addressed originally by Pancharatnam [13]. In order to decide whether two polarized beams are in phase or not, Pancharatnam defined the phase shift in this citation [13, p. 252]:

> *[...] the phase advance of one polarised beam over another (not necessarily in the same state of polarisation) is the amount by which its phase must be retarded relative to the second, in order that the intensity resulting from their mutual interference may be a maximum.*

According to this, two multivariate signals $\boldsymbol{x}(t)$ and $\boldsymbol{y}(t)$ can be said "*in-phase*" at instant t if:

$$\boldsymbol{x}^{\dagger}(t)\boldsymbol{y}(t) \geq 0, \tag{5}$$

which is a direct generalization of Pancharatnam's bivariate definition. Following these lines, the phase shift between two multivariate signals is simply $\arg(\boldsymbol{x}^{\dagger}(t)\boldsymbol{y}(t))$, which actually vanishes when $\boldsymbol{x}(t)$ and $\boldsymbol{y}(t)$ are *in phase*, as the inner product $\boldsymbol{x}^{\dagger}(t)\boldsymbol{y}(t)$ has a zero imaginary part in such a case.

2.3 Instantaneous phase of a multivariate signal

To define the *instantaneous* phase of a signal $\boldsymbol{x}(t)$, we thus need to define it as the phase between $\boldsymbol{x}(t)$ and a reference signal $\boldsymbol{y}(t)$. But introducing an extra signal is cumbersome, and it would be preferable to relate the phase of $\boldsymbol{x}$ at t to its phase at some reference time t_0, as done for scalar signals. The instantaneous phase of $\boldsymbol{x}$ at t relative to t_0, or instantaneous phase for short, of a multivariate signal is thus defined as

$$\phi_{\boldsymbol{x},t_0}(t) = \arg\left(\boldsymbol{x}^{\dagger}(t_0)\boldsymbol{x}(t)\right) \tag{6}$$

This operational definition corresponds to choosing a reference signal $\boldsymbol{y}(t)$ which is in phase with $\boldsymbol{x}(t_0)$ at all time.

2.4 General multivariate case

In the multivariate case, $\boldsymbol{x}(t)$ cannot be written any longer in a unique polar form. But any signal can be written as

$$\boldsymbol{x}(t) = a_{\boldsymbol{x}}(t)e^{\boldsymbol{j}\varphi_{\boldsymbol{x}}(t)}\boldsymbol{p}_{\boldsymbol{x}}(t) \tag{7}$$

where $\boldsymbol{p}_{\boldsymbol{x}}(t)$ is a unit vector for all t . If there is no ambiguity in defining the amplitude $\boldsymbol{x}(t)^\dagger\boldsymbol{x}(t) = a_{\boldsymbol{x}}(t) \in \mathbb{R}^+$, there are an infinity of choices for the couple $(\varphi_{\boldsymbol{x}}, \boldsymbol{p}_{\boldsymbol{x}})$. We will come back later on this fundamental problem.

For the moment, we choose one of these couples, and elaborate on the notion of instantaneous phase relative to t_0 as defined above. According to (6), the phase $\phi_{\boldsymbol{x},t_0}(t)$ of $\boldsymbol{x}$ reads

$$
\begin{aligned}
\phi_{\boldsymbol{x},t_0}(t) &= \arg\left(\boldsymbol{x}(t_0)^\dagger\boldsymbol{x}(t)\right) && (8)\\
&= \varphi_{\boldsymbol{x}}(t) - \varphi_{\boldsymbol{x}}(t_0) + \arg\left(\boldsymbol{p}_{\boldsymbol{x}}(t_0)^\dagger\boldsymbol{p}_{\boldsymbol{x}}(t)\right) && (9)
\end{aligned}
$$

where there is appearance of an extra-term compared to the univariate case.

As recalled in the the scalar case, we can define an instantaneous moment of a multivariate signal *via* the introduction of a function[2] of time $\nu_{\boldsymbol{x}}(t)$ such that

$$\langle \nu_{\boldsymbol{x}} \rangle = \int \nu \|\boldsymbol{X}(\nu)\|^2 d\nu = \int \nu_{\boldsymbol{x}}(t)\|\boldsymbol{x}(t)\|^2 dt$$

where $\langle \nu_{\boldsymbol{x}} \rangle$ is the mean frequency of the spectral density $\|\boldsymbol{X}(\nu)\|^2$. Simple properties of the Fourier transform (derivation properties and Parseval-Plancherel equalities) imply

$$\nu_{\boldsymbol{x}}(t) = \operatorname{Im}\left(\frac{\boldsymbol{x}^\dagger(t)\dot{\boldsymbol{x}}(t)}{2\pi||\boldsymbol{x}(t)||^2}\right)$$

This result is in accordance with what is known in the univariate case. However, even though the equality in (4) is always verified for univariate signals, it does not generalize to multivariate signals. Using parametrization (7), we get

$$\int_{t_0}^{t} \frac{\operatorname{Im}\left(\boldsymbol{x}^\dagger(s)\dot{\boldsymbol{x}}(s)\right)}{||\boldsymbol{x}(s)||^2}\, ds = \varphi_{\boldsymbol{x}}(t) - \varphi_{\boldsymbol{x}}(t_0) + \int_{t_0}^{t} \operatorname{Im}\left(\boldsymbol{p}_{\boldsymbol{x}}^\dagger(s)\dot{\boldsymbol{p}}_{\boldsymbol{x}}(s)\right)\, ds \qquad (10)$$

and therefore, combining with (9)

$$\phi_{\boldsymbol{x},t_0}(t) = \int_{t_0}^{t} \frac{\operatorname{Im}\left(\boldsymbol{x}^\dagger(s)\dot{\boldsymbol{x}}(s)\right)}{||\boldsymbol{x}(s)||^2}\, ds + \left(\arg\left(\boldsymbol{p}_{\boldsymbol{x}}(t_0)^\dagger\boldsymbol{p}_{\boldsymbol{x}}(t)\right) - \int_{t_0}^{t} \operatorname{Im}\left(\boldsymbol{p}_{\boldsymbol{x}}^\dagger(s)\dot{\boldsymbol{p}}_{\boldsymbol{x}}(s)\right)\, ds\right)$$

Compared to (4), a second term shows up. This term is independent of the particular parametrization $(\varphi_{\boldsymbol{x}}, \boldsymbol{p}_{\boldsymbol{x}})$ chosen. To verify this, choose another parametrization, say $(\psi_{\boldsymbol{x}}, \boldsymbol{q}_{\boldsymbol{x}})$. This means that $\boldsymbol{p}(t) = \exp(\boldsymbol{j}\alpha_{\boldsymbol{x}}(t))\boldsymbol{q}_{\boldsymbol{x}}$ where $\alpha_{\boldsymbol{x}} = \psi_{\boldsymbol{x}} - \varphi_{\boldsymbol{x}}$. Then direct calculations leads to $\boldsymbol{p}_{\boldsymbol{x}}^\dagger(s)\dot{\boldsymbol{p}}_{\boldsymbol{x}}(s) = \boldsymbol{j}\dot{\alpha}_{\boldsymbol{x}}(s) + \boldsymbol{q}_{\boldsymbol{x}}^\dagger(s)\dot{\boldsymbol{q}}_{\boldsymbol{x}}(s)$ and $\boldsymbol{p}_{\boldsymbol{x}}(t_0)^\dagger\boldsymbol{p}_{\boldsymbol{x}}(t) = \exp(\boldsymbol{j}(\alpha_{\boldsymbol{x}}(t) - \alpha_{\boldsymbol{x}}(t_0)))\boldsymbol{q}_{\boldsymbol{x}}(t_0)^\dagger\boldsymbol{q}_{\boldsymbol{x}}(t)$ and therefore

$$\arg\left(\boldsymbol{p}_{\boldsymbol{x}}(t_0)^\dagger\boldsymbol{p}_{\boldsymbol{x}}(t)\right) - \int_{t_0}^{t} \operatorname{Im}\left(\boldsymbol{p}_{\boldsymbol{x}}^\dagger(s)\dot{\boldsymbol{p}}_{\boldsymbol{x}}(s)\right) ds = \arg\left(\boldsymbol{q}_{\boldsymbol{x}}(t_0)^\dagger\boldsymbol{q}_{\boldsymbol{x}}(t)\right) - \int_{t_0}^{t} \operatorname{Im}\left(\boldsymbol{q}_{\boldsymbol{x}}^\dagger(s)\dot{\boldsymbol{q}}_{\boldsymbol{x}}(s)\right)\, ds$$

[2]We deliberatly do not define this function as an instantaneous frequency, contrary to the scalar case.

As claimed, this term depends only on $\boldsymbol{x}$ and not on the particular parametrization: It is called the *geometric phase* of the signal acquired betwen t_0 and t.

We have just shown that in the multivariate case, the instantaneous phase $\phi_{\boldsymbol{x},t_0}$ of a signal relative to a time t_0 is equal to the sum of two phases:

$$\phi_{d,\boldsymbol{x},t_0} = \int_{t_0}^{t} \frac{\operatorname{Im}\left(\boldsymbol{x}^\dagger(s)\dot{\boldsymbol{x}}(s)\right)}{||\boldsymbol{x}(s)||^2}\, ds \tag{11a}$$

$$\phi_{g,\boldsymbol{x},t_0} = \arg\left(\boldsymbol{p}_{\boldsymbol{x}}(t_0)^\dagger \boldsymbol{p}_{\boldsymbol{x}}(t)\right) - \int_{t_0}^{t} \operatorname{Im}\left(\boldsymbol{p}_{\boldsymbol{x}}^\dagger(s)\dot{\boldsymbol{p}}_{\boldsymbol{x}}(s)\right)\, ds \tag{11b}$$

$\phi_{d,\boldsymbol{x},t_0}$ is called the *dynamical phase*, and $\phi_{g,\boldsymbol{x},t_0}$ is the *geometric phase*. This second phase is purely due to geometry and can also be implicitely defined by

$$\phi_{\boldsymbol{x},t_0}(t) = \phi_{d,\boldsymbol{x},t_0}(t) + \phi_{g,\boldsymbol{x},t_0}(t)$$

This definition allows to evaluate $\phi_{g,\boldsymbol{x},t_0}(t)$ regardless of any parametric form of the signal $\boldsymbol{x}(t)$.

If $\boldsymbol{x}$ depends on time only through its scalar part, *i.e.* $\boldsymbol{x}(t) = a_{\boldsymbol{x}}(t)e^{\boldsymbol{j}\varphi_{\boldsymbol{x}}(t)}\boldsymbol{p}_0$ where $\boldsymbol{p}_0$ is a constant complex unit vector, equation (11b) shows that the geometric phase is equal to zero.

We can now try to generalize the concept of instantaneous frequency for multivariate signals. The developments above show that

$$\begin{aligned}\nu_{\boldsymbol{x}}(t) &= \frac{1}{2\pi}\dot{\phi}_{d,\boldsymbol{x},t_0}(t) \\ &= \frac{1}{2\pi}\dot{\phi}_{\boldsymbol{x},t_0}(t) - \frac{1}{2\pi}\dot{\phi}_{g,\boldsymbol{x},t_0}(t)\end{aligned}$$

Therefore, to remain consistant with what is known in the scalar case, we propose the following terminology:

- The ***instantaneous frequency*** $f_{\boldsymbol{x}}(t)$ of a multivariate signal is the derivative of the instantaneous phase at time t relative to t_0:

  $$f_{\boldsymbol{x}}(t) = \frac{1}{2\pi}\dot{\phi}_{\boldsymbol{x},t_0}(t)$$

  Since changing the reference time t_0 to t_1 modifies the instantaneous phase by the constant $\arg \boldsymbol{x}^\dagger(t_1)\boldsymbol{x}(t_0)$, this definition does indeed only depends on t. The instantaneous frequency is the sum of the two following frequencies:
- The ***dynamical frequency*** $\nu_{\boldsymbol{x}}(t)$ of a multivariate signal is the derivative of the dynamical phase. It is given by the instantaneous moment

  $$\nu_{\boldsymbol{x}}(t) = \frac{1}{2\pi}\dot{\phi}_{d,\boldsymbol{x},t_0}(t) = \operatorname{Im}\left(\frac{\boldsymbol{x}^\dagger(t)\dot{\boldsymbol{x}}(t)}{2\pi||\boldsymbol{x}(t)||^2}\right)$$

- The ***geometric frequency*** $f_{g,\boldsymbol{x}}(t)$ of a multivariate signal is the derivative of the geometric phase. It writes

$$\begin{aligned} f_{g,\boldsymbol{x}}(t) &= \frac{1}{2\pi}\dot{\phi}_{g,\boldsymbol{x},t_0}(t) \\ &= f_{\boldsymbol{x}}(t) - \nu_{\boldsymbol{x}}(t) \end{aligned}$$

These notions will be illustrated in Section 6.

Having shown the existence of a geometric phase for multivariate signals, we now turn to describing the mathematical causes of why this occurs. After a detailed derivation in Section 3, understanding of the geometry induced by the the peculiar amplitude-modulation multivariate signal model is helpful and will be developed in Section 4.

3 A geometric description of the concept of phase

As recalled and exploited in the last section, phase is a relative notion. The phase of a complex number is evidently dependent on the basis chosen to define the real and imaginary parts. This means that once the basis is chosen, the phase of a complex number is covariant under a multiplication by a unit norm complex number, or equivalently covariant under an action of the unitary group $U(1)$. This remains valid in the multivariate case: The definition adopted after Pantcharatnam of the instantaneous phase $\phi_{\boldsymbol{x},t_0}(t)$ relative to an origin t_0 is covariant under the action of $U(1)$ since if $\boldsymbol{y}(t) = \exp(\boldsymbol{j}\varphi(t))\boldsymbol{x}(t)$ then

$$\phi_{\boldsymbol{y},t_0}(t) = \varphi(t) - \varphi(t_0) + \phi_{\boldsymbol{x},t_0}(t)$$

This allows to conclude that the theory of phase in multivariate signals can only be done up to a global phase, and that solving this ambiguity requires to work in the quotient space $\mathbb{C}^n/U(1)$, as developed below.

3.1 Ambiguity and Equivalent classes

Let us decompose $\boldsymbol{x}(t)$ as $\boldsymbol{x}(t) = a_{\boldsymbol{x}}(t)\boldsymbol{p}_{\boldsymbol{x}}(t)$, where $a_{\boldsymbol{x}}(t) = (\boldsymbol{x}(t)^\dagger\boldsymbol{x}(t))^{1/2} \in \mathbb{R}^+$ is the instantaneous amplitude (unambiguously defined), and $\boldsymbol{p}_{\boldsymbol{x}}(t) \in \mathcal{S}^{2n-1}$ is a unit norm complex vector.

The decomposition above is ambiguous because vector $\boldsymbol{p}(t)$ is defined up to a global phase. In fact, the equivalence relation $\sim$ defined on $\mathcal{S}^{2n-1}$ by

$$(\boldsymbol{p}(t) \sim \mathbf{q}(t)) \Longleftrightarrow \Big(\boldsymbol{p}(t) = e^{\boldsymbol{j}\varphi(t)}\mathbf{q}(t)\Big)$$

implies that $\boldsymbol{x}(t)$ is as well written $a_{\boldsymbol{x}}(t)\boldsymbol{p}(t)$ as $a_{\boldsymbol{x}}(t)\mathbf{q}(t)$ if $(\boldsymbol{p}(t) \sim \mathbf{q}(t))$.

The decomposition becomes unambiguous if we impose to $\boldsymbol{p}_{\boldsymbol{x}}(t)$ to belong to the quotient $\mathcal{S}^{2n-1}/U(1)$. The space $\mathcal{S}^{2n-1}/U(1)$ is known to be the complex projective space $\mathbb{C}\mathbf{P}^{n-1}$.

3.2 Principal fiber bundle structure of $\mathcal{S}^{2n-1}$ and $\mathbb{C}^n$

Fiber bundles appear in many physical models [10]. Here, we describe the space $\mathcal{S}^{2n-1}$ of normalized complex signals as a fiber bundle. This fiber bundle is denoted $\mathcal{S}^{2n-1}\left(\mathbf{U}(1), \mathbb{C}\mathbf{P}^{n-1}\right)$. $\mathcal{S}^{2n-1}$ is called the *total space*; $\mathbb{C}\mathbf{P}^{n-1}$ is the *base space* and is the set of all equivalent classes of $\mathcal{S}^{2n-1}$ under the equivalence relation $\sim$. A *fiber* above an element $[\boldsymbol{p}] \in \mathbb{C}\mathbf{P}^{n-1}$ is an orbit of $\boldsymbol{p}$ under the action of the group $U(1)$, that is the set $\mathcal{F}_{\boldsymbol{p}} = \{\exp(\boldsymbol{j}\psi)\boldsymbol{p}, \forall \exp(\boldsymbol{j}\psi) \in U(1)\}$.

The bundle structure comes with a so-called projection π : $\mathcal{S}^{2n-1}\left(\mathbf{U}(1), \mathbb{C}\mathbf{P}^{n-1}\right) \longrightarrow \mathbb{C}\mathbf{P}^{n-1}$ which associates to an element of the total space an element of the base space, with the property that $\mathcal{F}_{\boldsymbol{p}} = \pi^{-1}([\boldsymbol{p}])$ for any $[\boldsymbol{p}] \in \mathbb{C}\mathbf{P}^{n-1}$. Since a fiber is a subset of a vector space, we can use the usual projector for $\boldsymbol{p} \in \mathcal{S}^{2n-1}$ to find its equivalence class in $\mathbb{C}\mathbf{P}^{n-1}$, or $\pi(\boldsymbol{p}) = \boldsymbol{p}\boldsymbol{p}^\dagger = [\boldsymbol{p}]$ since obviously it is invariant under action of $U(1)$. We will use the notation $\rho_{\boldsymbol{p}} = [\boldsymbol{p}]$ from now on, and mention the fact that $\rho_{\boldsymbol{p}}$ is the coherence matrix of vector $\boldsymbol{p}$ (also called the density operator in quantum mechanics [14]).

Two important remarks must be made here:

- Firstly, for technical reasons, $\{0\}$ must be excluded from the total space: This ensures the manifold structures we are going to consider ([15, 16] for details–freeness of group actions). This is assumed in the sequel, but is not explicetly mentionned to avoid heavier notations.
- Secondly, the total space considered up to here is the $2n-1$ dimensional sphere, and the fiber at $\rho_{\boldsymbol{p}}$ is the set of all $\exp(\boldsymbol{j}\psi)\boldsymbol{p}$. But the construction can be easily generalized by considering $\mathbb{C}^n$ as the total space and $\mathbb{C}\mathbf{P}^{n-1}$ as the *base space*, the fibers being defined as $\mathcal{F}_{\boldsymbol{x}} = \{\lambda\boldsymbol{x}, \forall\lambda \in \mathbb{C}\}$ for any $\boldsymbol{x} \in \mathbb{C}^n$ along with the projection

$$\rho_{\boldsymbol{x}} = \frac{\boldsymbol{x}\boldsymbol{x}^\dagger}{\boldsymbol{x}^\dagger\boldsymbol{x}} = \boldsymbol{p}_{\boldsymbol{x}}\boldsymbol{p}_{\boldsymbol{x}}^\dagger \in \mathbb{C}\mathbf{P}^{n-1} \tag{12}$$

  However, in the sequel, since we can unbiguously extract the instantaneous amplitude, we only deal with the bundle structure of $\mathcal{S}^{2n-1}$ and concentrates on the properties of $\boldsymbol{p}_{\boldsymbol{x}}(t)$ in the decomposition $\boldsymbol{x}(t) = a_{\boldsymbol{x}}(t)\boldsymbol{p}_{\boldsymbol{x}}(t)$.

A signal $\boldsymbol{x}(t)$ is a trivialized section of the bundle, it is to say an application from the base $\mathbb{C}\mathbf{P}^{n-1}$ to the total space $\mathbb{C}^n$ together with a continuous basis of $\mathcal{F}_{\boldsymbol{x}}$. The particular sections we consider here are continuous, differentiable and of finite energy. They are parametrized by time $t \in [t_0, T]$ (each bound may be infinite), by an instantaneous positive amplitude $a_{\boldsymbol{x}}(t) \in L^2(\mathbb{R})$ and a unit norm complex vector $\boldsymbol{p}_{\boldsymbol{x}}(t)$. We assume both to be continuous and differentiable. $\pi(\boldsymbol{x}(t))$ then draws a curve on the manifold $\mathbb{C}\mathbf{P}^{n-1}$. Conversely, to each continuous and differentiable curve $\rho_{\boldsymbol{x}}(t) \in \mathbb{C}\mathbf{P}^{n-1}$ corresponds an infinite number of possible signals in the total space, each differing by some unit norm complex multiplicative function $\exp(\boldsymbol{j}\lambda(t))$: Each of these indeed project onto $\rho_{\boldsymbol{x}}(t)$ by π. These signals are called *liftings* of the curve $\rho_{\boldsymbol{x}}(t)$. Likewise, a curve $\rho_{\boldsymbol{x}}(t)$ may be lifted in different sections of $\mathcal{S}^{2n-1}$, each differing by some complex multiplicative function $\exp(\boldsymbol{j}\alpha(t))$.

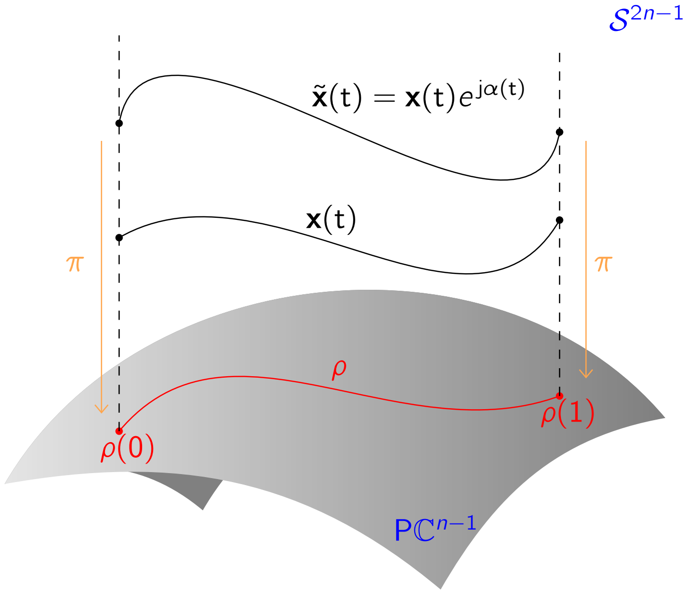

Fig. 1 Representation of the fiber bundle structure of $\mathcal{S}^{2n-1}$ for multivariate normalized signals. At each point of the base space is attached a fiber ($\mathcal{S}^1$) NLB:Finir le caption et changer les notations dans la figure

The existence of different possible liftings is at the root of the existence of the geometrical phase.

3.3 Ehresmann connection

In order to study the dynamics of multivariate signals, and to understand and describe the time evolution of their instantaneous attributes, a key concept is the tangent space to the total space $\mathcal{S}^{2n-1}$. The Ehresmann connection provides a decomposition of this tangent space into two orthogonal vector spaces, the so-called *horizontal* and *vertical* spaces. It also provides a 1-form (or linear form) allowing to connect the fibers.

All the spaces considered up to here are manifolds. Obviously $\mathcal{S}^{2n-1}$ is a submanifold of $\mathbb{C}^n/\{0\}$, itself a submanifold of $\mathbb{C}^n$. Since $\mathbb{C}^n$ is of finite dimension n, its tangent space at $\boldsymbol{x}$ is also of finite dimension, and is thus isomorphic to $\mathbb{C}^n$: It is therefore identified to it. Likewise, $\mathbb{C}^n/\{0\}$ is of dimension n and its tangent space $T_{\boldsymbol{x}}\mathbb{C}^n/\{0\}$ at $\boldsymbol{x}$ is identified with $\mathbb{C}^n$. However, $\mathcal{S}^{2n-1}$ is of dimension $n-1$ and its tangent space $T_{\boldsymbol{x}}\mathcal{S}^{2n-1}$ at $\boldsymbol{x}$ is of dimension $n-1$. Furthemore, a fiber at $[\boldsymbol{x}]$ is identified with the circle $\mathcal{S}^1$ and is also a manifold.

Consider a smooth curve $\boldsymbol{x}(t) \in \mathcal{S}^{2n-1}$. Since $\boldsymbol{x}(t)^\dagger \boldsymbol{x}(t) = 1 \,\forall t$, derivating with respect to t shows that $\mathrm{Re}(\boldsymbol{x}(t)^\dagger \dot{\boldsymbol{x}}(t)) = 0$ and thus that $T_{\boldsymbol{x}}\mathcal{S}^{2n-1}$ at $\boldsymbol{x}$ is the set of vectors $\boldsymbol{v}$ of $\mathbb{C}^n$ such that $\mathrm{Re}(\boldsymbol{x}^\dagger \boldsymbol{v}) = 0$.

We want to link the tangent space at $\boldsymbol{x} \in \mathcal{S}^{2n-1}$ with the tangent space at $[\boldsymbol{x}] = \pi(\boldsymbol{x}) \in \mathbb{C}\mathbf{P}^{n-1}$. However, there is a difficulty since the tangent space at $\boldsymbol{x}$ can be given two definitions, depending on the choice $\boldsymbol{x} \in \mathcal{S}^{2n-1}$ or $\boldsymbol{x} \in \mathcal{F}_{[\boldsymbol{x}]}$. Indeed, to define a tangent vector at $\boldsymbol{x}$, we need a curve lying in the space to which $\boldsymbol{x}$ belongs. But obviously the fiber at $\boldsymbol{x}$ is included in $\mathcal{S}^{2n-1}$ so that $T_{\boldsymbol{x}}\mathcal{F}_{[\boldsymbol{x}]} \subset T_{\boldsymbol{x}}\mathcal{S}^{2n-1}$. Thus, the tangent space at $\boldsymbol{x}$ in $\mathcal{S}^{2n-1}$ can be decomposed as a direct sum, the tangent space $T_{\boldsymbol{x}}\mathcal{F}_{[\boldsymbol{x}]}$ plus its orthogonal complement.

In the literature, the tangent space to the fiber at $\boldsymbol{x}$ is called the vertical space $V_{\boldsymbol{x}}$. In the case studied here, it is the set $V_{\boldsymbol{x}}\mathcal{S}^{2n-1} = \{\boldsymbol{j}a\boldsymbol{x}, a \in \mathbb{R}\}$. Indeed, consider a curve $\boldsymbol{y}(t) \in \mathcal{F}_{[\boldsymbol{x}]}$ passing through $\boldsymbol{x}$ at $t = 0$. Since the fiber is the set of all equivalent vectors to $\boldsymbol{x}$, we can parametrize $\boldsymbol{y}(t)$ by $\boldsymbol{y}(t) = e^{\boldsymbol{j}\alpha(t)}\boldsymbol{x}(t)$ with $\alpha(0) = 0$. Deriving $\boldsymbol{y}$ at $t = 0$ gives the tangent vector $\boldsymbol{j}\dot{\alpha}_0\boldsymbol{x}$, which proves the assertion. The orthogonal complement $H_{\boldsymbol{x}}\mathcal{S}^{2n-1}$ to $V_{\boldsymbol{x}}\mathcal{S}^{2n-1}$ in $T_{\boldsymbol{x}}\mathcal{S}^{2n-1}$ is called the horizontal space. Note that $V_{\boldsymbol{x}}\mathcal{S}^{2n-1}$ is also the kernel of the differential of π at $\boldsymbol{x}$. $D\pi_{\boldsymbol{x}} : T_{\boldsymbol{x}}\mathcal{S}^{2n-1} \longrightarrow T_{[\boldsymbol{x}]}\mathbb{C}\mathbf{P}^{n-1}$ associates a tangent vector $\boldsymbol{v}_{\boldsymbol{x}}$ at $\boldsymbol{x}$ to a tangent vector $\boldsymbol{w}_{[\boldsymbol{x}]}$ at $[\boldsymbol{x}] \in \mathbb{C}\mathbf{P}^{n-1}$. $\boldsymbol{v}_{\boldsymbol{x}}$ is by definition the velocity vector of a curve in $\mathcal{S}^{2n-1}$, and $\boldsymbol{w}_{[\boldsymbol{x}]}$ is a velocity vector of the image of the curve by π. Now is $\boldsymbol{v}_{\boldsymbol{x}}$ lies in $V_{\boldsymbol{x}}\mathcal{S}^{2n-1}$, then the curve is included in $\mathcal{F}_{[\boldsymbol{x}]}$, and its image by π is a constant curve, with zero velocity. Thus if $\boldsymbol{v}_{\boldsymbol{x}} \in V_{\boldsymbol{x}}\mathcal{S}^{2n-1}$, $D\pi_{\boldsymbol{x}}(\boldsymbol{v}_{\boldsymbol{x}}) = 0$. This explains the term vertical space.

To characterize $H_{\boldsymbol{x}}\mathcal{S}^{2n-1}$, let us consider $\boldsymbol{y} \in T_{\boldsymbol{x}}\mathcal{S}^{2n-1}$. It can be approximated by a vector of $V_{\boldsymbol{x}}\mathcal{S}^{2n-1}$ plus an error term χ orthogonal to $V_{\boldsymbol{x}}\mathcal{S}^{2n-1}$, whence in $H_{\boldsymbol{x}}\mathcal{S}^{2n-1}$. Thus $\boldsymbol{y} = \boldsymbol{j}b\boldsymbol{x} + \chi$, $b \in \mathbb{R}$, such that χ is orthogonal to $V_{\boldsymbol{x}}\mathcal{S}^{2n-1}$, that is to all $\boldsymbol{j}a\boldsymbol{x}$ with a real. Therefore $-\boldsymbol{j}a\boldsymbol{x}^\dagger\chi = 0 = -\boldsymbol{j}a\boldsymbol{x}^\dagger\boldsymbol{y} - ab\boldsymbol{x}^\dagger\boldsymbol{x}$ and then $b = -\boldsymbol{j}\boldsymbol{x}^\dagger\boldsymbol{y}$ which must be real by hypothesis. Hence $\boldsymbol{y}$ can be written as $\boldsymbol{y} = \boldsymbol{j}\mathrm{Im}(\boldsymbol{x}^\dagger\boldsymbol{y})\boldsymbol{x} + \chi$ and $\boldsymbol{y}$ lies in $H_{\boldsymbol{x}}\mathcal{S}^{2n-1}$ if and only if $\mathrm{Im}(\boldsymbol{x}^\dagger\boldsymbol{y}) = 0$. Consider the following linear form

$$
\begin{aligned}
A_{\boldsymbol{x}} : T_{\boldsymbol{x}}\mathcal{S}^{2n-1} &\longrightarrow \mathbb{R} \\
y &\longmapsto A_{\boldsymbol{x}}(\boldsymbol{y}) = \mathrm{Im}(\boldsymbol{x}^\dagger\boldsymbol{y})
\end{aligned} \tag{13}
$$

called Berry connection in the physics literature. The preceding discussion shows that $H_{\boldsymbol{x}}\mathcal{S}^{2n-1} = \ker A_{\boldsymbol{x}}$.

To sum up, the Erhesmann connexion can be defined as follows: For all $\boldsymbol{x} \in \mathcal{S}^{2n-1} = \big(\mathbf{U}(1), \mathbb{C}\mathbf{P}^{n-1}\big)$:

- $A_{\boldsymbol{x}} : T_{\boldsymbol{x}}\mathcal{S}^{2n-1} \longrightarrow \mathbb{R}$ defined by $A_{\boldsymbol{x}}(\boldsymbol{y}) = \mathrm{Im}(\boldsymbol{x}^\dagger\boldsymbol{y})$
- $T_{\boldsymbol{x}}\mathcal{S}^{2n-1} = V_{\boldsymbol{x}}\mathcal{S}^{2n-1} \oplus H_{\boldsymbol{x}}\mathcal{S}^{2n-1}$ where

$$
\begin{aligned}
V_{\boldsymbol{x}}\mathcal{S}^{2n-1} &= \{\boldsymbol{j}a\boldsymbol{x}, a \in \mathbb{R}\} = \ker D\pi_{\boldsymbol{x}} \\
H_{\boldsymbol{x}}\mathcal{S}^{2n-1} &= \big\{\boldsymbol{v} \in T_{\boldsymbol{x}}\mathcal{S}^{2n-1} / \mathrm{Re}(\boldsymbol{x}^\dagger\boldsymbol{v}) = 0\big\} = \ker A_{\boldsymbol{x}}
\end{aligned}
$$

3.4 Horizontal lifting

We just saw that the differential $D\pi_{\boldsymbol{x}}$ of π at $\boldsymbol{x}$ has the vertical space at $\boldsymbol{x}$ as kernel. Therefore, the restriction of $D\pi_{\boldsymbol{x}}$ to $H_{\boldsymbol{x}}$ is a bijective application from $H_{\boldsymbol{x}}\mathcal{S}^{2n-1}$ to $T_{[\boldsymbol{x}]}\mathbb{C}\mathbf{P}^{n-1}$. Its inverse $D\pi_{\boldsymbol{x}}^{-1}$ uniquely associates a tangent vector $\boldsymbol{v}_{[\boldsymbol{x}]}$ at $[\boldsymbol{x}]$ to a tangent vector $\boldsymbol{w}_{\boldsymbol{x}}$ at $\boldsymbol{x}$, called the *horizontal lift* of $\boldsymbol{v}_{[\boldsymbol{x}]}$. Note that $\boldsymbol{v}_{[\boldsymbol{x}]}$ can be lifted to a unique vector $\boldsymbol{w}_{\boldsymbol{y}}$ at any $\boldsymbol{y}$ in the fiber $\mathcal{F}_{[\boldsymbol{x}]}$.

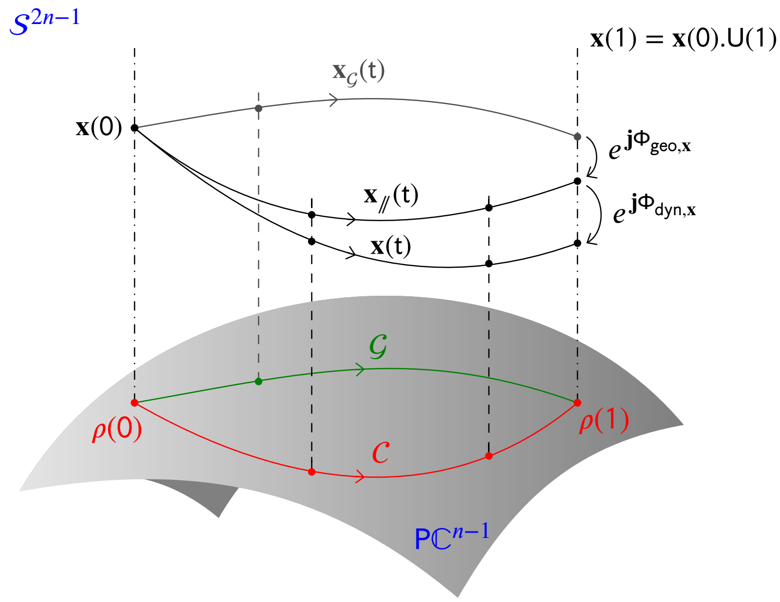

Fig. 2 to be completed. The path of $\boldsymbol{x}_{\mathcal{G}}(t)$ in $\mathcal{S}^{2n-1}$ projects as a geodesic $\mathcal{G}$ in the base space. It is also a geodesic in the total space. Among all these geodesic, the horizontal geodesic remains in phase with $\boldsymbol{x}(0)$. The trajectory $\boldsymbol{x}_{/\!/}(t)$ is the *horizontal lift* of $\mathcal{C}$. The initial point $\boldsymbol{x}(0)$ lives in the fiber above $\rho(0)$. The final point $\boldsymbol{x}(1)$ lives in the fiber above $\rho(1)$. It is related to $\boldsymbol{x}(0)$ by a pure phase term (en element of $U(1)$) which depends on the trajectory followed during the time interval.

Consider now a path $\mathcal{C} = \{\rho(t), t \in [0,T]\}$ in the base space $\mathbb{C}\mathbf{P}^{n-1}$, starting at point $\rho(0)$, as represented in Figure 1. To this path in the base space corresponds many possible paths in the total space $\mathcal{S}^{2n-1}$. This means that there exists infinitely many multivariate signals that may have their projected trajectory over time that is exactly the one followed by $\rho(t)$. However, some peculiar trajectories exist. Using for example the fact that $D\pi_{\boldsymbol{x}}^{-1}$ uniquely associates a tangent vector $\boldsymbol{v}_{[\boldsymbol{x}]}$ at $[\boldsymbol{x}]$ to a tangent vector $\boldsymbol{w}_{\boldsymbol{x}}$ at $\boldsymbol{x}$, $D\pi_{\boldsymbol{x}}^{-1}(\mathcal{C}) \in \mathcal{S}^{2n-1}$ defines a curve such that at each time, the velocity is the horizontal lift of the corresponding velocity $\dot{\rho}(t)$ tangent to the curve $\mathcal{C} \in \mathbb{C}\mathbf{P}^{n-1}$. Equivalently, using Erhesmann connection, this curve is defined by the signal $\boldsymbol{x}(t)$ such that $A_{\boldsymbol{x}(t)}(\dot{\boldsymbol{x}}(t)) = \operatorname{Im}\left(\boldsymbol{x}^H(t)\dot{\boldsymbol{x}}(t)\right) = 0, \forall t$. It is called the *horizontal lift* of $\mathcal{C}$.

In Figure 2, the horizontal lift of the path $\mathcal{C}$ is the trajectory denoted $\boldsymbol{x}_{/\!/}(t)$. It corresponds to a trajectory where the vector $\dot{\boldsymbol{x}}_{/\!/}(t)$ tangent to the trajectory is parallel transported.

3.5 Turning manifolds into Riemanian manifolds, geodesics

We have already encountered a metric above when dealing with unit norm vectors in $\mathbb{C}^n$. The metric considered at $\boldsymbol{x}$ in $T_{\boldsymbol{x}}\mathcal{S}^{2n-1}$ is provided by the usual euclidean inner product of $\mathbb{C}^n$, $\boldsymbol{v}^\dagger\boldsymbol{w}$ where $\boldsymbol{v}$ and $\boldsymbol{w}$ are tangent vectors at $\boldsymbol{x} \in \mathcal{S}^{2n-1}$. This is provided by the identification of $T_{\boldsymbol{x}}\mathcal{S}^{2n-1}$ with $\mathbb{C}^{n-1}$. The metric allows to define the Levi-Civita connection, which in this situation, is simply the orthogonal projection of the usual differential onto the tangent space of the fiber, that is the vertical space.

Precisely, if $\boldsymbol{x}(t)$ is a curve for t in some interval, the covariant derivative of a tangent vector $\boldsymbol{v}(t)$ along the curve is given by

$$\frac{D\boldsymbol{v}(t)}{dt} = \boldsymbol{x}(t)\boldsymbol{x}^{\dagger}(t)\boldsymbol{v}(t)$$

But the euclidean metric can not be used directly for the base space (or quotient) $\mathbb{C}\mathbf{P}^{n-1}$ because it is not invariant under $\sim$. However, extracting from a tangent vector at $\boldsymbol{x}$ its horizontal part may provide a solution. Indeed, let $\boldsymbol{v} \in T_{\boldsymbol{x}}\mathcal{S}^{2n-1}$. Then, its projection onto $H_{\boldsymbol{x}}\mathcal{S}^{2n-1}$ writes $\boldsymbol{v}_{\perp} = \boldsymbol{v} - \boldsymbol{j}A_{\boldsymbol{x}}(v)\boldsymbol{x}$ and can be shown to be covariant to $\sim$. Thus the metric $< \boldsymbol{v}|\boldsymbol{w} >_{\boldsymbol{x}} = \boldsymbol{v}_{\perp}^{\dagger}\boldsymbol{w}_{\perp}$ is invariant to $\sim$ and can be used as a metric in the quotient manifold $\mathbb{C}\mathbf{P}^{n-1}$ to compare the tangent vectors at $[\boldsymbol{x}]$, $d\pi_{\boldsymbol{x}}(\boldsymbol{v})$ and $d\pi_{\boldsymbol{x}}(\boldsymbol{w})$ [14, 15].

This also allows to study geodesics in the quotient manifold. Since the metric in the quotient manifold is evaluated by the metric in the horizontal lifted space, then a geodesic in the quotient manifold will be lifted to an horizontal geodesic in the lifted space. Importantly, a geodesic in the quotient manifold can be lifted to another geodesic which is not horizontal, but whose vertical component does not count for the length in the quotient space.

The expression of the geodesics in $\mathcal{S}^{2n-1}$ are known [14, 15]. Since this manifold is the sphere in $\mathbb{C}^n$, the geodesics can be shown to be

$$\boldsymbol{x}_t = \boldsymbol{x}_0 \cos(\omega t) + \dot{\boldsymbol{x}}_0 \frac{\sin(\omega t)}{\omega}$$

where $\omega^2 = \|\dot{\boldsymbol{x}}_0\|^2$, and with $\boldsymbol{x}_0^{\dagger}\boldsymbol{x}_0 = 1$. It remains horizontal if $\boldsymbol{x}_0^{\dagger}\dot{\boldsymbol{x}}_0 = 0$. Projecting this equation using π would allow to write down the equation of geodesics in the projective space.

Figure 2 displays the geodesic $\mathcal{G}$ as an alternate trajectory between points $\rho(0)$ and $\rho(1)$ in the base space. This trajectory is the equivalent of the straight line for flat spaces. Here, the base space is curved, the geodesic $\mathcal{G}$ between initial and final point is the curve with zero acceleration [10]. Its horizontal lift in the total space is denoted $\boldsymbol{x}_{\mathcal{G}}(t)$ and has no contribution in the vertical space $V_{\boldsymbol{x}}\mathcal{S}^{2n-1}$ while the velocity vector is also parallel transported along the path.

3.6 A geometric view of the phases

As shown using signal processing arguments, the main point of the paper is to show that the phase $\phi_{\boldsymbol{x},t_0}(t) = \arg \boldsymbol{x}(t_0)^{\dagger}\boldsymbol{x}(t)$ is the sum of two terms, the first one being the *dynamical phase*, a phase acquired along time with respect to a particular reference signal; The second one, overlooked in signal theory up to now, is the *geometrical phase*, a phase existing only for multivariate signals, reflecting purely geometrical aspects of the signals and reminiscent of the underlying base space.

We now provide a differential geometry proof of the result, using the developments presented above. Even if inspired by proofs of the existence of the Berry phase in quantum mechanics ([14] for example), the proof presented here is new.

Basically the dynamical phase comes from the behavior of the signal in the total space $\mathcal{S}^{2n-1}$ whereas the geometrical phase shows up from the behavior of the equivalent class of the signal in the base space, *i.e.* the quotient $\mathbb{C}\mathbf{P}^{n-1}$.

To develop our presentation, we consider as in figure (2), two states of a physical system in $\mathbb{C}\mathbf{P}^{n-1}$, and study what can happen for the physical measurements in $\mathcal{S}^{2n-1}$, depending on how the states are joined by a curve along time and by how we lift this curve to the physical measurement $\boldsymbol{x}(t)$.

Thus, we select two times, $t_0 = 0$ and t, such that the physical states at those two times are $\rho(0) \in \mathbb{C}\mathbf{P}^{n-1}$ and $\rho(t) \in \mathbb{C}\mathbf{P}^{n-1}$. Then the physical measurements at 0 and t are elements of the fibers $\mathcal{F}_{[\rho(0)]}$ and $\mathcal{F}_{[\rho(t)]}$ respectively. But these measurements are not unique in their fiber and differ within a fiber by a pure phase term. In other words, if $\mathcal{C} = \{\rho(u), u \in [0,t]\}$ is the curve in $\mathbb{C}\mathbf{P}^{n-1}$ followed by the state of the system, a possible measurement consists in a lifted curve $\overline{\mathcal{C}} \in \mathcal{S}^{2n-1}$, leaving $\mathcal{F}_{[\rho(0)]}$ at time 0 and arriving at $\mathcal{F}_{[\rho(t)]}$ at time t. Any signal $\boldsymbol{x}(t)$ such that $\{\pi(\boldsymbol{x}(u)) = \rho(u)), u \in [0,t], \pi(\boldsymbol{x}(0)) = \rho(0), \pi(\boldsymbol{x}(t)) = \rho(t)\}$ is a possible physical measurement.

We will however fix the initial condition of the measurement to $\boldsymbol{x}(0) \in \mathcal{F}_{[c(0)]}$, so-that all possible signals mentionned above start at $\boldsymbol{x}(0)$ at time 0.

We now show that two different phases appear in a generic physical measurement: The first one is purely geometric and comes from the geometry of the curve $\mathcal{C}$ traced in the base space from $\rho(0)$ to $\rho(t)$; The second one is purely dynamical and reflects the behavior of the lifted curve $\overline{\mathcal{C}}$ in the physical space.

1. Let $\mathcal{C}_g$ be a geodesic in $\mathbb{C}\mathbf{P}^{n-1}$ joining $\rho(0)$ to $\rho(t)$. Then we saw that a lifting of this curve is a geodesic and hence will be described by $\boldsymbol{x}_g(u), \mathcal{S}^{2n-1}$ for $u \in [0,t]$, and thus writes $\boldsymbol{x}_g(u) = \boldsymbol{x}(0)\cos(\omega u) + \dot{\boldsymbol{x}}_0 \frac{\sin(\omega u)}{\omega}$ with $\mathrm{Re}(\boldsymbol{x}(0)^\dagger \dot{\boldsymbol{x}}(0)) = 0$. Note that in general this measurement is valid only at time 0 and time t, since there is not reason that $\pi(\boldsymbol{x}_g(u)) = \rho(u)$ for $u \in (0,t)$. We will see later why this is the root of the geometrical aspect of the geometrical phase.

   As we saw above, $\boldsymbol{x}_g(u)$ can be chosen to remain horizontal if $\mathrm{Im}(\boldsymbol{x}(0)^\dagger \dot{\boldsymbol{x}}(0)) = 0$. In this situation we denote it as $\boldsymbol{x}_{g/\!/}(u)$, and the curve $\mathcal{C}_g$ is lifted to the horizontal geodesic $\overline{\mathcal{C}}_{g/\!/}$ traced out by the section $\boldsymbol{x}_{g/\!/u}, u \in [0,t]$. Therefore, using the equation of $\boldsymbol{x}_{g/\!/u}$ we obtain $\boldsymbol{x}^\dagger_{g/\!/}(u)\boldsymbol{x}_{g/\!/}(0) \in \mathbb{R}^+, \forall u \in [0,t]$ and $\boldsymbol{x}_{g/\!/}(t)$ remains in phase with $\boldsymbol{x}_{g/\!/}(0) = \boldsymbol{x}(0)$ or $\phi_{\boldsymbol{x}_{g/\!/},0}(u) = 0$, $\forall u \in [0,t]$. Thus $\boldsymbol{x}_{g/\!/}(u), u \in [0,t]$ provides a perfect reference signal to compare measurements having a same physical state.
2. Suppose that $\mathcal{C}$ is not a geodesic in $\mathbb{C}\mathbf{P}^{n-1}$ joining $\rho(0)$ to $\rho(t)$.

   Consider $\boldsymbol{x}_{/\!/}(u), u \in [0,t]$ the horizontal lift such that $\boldsymbol{x}_{/\!/}(0) = \boldsymbol{x}_0 = \boldsymbol{x}_{g/\!/}(0) = \boldsymbol{x}(0)$, which traces out the curve $\overline{\mathcal{C}}_{/\!/}$. Then at time t, $\boldsymbol{x}_{/\!/}(t)$ and $\boldsymbol{x}_{g/\!/}(t)$ are in the same fiber $\mathcal{F}_{[\rho(t)]}$ since they project down to the same point $\rho(t)$ in $\mathbb{C}\mathbf{P}^{n-1}$. Thus there exist a phase $\varphi_g(t)$ such that $\boldsymbol{x}_{/\!/}(t) = e^{\boldsymbol{j}\varphi_g(t)}\boldsymbol{x}_{g/\!/}(t)$. We call $\varphi_g(t)$ the geometrical phase at time t.
3. Consider now $\boldsymbol{x}(u), u \in [0,t]$ a non horizontal lift of $\mathcal{C}$. $\boldsymbol{x}(u)$ and $\boldsymbol{x}_{/\!/}(u)$ have thus the same projection by π, the same initial condition, but differ by a pure phase term (unit complex number) in the total space at all time $u \in [0,t]$, or $\boldsymbol{x}(u) = e^{\boldsymbol{j}\psi(u)}\boldsymbol{x}_{/\!/}(u)$, with $\psi_0 = 0$. If we now evaluate the covariant derivative of $\boldsymbol{x}(u)$ we

get

$$\frac{D\boldsymbol{x}(u)}{du} \underbrace{=}_{\text{def.}} \boldsymbol{x}(u)\boldsymbol{x}(u)^\dagger \dot{\boldsymbol{x}}(u)$$

$$\underbrace{=}_{\text{Leibnitz rule}} \boldsymbol{j}\dot{\psi}(u)\boldsymbol{x}(u) + e^{\boldsymbol{j}\psi(u)}\frac{D\boldsymbol{x}_{/\!/}(u)}{du}$$

Since $\boldsymbol{x}_{/\!/}(u)$ follows the *parallel transport*, its covariant derivative is zero

$$\frac{D\boldsymbol{x}_{/\!/}(u)}{du} = 0$$

and $\psi(u) \in \mathbb{R}$ is then determined by the differential equation

$$\dot{\psi}(u) = -\boldsymbol{j}\boldsymbol{x}(u)^\dagger \dot{\boldsymbol{x}}(u) = A_{\boldsymbol{x}(u)}(\dot{\boldsymbol{x}}(u))$$

whose solution reads

$$\psi(t) = \int_0^t A_{\boldsymbol{x}(u)}(\dot{\boldsymbol{x}}(u))du$$

We insist again on the fact that $\boldsymbol{x}(u)$ and $\boldsymbol{x}_{/\!/}(u)$ are equivalent according to $\sim$: In the bivariate setting for example, these signals could be two electromagnetics waves having the same polarization state over time.

$\psi(t)$ only depends on the curve followed, not on the speed at which it is traced out. Indeed, the previous integral is invariant under a reparametrization $\boldsymbol{x} \mapsto \boldsymbol{y}$ given by any bijection $u \mapsto s(u)$ such that $\boldsymbol{y}(s) = \boldsymbol{x}(s(u))$. Hence, we call this phase *the dynamical phase* and change a bit the notation to $\varphi_d(t) = \psi(t)$. *The dynamical phase hence measures the departure of the path followed by $\boldsymbol{x}(t)$ from the path followed by the parallel transport $\boldsymbol{x}_{/\!/}(t)$.*

Back to the instantaneous phase. We previously showed that at all time $u \in [0,t]$, $\boldsymbol{x}(u) = e^{\boldsymbol{j}\varphi_d(u)}\boldsymbol{x}_{/\!/}(u)$ and also that $\boldsymbol{x}_{/\!/}(t) = e^{\boldsymbol{j}\varphi_g(t)}\boldsymbol{x}_{g/\!/}(t)$. We thus end up with

$$\boldsymbol{x}(t) = e^{\boldsymbol{j}(\varphi_d(t)+\varphi_g(t))}\boldsymbol{x}_{g/\!/}(t)$$

Therefore, the instantaneous phase of $\boldsymbol{x}(t)$ relative to $t=0$ is given by

$$\begin{aligned}\phi_{\boldsymbol{x},0}(t) = \arg(\boldsymbol{x}^\dagger(0)\boldsymbol{x}(t)) &= \varphi_g(t) + \varphi_d(t) + \arg(\boldsymbol{x}^\dagger_{g/\!/}(0)\boldsymbol{x}_{g/\!/}(t)) \\ &= \varphi_g(t) + \varphi_d(t)\end{aligned}$$

because we saw $\boldsymbol{x}^\dagger_{g/\!/}(0)\boldsymbol{x}_{g/\!/}(u) \in \mathbb{R}^+, \forall u \in [0,t]$.

This concludes the proof of the main statement. It remains to give an explicit form to the geometric phase. As already shown in the previous section, we can have two definitions for it. Let $\boldsymbol{x}(t)$ be factored as $a_{\boldsymbol{x}}(t)\boldsymbol{p}_{\boldsymbol{x}}(t)$ where $\boldsymbol{p}_{\boldsymbol{x}}(t)$ is a unit norm vector in $\mathbb{C}^n$ at all time. Let signal $\boldsymbol{x}$ be observed on the interval $[t_0, t]$. Then we have

$$\begin{aligned}\varphi_d(t) = \phi_{d,\boldsymbol{x},t_0}(t) \;&=\; \int_{t_0}^{t} A_{\boldsymbol{x}(u)}(\dot{\boldsymbol{x}}(u))du = \int_{t_0}^{t} \operatorname{Im}\left(\frac{\boldsymbol{x}^\dagger(u)\dot{\boldsymbol{x}}(u)}{\boldsymbol{x}^\dagger(u)\boldsymbol{x}(u)}\right) du \\ \varphi_g(t) = \phi_{g,\boldsymbol{x},t_0}(t) \;&=\; \phi_{\boldsymbol{x},t_0}(t) - \phi_{d,\boldsymbol{x},t_0}(t)\end{aligned}$$

which are definitions in agreement with equations (11).

3.7 The geometrical phase as the holonomy of a closed curve

Physicist readers familiar with the geometric or Berry phase may complain that usually the geometric phase is defined as the holonomy of a loop traced by a physical system in the base space. However, our proof above apparently does not close any loop. In fact, the loop is closed by the choice of the reference horizontal lifted geodesic. We develop this now.

Let $\overline{\mathcal{C}}_t$ be any curve in the fiber $\mathcal{F}_{[\rho(t)]}$ joining $\boldsymbol{x}_t \in \mathcal{F}_{[\rho(t)]}$ to $\boldsymbol{x}_{g/\!/}(t) \in \mathcal{F}_{[\rho(t)]}$. Let the closed curve $\overline{\mathcal{C}}_\circ = \overline{\mathcal{C}} \cup \overline{\mathcal{C}}_t \cup \overline{\mathcal{C}}_{g/\!/}$. We now relate the geometric phase to $\oint_{\overline{\mathcal{C}}_\circ} A_{\boldsymbol{x}}$.

The curve $\overline{\mathcal{C}}_\circ$ can be parametrized by a function $\boldsymbol{y}(u)$ from an interval $I_T = [0, T]$ in $\mathbb{R}$ to $\mathcal{S}^{2n-1}$ such that : $t \in I_T$, $\boldsymbol{y}(u) = \boldsymbol{x}(u), \forall u \in [0, t]$, $\boldsymbol{y}(u) = e^{\boldsymbol{j}\alpha(u)}\boldsymbol{x}(t)$ for $u \in [t, s]$ (for some $T > s \geq t$ and some $\alpha(t)$ ensuring $\boldsymbol{y}$ to remain continuous and differentiable), and finally $\boldsymbol{y}(t) = \boldsymbol{x}_{g/\!/}(T - u)$ for $u \in [s, T]$.

In other words, $\boldsymbol{y}(u)$ traces the closed curve following the physical measurement $\boldsymbol{x}(t)$ from 0 to t, then evolving from t to s in the fiber $\mathcal{F}_{[\rho(t)]}$, joining $\boldsymbol{x}(t)$ to $\boldsymbol{x}_{g/\!/}(t)$, and then going back to the initial measurement $\boldsymbol{x}(0)$ at time T.

Integrating the 1-form along the close curve $\overline{\mathcal{C}}_\circ$ we get

$$\oint_{\overline{\mathcal{C}}_\circ} A_{\boldsymbol{y}}(\boldsymbol{u}) \;=\; \int_{\overline{\mathcal{C}}} A_{\boldsymbol{y}}(\boldsymbol{u}) + \int_{\overline{\mathcal{C}}_t} A_{\boldsymbol{y}}(\boldsymbol{u}) + \int_{-\overline{\mathcal{C}}_g} A_{\boldsymbol{y}}(\boldsymbol{u})$$

As shown before $\int_{\overline{\mathcal{C}}} A_{\boldsymbol{y}} = \varphi_d(t)$ and $\int_{-\overline{\mathcal{C}}_g} A_{\boldsymbol{y}} = 0$. For the contribution on $\overline{\mathcal{C}}_t$ in the fiber, the signal reads $\boldsymbol{y}(u) = e^{\boldsymbol{j}\alpha(u)}\boldsymbol{x}(t)$ for $u \in [t, s]$ for some phase $\alpha(u)$. Since $\boldsymbol{y}(t) = \boldsymbol{x}(t)$ we need $\alpha(t) = 0$. Furthermore, since $\boldsymbol{y}(s) = e^{\boldsymbol{j}\alpha(s)}\boldsymbol{x}(s) = \boldsymbol{x}_{g/\!/}(t)$ we also need $\alpha(s) = -(\varphi_g(t) + \varphi_d(t))$.

Now, we have $\dot{\boldsymbol{y}}(u) = \boldsymbol{j}\dot{\alpha}(u)\boldsymbol{y}(u)$ so that $\alpha(s) - \alpha(t) = \int_{\overline{\mathcal{C}}_t} A_{\boldsymbol{y}}$. Adding up the three integrals then leads to

$$\varphi_g(t) = -\oint_{\overline{\mathcal{C}}_\circ} A_{\boldsymbol{x}}$$

It is evident that the sign above comes from the direction of time: Reversing the direction, *i.e.* tracing $\overline{\mathcal{C}}_\circ$ in the opposite direction, would reverse the sign.

We would like the same kind of result in $\mathbb{C}\mathbf{P}^{n-1}$. Using results on integration in manifolds from [16], we would have an immediate result if $A_{\boldsymbol{x}}$ was the pullback by π of a 1-form on $\mathbb{C}\mathbf{P}^{n-1}$. But, as noticed by Mukunda & Simon [14], this is not the case.

In fact, as shown below, φ_g can be calculated along curves in $\mathbb{C}\mathbf{P}^{n-1}$ but not in an additive way (with respect to curve splitting), excluding the possibility to calculate it as the integral of any 1-form along a curve.

However, this conclusion is reversed when considering 2-forms. Stokes theorem [16] stipulates that

$$\varphi_g(t) = -\oint_{\overline{\mathcal{C}}_\circ} A_{\boldsymbol{x}} = -\iint_{\overline{\mathcal{S}}_\circ} dA_{\boldsymbol{x}}$$

where $\overline{\mathcal{S}}_\circ$ is any surface such that $\overline{\mathcal{C}}_\circ = \partial\overline{\mathcal{S}}_\circ$, and where $dA_{\boldsymbol{x}}$ is the exterior derivative of $A_{\boldsymbol{x}}$. It is a 2-form, that is an alternating second order differential form or equivalently a second order antisymmetric covariant tensor. As shown by Mukunda & Simon [14], there exists a 2-form Ω on $\mathbb{C}\mathbf{P}^{n-1}$ such that $dA_{\boldsymbol{x}}$ is the pullback by the natural projection π of Ω. Using results from Lee [16] allows to conclude that

$$\varphi_g(t) = \phi_{g,\boldsymbol{x},t_0}(t) = -\iint_{\overline{\mathcal{S}}_\circ} dA_{\boldsymbol{x}} = -\iint_{\mathcal{S}_\circ} \Omega$$

where $\mathcal{S}_\circ$ is any surface in $\mathbb{C}\mathbf{P}^{n-1}$ such that $\partial\mathcal{S}_\circ = \pi(\overline{\mathcal{C}}_\circ) = \mathcal{C}_\circ$. This quantity is called the holonomy of the curve.

3.8 To sum up...

To end these general theoretical sections, we recall and emphasize some of the main properties of the geometric phase already known in quantum mechanics [14], but here presented in a signal processing language:

- *Reference and time interval*: Geometric phase is defined with respect to a reference time t_0 and is a function of time t (integration from t_0 to t). The geometric phase is a quantity accumulated over a time period.
- *Gauge invariance* : Signals $\boldsymbol{x}(t)$ and $\boldsymbol{x}(t)e^{\boldsymbol{j}\alpha(t)}$ may accumulate the same geometric phase over a time interval. It is a pure geometric quantity as it only depends on the path in the base space.
- *Time reparametrization*: For a given smooth bijective function $s : [t_0, t] \to s([t_0, t])$, $\boldsymbol{x}(t)$ and $\boldsymbol{x}(s(t))$ have the same geometric phase, meaning that it does not matter how fast or slow the path is followed. This means that the geometric phase is independent of the carrying frequency of the signal. Said differently, two signals having different carrying frequency accumulate the same geometric phase if their initial and final points are the same in the projective space.
- *Geodesic path*: The geometric phase accumulated along a geodesic path is null. For a geodesic, the tangent vector to $\overline{\mathcal{C}}_g$ has no component in the vertical space (it is an horizontal lift), and also $\mathrm{Im}\left(\boldsymbol{p}^\dagger(t)\dot{\boldsymbol{p}}(t)\right) = 0\,\forall t$. For a signal $\boldsymbol{x}(t)$, it means that there exists some paths in the projected space that leave the phase unchanged, however there is a change in the values of $\rho(t)$. In the bivariate case, this means that if $\rho(t)$ follows a a great circle on the sphere, then the signal has no geometric phase accumulated along time, as soon as the great circle does not cross the opposite pole.

In this case, the closing geodesic is itself and the closed loop encloses a zero surface. In the contrary, the closing geodesic completes the great circle, and the closed loop encloses a non zero surface (in the bivariate case, the bivariate case, this surface will be shown to be a half sphere, and the geometric phase experiences a π jump at each crossing of an antipodal point). PO:A AMELIORER.

- Single point path: If the path in the projective space reduces to a single point, then there is no geometric phase accumulated. The point like trajectory can be due to: *i)* a time interval that is vanishing, such that $\lim_{\Delta t \to 0} \Phi_{geo,\boldsymbol{x}}(\Delta t) = 0$ (one can then see directly from the definition in (10)), *ii)* a constant value for $\rho(t)$, meaning that the geometric phase term $\operatorname{Im}\left(\boldsymbol{p}^H(t)\dot{\boldsymbol{p}}(t)\right)$ vanishes as $\boldsymbol{p}(t)$ is actually a constant. In the bivariate case, it means that the polarization state of the signal is constant, meaning that a geometric phase can only appear for a bivariate signal if a change occurs in its polarization state $\boldsymbol{p}(t)$ along time.
- Area surrounded by $\mathcal{C}$: The geometric phase is related to the area surrounded by the path $\mathcal{C}$. If the path $\mathcal{C}$ between $t = 0$ and $t = 1$ is a *loop* ($\rho(0) = \rho(1)$), then the geometric phase can be obtained from the area inside the loop [10, chapter 10]. If the path is *open* (such that $\rho(1) \neq \rho(0)$), then it is possible to close the loop by a geodesic from $\rho(1)$ and $\rho(0)$ as geodesic paths has null geodesic contribution, so that the area surrounded is still connected to the geometric phase.

We turn in the next section to the application of all this theory to the case of modulated mulivariate signals, make the link with the work of Lilly.

4 The geometric phase of modulated multivariate signals

The signals we are mostly interested in are measurements by sensors of some two or three dimensional waves propagating in some medium. These can be seismic waves, electromagnetic waves such as light, or even gravitational waves.

These waves are described in physics by complex fields in space and time. We focus here on a measurement at one point in space, and therefore the fields are represented by multivariate signals taking complex values. Furthermore we assume that the evolution is mostly related to an amplitude modulated description, meaning that the fields can be represented as a complex time dependent vector modulated by a complex scalar signal. The aim of the section is to described in this particular case the occurence and meaning of a geometric phase. We begin however by a brief overview of an important existing framework developed by Lilly and co-workers.

4.1 Modulated multivariate signals: Lilly's approach

The AM-FM case was developed in several papers by Lilly and his co-authors and we refer to these papers for a nice presentation. However, Lilly restricted the study (more precisely the examples) to the cases in which the unit vector is constant in time. This leads to his missing of the existence of the geometrical phase for multivariate signals. However, his interesting approach leaded to a nice definition of AM-FM signal in the multivariate setting. We recall his approach here and show the link with the developments above.

Basically, he studied multidimensional measurements of AM-FM signals, *e.g.* $(b_1(t)\cos(f_1(t)), \dots, b_n(t)\cos(f_n(t)))^\top$ where the Bedrosian conditions[3] are satisfied for all components. A complex signal can be created by using the analytic signal component wise and writes $\boldsymbol{x}(t) = (a_1(t)\exp \boldsymbol{j}\varphi_1(t)) \dots a_n(t)\exp \boldsymbol{j}\varphi_n(t))^\top$. The pairs (a_k, φ_k) are called canonical pairs ([17]) for the components.

Lilly exposes how to extract from these multiple measurements a common phase which allow to write the analytic signal as

$$\boldsymbol{x}(t) = a_{\boldsymbol{x}}(t) e^{\boldsymbol{j}\varphi_{\boldsymbol{x}}(t)} \boldsymbol{p}_{\boldsymbol{x}}(t) \tag{14}$$

where the extraction means that the parameterization is (almost) unambiguous. The idea is to consider the notion of a modulated multivariate oscillation as multicomponents sharing locally a same carrier.

A typical signal is $\boldsymbol{x}(t) = \exp(\boldsymbol{j}\omega_0 t)\boldsymbol{x}_0$, $\boldsymbol{x}_0$ being a constant vector. Then $\boldsymbol{x}(t+\tau) = \exp(\boldsymbol{j}\omega_0\tau)\boldsymbol{x}(t)$ such that $\boldsymbol{x}(t+\tau)$ is obtained from $\boldsymbol{x}(t)$ by a phase shift. This obviously cannot be true for general multivariate oscillations, but gives a hint to approach locally $\boldsymbol{x}(t+\tau)$.

Write $\boldsymbol{x}(t+\tau) = \exp(\boldsymbol{j}\omega(t)\tau)\tilde{\boldsymbol{x}}(t,\tau)$ where $\tilde{\boldsymbol{x}}(t,\tau) = \exp(-\boldsymbol{j}\omega(t)\tau)\boldsymbol{x}(t+\tau)$ is a local demodulation (or phase shift) of $\boldsymbol{x}(t+\tau)$. $\omega(t)$ is *a priori* chosen arbitrarily for the moment. Lilly then proposes that locally (*i.e.* for small τ) $\tilde{\boldsymbol{x}}(t,\tau)$ should be close to $\boldsymbol{x}(t)$. Defining $\boldsymbol{x}_i(t,\omega_t) = \partial\left[\tilde{\boldsymbol{x}}(t,\tau)\right]/\partial\tau^i|_{\tau=0}$, Taylor expanding $\tilde{\boldsymbol{x}}(t,\tau)$ leads to the approximation

$$\boldsymbol{x}(t+\tau) \,=\, e^{\boldsymbol{j}\omega(t)\tau}\boldsymbol{x}_t + e^{\boldsymbol{j}\omega(t)\tau}(\boldsymbol{x}_1(t,\omega(t)) + \frac{\tau^2}{2}\boldsymbol{x}_2(t,\omega(t)) + O(\tau^3))$$

If $\omega(t)$ is chosen equal to the constant ω_0, the approximation is perfect if $\boldsymbol{x}_1(t,\omega(t)) = \dot{\boldsymbol{x}}(t) - \boldsymbol{j}\omega(t)\boldsymbol{x}(t) = 0$, an ODE whose solution is hopefully $e^{\boldsymbol{j}\omega_0\tau}$!

For general multivariate oscillations, the idea of Lilly is to choose $\omega(t)$ so as to minimize the first order reminder. The relative quadratic error to minimize is given by

$$v_{\boldsymbol{x}}^2 \,=\, \frac{\|\boldsymbol{x}_1(t,\omega(t))\|^2}{\boldsymbol{x}(t)^\dagger\boldsymbol{x}(t)} = \frac{\|\dot{\boldsymbol{x}}(t) - \boldsymbol{j}\omega_t\boldsymbol{x}(t)\|^2}{\boldsymbol{x}(t)^\dagger\boldsymbol{x}(t)}$$

Note this is the leading order error in τ when approximating $\boldsymbol{x}(t+\tau)$ by $e^{-\boldsymbol{j}\omega(t)\tau}\boldsymbol{x}(t)$. The optimal phase (in the sense considered) is shown to be

$$\omega_{\boldsymbol{x}}(t) = \frac{\mathrm{Im}(\boldsymbol{x}^\dagger(t)\dot{\boldsymbol{x}}(t))}{\boldsymbol{x}(t)^\dagger\boldsymbol{x}(t)}$$

[3] Recall that for one component $y(t) = b(t)\cos(f(t))$ Bedrosian conditions mean that $b(t)$ is a low frequency signal, whereas $\cos(f(t))$ is a high frequency signal, and that the frequency bands of the two signals do not overlap. This implies that the Hilbert transform $\mathcal{H}(y(t))$ can be written as $b(t)\mathcal{H}(\cos(f(t)))$ and thus the analytic signal of $y(t)$ written as $b(t)\left(\cos(f(t)) + \boldsymbol{j}\mathcal{H}(\cos(f(t)))\right) = \rho(t)\exp(\boldsymbol{j}\varphi(t))$. As recalled in [17] there is no reason for having $\mathcal{H}(\cos(f(t))) = \sin(f(t))$ a nd hence having $\varphi(t) = f(t)$ and $\rho(t) = b(t)$. $(\rho(t), \varphi(t))$ is called a canonical pair.

This extension comes from a direct generalization of what is known for the scalar case to the bivariate case. Indeed, the idea is to keep the duality between the time and the frequency domains in the definition of a mean frequency. This can be defined as

$$\langle \nu_{\boldsymbol{x}} \rangle = E_{\boldsymbol{x}}^{-1} \int \nu \|\boldsymbol{X}(\nu)\|^2 d\nu$$

where $E_{\boldsymbol{x}} = \int \|\boldsymbol{X}(\nu)\|^2 d\nu = \int \|\boldsymbol{x}(t)\|^2 dt$ is the energy of the signal. An instantaneous frequency $\nu_{\boldsymbol{x}}(t)$ can be defined a time dependent quantity such that

$$\langle \nu_{\boldsymbol{x}} \rangle = E_{\boldsymbol{x}}^{-1} \int \nu_{\boldsymbol{x}}(t) \|\boldsymbol{x}(t)\|^2 dt$$

In the scalar case, the instantaneous frequency is the imaginary part of the derivative of the log of the signal. It can also be written as the imaginary part of $x^*(t)\dot{x}(t)/|x|^2$, whence the generalisation which satisfies the previous equation

$$2\pi \nu_{\boldsymbol{x}}(t) = \frac{\mathrm{Im}(\boldsymbol{x}^\dagger(t)\dot{\boldsymbol{x}}(t))}{\boldsymbol{x}(t)^\dagger \boldsymbol{x}(t)} = A_{\boldsymbol{x}(t)}(\dot{\boldsymbol{x}}(t))$$

where $A_{\boldsymbol{x}}$ is the linear form defined in equation (13). This is really done by generalizing the scalar to the multivariate case. Furthermore, the dispersion of energy around the mean frequency can also have a time counterpart. An instantaneous energy can be defined as the function $\sigma_{\boldsymbol{x}}^2(t)$ which satisfies

$$\overline{\sigma_{\boldsymbol{x}}}^2 = E_{\boldsymbol{x}}^{-1} \int (\nu - \overline{\nu_{\boldsymbol{x}}})^2 \|\boldsymbol{X}(\nu)\|^2 d\nu = E_{\boldsymbol{x}}^{-1} \int \sigma_{\boldsymbol{x}}^2(t) \|\boldsymbol{x}(t)\|^2 dt$$

A solution to this is

$$\sigma_{\boldsymbol{x}}^2(t) = \frac{\|\dot{\boldsymbol{x}}_t - 2\pi \boldsymbol{j} \overline{\nu_{\boldsymbol{x}}} \boldsymbol{x}(t)\|^2}{\boldsymbol{x}(t)^\dagger \boldsymbol{x}(t)}$$

which is linked to the above $v_{\boldsymbol{x}}^2$ by

$$\min_{\omega_t} v_{\boldsymbol{x}}^2 = \sigma_{\boldsymbol{x}}^2(t) - (2\pi\nu_{\boldsymbol{x}}(t) - 2\pi\langle \nu_{\boldsymbol{x}} \rangle)^2$$

Lilly calls this the instantaneous bandwidth which corresponds to the part of the second order moments not accounted for by the deviation of the instantaneous frequency from the mean frequency. Or as seen as an estimation measure, it relates the MMSE to the variance and the bias.

The interesting point made here, is the fact that the instantaneous frequency as defined by Lilly corresponds to the definition coming from differential geometry which leads to the notion of dynamical phase. Since the approach of Lilly considers constant vectors $\boldsymbol{p}_x$ in equation (14), the geometric phase is zero and Lilly was not able to put it in light in full generality. Note however than in his 2011 paper [5, eq (43)] for the case $n = 3$, Lilly decomposes the phase[4] into "two terms, (i) phase progression of the "particle" along the ellipse periphery and (ii) the combined effect of internal precession and external precession of the ellipse.", which clearly corresponds to the dynamical phase (i) and the geometrical phase (ii), giving by the way an alternate form to the analytical description to be presented in section 4.5 below.

4.2 Modulated multivariate signals: instantaneous frequency and geometric phase.

We now apply our general setting to the particular case of AM-FM modulated signal. As did by Lilly, we thus suppose that the analytic signal measured writes

$$\boldsymbol{x}(t) = a_{\boldsymbol{x}}(t) e^{\boldsymbol{j}\varphi_{\boldsymbol{x}}(t)} \boldsymbol{p}_{\boldsymbol{x}}(t) \tag{15}$$

where $\boldsymbol{p}_{\boldsymbol{x}}(t)$ is a unit norm complex vector. Note that the phase ambiguity remains, and this allows to choose the parametrization of $\boldsymbol{p}_{\boldsymbol{x}}(t)$ as we whish as soon as it remains in $\mathcal{S}^{2n-1}$. We can for example decide that its first component is real. Any other choice leads in a modification of $\varphi_{\boldsymbol{x}}(t)$.

As shown in section 2, the dynamical and geometrical phases of $\boldsymbol{x}(t)$ read

$$\begin{aligned} \phi_{d,\boldsymbol{x},t_0}(t) &= \int_{t_0}^{t} \frac{\operatorname{Im}\left(\boldsymbol{x}^\dagger(s)\dot{\boldsymbol{x}}(s)\right)}{||\boldsymbol{x}(s)||^2}\, ds \\ \phi_{g,\boldsymbol{x},t_0}(t) &= \arg\left(\boldsymbol{p}_{\boldsymbol{x}}(t_0)^\dagger \boldsymbol{p}_{\boldsymbol{x}}(t)\right) - \int_{t_0}^{t} \operatorname{Im}\left(\boldsymbol{p}_{\boldsymbol{x}}^\dagger(s)\dot{\boldsymbol{p}}_{\boldsymbol{x}}(s)\right) ds \end{aligned}$$

and the instantaneous phase relative to t_0 is their sum $\phi_{\boldsymbol{x},t_0}(t) = \phi_{d,\boldsymbol{x},t_0}(t) + \phi_{g,\boldsymbol{x},t_0}(t)$.

Note that the result in equation (10) suggests that the definition of the instantaneous frequency for a multivariate signal should be discussed. Any fluctuation over time of the vector $\boldsymbol{p}_{\boldsymbol{x}}(t)$ is likely to add a contribution to the instantaneous frequency. This was actually noted by Cano in [18, Section 1.4] while defining the moments of a quaternion time-frequency density for AM-FM-PM bivariate signals, but not identified as a geometric phase at that time. A detailed derivation of the result in [18, Section 1.4] is given in C to highlight the fact that the geometric phase term can be obtained through an alternative approach, close to the one in [4].

4.3 Hyperspherical parametrization

Explicit calculations may have an interest in some situations, but lead to cumbersome interpretations. An example of applications where this could be useful is array processing in order to study how fluctuations of the sensors may be monitored by tracking geometrical phases. We keep this idea for further work, even if we propose a simple example of the idea in section 6 where we detect a fluctuation of one sensor in an array using the geometric phase.

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[4] The instantaneous frequency in Lilly's paper.

For the moment we illustrate how we can parametrize vector $\boldsymbol{p_x}$ to have explicit expressions of the different phases and of the instantaneous frequency of a n-dimensional AM-FM signals. A simple parametrization uses a generalization to complex spaces of hyperspherical coordinates, generalizing Jones vector to dimensions higher than 2 (see *e.g.* [19]).

In dimension n, vector $\boldsymbol{p_x}$ has components $p_i, i = 1, \dots, n$ which in an hyperspherical parametrization may be written as

$$\begin{aligned} p_i(t) &= \cos(\theta_i(t)) e^{\boldsymbol{j}\chi_i(t)} \prod_{k=1}^{i-1} \sin(\theta_i(t)) \\ \chi_1(t) &= 0 = \theta_n(t) \end{aligned}$$

where the product term is intended to be equal to 1 for $i = 1$. Note that since we set $\chi_1(t) = 0 = \theta_n(t)$ the vector is in dimension $2n - 2$, is of unit norm and hence an element of $\mathbb{C}\mathbf{P}^{n-1}$.

For $n = 2$ we recover Jones vector $(\cos(\theta); \sin(\theta)\exp(\boldsymbol{j}\chi)), \theta \in [0, \pi], \chi \in [0, 2\pi]$: θ is the latitude measured from the North pole, χ the longitude.

The berry connection (equivalently the 1-form $A_{\boldsymbol{x}}$) reads

$$\operatorname{Im}\left(\boldsymbol{p}_{\boldsymbol{x}}^{\dagger}(t)\dot{\boldsymbol{p}}_{\boldsymbol{x}}(t)\right) = \sum_{i=1}^{n} \dot{\chi}_i \cos^2(\theta_i(t)) \prod_{k=1}^{i-1} \sin^2(\theta_k(t))$$

The second term needed to evaluate the geometrical phase is $\arg\left(\boldsymbol{p_x}(t_0)^{\dagger}\boldsymbol{p_x}(t)\right)$. A general form for the inner product is

$$\boldsymbol{p}_{\boldsymbol{x}}^{\dagger}(t_0)\boldsymbol{p}_{\boldsymbol{x}}(t) = \sum_{i=1}^{n} \cos(\theta_i(t_0)) \cos(\theta_i(t)) e^{\boldsymbol{j}(\chi_i(t) - \chi_i(t_0))} \prod_{k=1}^{i-1} \sin(\theta_k(t_0)) \sin(\theta_k(t))$$

from which the argument can be extracted. A general formula for it is awfull, and we will concentrates only on $n = 2$ and 3 since the main applications we focus on in this paper are the analysis of polarization of waves in two or three dimensions. Furthermore, the representation of vector $\boldsymbol{p_x}$ can be studied in Euclidean geometry by identification of $\mathbb{C}\mathbf{P}^1$ as the sphere $\mathcal{S}^2 \in \mathbb{R}^3$. For $n = 2$, the sphere is also called the Poincaré sphere (polarization literature) of the Bloch sphere (quantum mechanics literature). For $n > 2$, as we illustrate for the trivariate case, $\mathcal{S}^2 \in \mathbb{R}^3$ remains of interest, but the evolution of $\boldsymbol{p_x}$ is represented by the evolution of a constellation of $n - 1$ dots on the sphere, a representation called the Majorana stellar representation. The two following sections detail the bivariate and the trivariate case.

4.4 The bivariate case

Several models for bivariate modulated signals have been proposed, for example in [4], [20], [9]. Many parametrization exists for vector $\boldsymbol{p_x}$ and as a consequence for $\varphi_{\boldsymbol{x}}$. In essence, the models are equivalent and consist in a set of four parameters.

The first parametrization directly models the ellipse of the polarization vector. The four parameter are denoted as $\{a_{\boldsymbol{x}}(t), \varphi_{\boldsymbol{x}}(t), \theta_{\boldsymbol{x}}(t), \chi_{\boldsymbol{x}}(t)\}$ such that the complex modulated bivariate signal $\boldsymbol{x}(t)$, as given in [21], reads:

$$\begin{aligned}\boldsymbol{x}(t) &= a_{\boldsymbol{x}}(t)e^{\boldsymbol{j}\varphi_{\boldsymbol{x}}(t)}\begin{pmatrix}\cos\theta_{\boldsymbol{x}}(t)\cos\chi_{\boldsymbol{x}}(t)+\boldsymbol{j}\sin\theta_{\boldsymbol{x}}(t)\sin\chi_{\boldsymbol{x}}(t)\\ \sin\theta_{\boldsymbol{x}}(t)\cos\chi_{\boldsymbol{x}}(t)-\boldsymbol{j}\cos\theta_{\boldsymbol{x}}(t)\sin\chi_{\boldsymbol{x}}(t)\end{pmatrix}\\ &= a_{\boldsymbol{x}}(t)e^{\boldsymbol{j}\varphi_{\boldsymbol{x}}(t)}J_3(\theta_{\boldsymbol{x}}(t))\begin{pmatrix}\cos\chi_{\boldsymbol{x}}(t)\\ -\boldsymbol{j}\sin\chi_{\boldsymbol{x}}(t)\end{pmatrix}\end{aligned} \tag{16}$$

Vector $\mathbf{p}_{\boldsymbol{x}}(t) \in \mathbb{C}^2$ is completely determined by the two angles $\theta_{\boldsymbol{x}}(t) \in [-\pi/2; \pi/2]$ and $\chi_{\boldsymbol{x}}(t) \in [-\pi/4; \pi/4]$, and where $J_3(\theta)$ is the matrix rotation of angle θ. Equation (16) is a $\mathbb{C}^2$ version of the quaternion embedding proposed in [9], so that $\theta_{\boldsymbol{x}}(t)$ and $\chi_{\boldsymbol{x}}(t)$ are the parameters of the ellipse described by the bivariate signal $\boldsymbol{x}(t)$ over time (see figure 2 in [9]). $\theta_{\boldsymbol{x}}(t)$ is the orientation of the ellipse, and $\chi_{\boldsymbol{x}}(t)$ is the ellipticity of the ellipse.

With parametrization (16), one gets[5]:

$$\phi_{d,\boldsymbol{x},t_0}(t) = \varphi_{\boldsymbol{x}}(t) - \varphi_{\boldsymbol{x}}(t_0) + \int_{t_0}^{t}\dot{\theta}_{\boldsymbol{x}}(s)\sin 2\chi_{\boldsymbol{x}}(s)\,ds \tag{17}$$

$$\phi_{g,\boldsymbol{x},t_0} = \arctan\left(\frac{\sin(\theta_{\boldsymbol{x}}(t)-\theta_{\boldsymbol{x}}(t_0))\sin(\chi_{\boldsymbol{x}}(t)+\chi_{\boldsymbol{x}}(t_0))}{\cos(\theta_{\boldsymbol{x}}(t)-\theta_{\boldsymbol{x}}(t_0))\cos(\chi_{\boldsymbol{x}}(t)+\chi_{\boldsymbol{x}}(t_0))}\right) - \int_{t_0}^{t}\dot{\theta}_{\boldsymbol{x}}(s)\sin 2\chi_{\boldsymbol{x}}(s)\,ds \tag{18}$$

for values of the parameters such that the denominator is not zero.

From this result, some comments can be made. If $\theta_{\boldsymbol{x}}(t)$ remains constant equal to $\theta_{\boldsymbol{x}}(t_0)$, the geometric phase is constant and equal to zero. In this parametrization, $2\theta_{\boldsymbol{x}}$ represents the longitude on Poincaré sphere, and $\theta_{\boldsymbol{x}}$ being constant means that vector $\boldsymbol{p}_{\boldsymbol{x}}$ follows a vertical great circle on the sphere: The geometric phase is zero along geodesics. If the elevation (or latitude) $\chi_{\boldsymbol{x}}(t)$ is constant, then the geometric phase reads

$$\phi_{g,\mathbf{x},\mathbf{t_0}}(t) = \arctan\left(\tan(2\chi_{\boldsymbol{x}}(t_0)\tan(\theta_{\boldsymbol{x}}(t)-\theta_{\boldsymbol{x}}(t_0)\right) - \sin(2\chi_{\boldsymbol{x}}(t_0))(\theta_{\boldsymbol{x}}(t)-\theta_{\boldsymbol{x}}(t_0))$$

It is thus never equal to zero, except except if $\chi(t_0) = 0$, in which case vector $\boldsymbol{p}_{\boldsymbol{x}}$ moves along the equator (a great circle) of Poincaré sphere, or if $\chi(t_0) = \pm\pi/4$ meaning that vector $\boldsymbol{p}_{\boldsymbol{x}}$ points either the north or the south pole and is static (in this situation last formula is even indefinite). Finally, suppose that $\chi(t) = \chi(t_0) \neq 0$ or $\pm\pi/4$, then for all t_k such that $\theta_{\boldsymbol{x}}(t_k) = \theta_{\boldsymbol{x}}(t_0) + k\pi, k \in \mathbb{Z}$, then

$$\phi_{g,\mathbf{x},\mathbf{t_0}}(t_k) = k\pi\sin(2\chi_{\boldsymbol{x}}(t_0))$$

A famous example of a geometric phase, namely the deviation of the direction of oscillation of the Foucault pendulum [12, Chapter 3], has a similar expression: its value accumulated over one day ($t - t_0 = 24$ hours) is $2\pi\sin\ell$, where ℓ is the lattitude of the

[5] See A for detailed calculation of this equality.

location of the pendulum (2χ in the Poincaré sphere parametrization). In Paris, where $\ell = 48°52'$, the deviation is thus of $271°$ after a complete revolution of the Earth, *i.e.* in 24 hours.

In the hyperspherical parametrization described earlier, vector $\boldsymbol{p_x}$ is a Jones vector and the signal reads

$$\boldsymbol{x}(t) = a_{\boldsymbol{x}}(t) e^{\boldsymbol{j}\psi_{\boldsymbol{x}}(t)} \begin{pmatrix} \cos\theta_1(t) \\ \sin\theta_1(t) e^{\boldsymbol{j}\chi_2(t)} \end{pmatrix} \tag{19}$$

where $\theta_1(t) \in [0, \pi], \chi_2(t) \in [0, 2\pi]$ With the Jones vector parametrization, the dynamical and geometrical phases write

$$\begin{aligned} \phi_{d,\boldsymbol{x},t_0}(t) &= \psi_{\boldsymbol{x}}(t) - \psi_{\boldsymbol{x}}(t_0) + \int_{t_0}^{t} \dot{\chi}_2(s) \sin^2\theta_1(s)\, ds \\ \phi_{g,\mathbf{x},\mathbf{t_0}}(t_k) &= \frac{\chi_2(t) - \chi_2(t_0)}{2} - \arctan\left(\tan(\frac{\chi_2(t) - \chi_2(t_0)}{2}) \frac{\cos(\theta_1(t) + \theta_1(t_0))}{\cos(\theta_1(t) - \theta_1(t_0))}\right) - \int_{t_0}^{t} \dot{\chi}_2(s) \sin^2\theta_1(s)\, ds \end{aligned}$$

In this parametrization, $2\theta_1$ is the latitude measured from the North Pole, and χ_2 the longitude. We recover that $\phi_g = 0$ if χ_2 is constant (vector $\boldsymbol{p_x}$ follows a geodesic), that $\phi_g = 2\pi \sin^2\theta_1(t_0)$ if θ_1 is constant and χ_2 does a 2π excursion (vector $\boldsymbol{p_x}$ traces an isolatitude circle).

At this point and to prepare next section, it is interesting to link $\mathbb{C}\mathbf{P}^1$ to $\mathcal{S}^2$ explicitely. Let $\boldsymbol{p} \in \mathbb{C}^2$ with unit norm. Then $\pi(\boldsymbol{p}) = \boldsymbol{p}\boldsymbol{p}^\dagger$ lies in $\mathbb{C}\mathbf{P}^1$ and can be parametrized with Pauli matrices: There exists a vector $\boldsymbol{u} \in \mathbb{R}^3$ with unit norm such that $\pi(\boldsymbol{p}) = (\boldsymbol{I} + \boldsymbol{u} \cdot \sigma)/2$ where $\boldsymbol{u} \cdot \sigma = \sum_i u_i \sigma_i$, and where the Pauli matrices are defined by

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -\boldsymbol{j} \\ \boldsymbol{j} & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

These matrices are traceless, orthogonal with respect to $\mathrm{Tr}(\sigma_i \sigma_j)$ and satisfy $\sigma_1\sigma_2 = \boldsymbol{j}\sigma_3, \sigma_1\sigma_3 = \boldsymbol{j}\sigma_2, \sigma_2\sigma_3 = \boldsymbol{j}\sigma_2$. Vector $\boldsymbol{u}$ defines a dot on the Poincaré sphere. For the Jones vector above, $\boldsymbol{u}$ is written in spherical coordinate as $(\sin 2\theta \cos\chi, \sin(2\theta)\sin\chi, \cos(2\theta))$. Therefore when the two dimensional complex vector $\boldsymbol{p}(t)$ traces a curve in $\mathbb{C}^2$, its projection by π traces a curve in $\mathbb{C}\mathbf{P}^1$ which can be identified with the curve traced on the sphere by the associated vector $\boldsymbol{u}(t)$.

4.5 Trivariate case: some results

4.5.1 Three dimensional Jones vector and associated geometric phase

In three dimension, the study may be important for polarization studies. We thus consider the signal

$$\boldsymbol{x}(t) = a_{\boldsymbol{x}}(t)e^{\boldsymbol{j}\varphi_{\boldsymbol{x}}(t)} \begin{pmatrix} \cos\theta_1(t) \\ \sin\theta_1(t)\cos\theta_2(t)e^{\boldsymbol{j}\chi_2(t)} \\ \sin\theta_1(t)\sin\theta_2(t)e^{\boldsymbol{j}\chi_3(t)} \end{pmatrix} \tag{20}$$

We exclude the cases $\theta_1(t) = \pi/2$ and $\theta_2(t) = 0$ since they correspond to a two dimensional problem embedded in three dimensions. The angles θ_i vary in $[0, \pi]$ and χ_i in $[0, 2\pi]$.

The geometric phase can be written as

$$\begin{aligned} \phi_{g,\mathbf{x},\mathbf{t_0}}(t_k) \; = \; & \frac{\delta\chi_2(t)+\delta\chi_3(t)}{4} - \arctan\left(\tan(\frac{\delta\chi_2(t)+\delta\chi_3(t)}{4})\frac{\cos(\theta_1(t)+\theta_1(t_0))}{\cos(\theta_1(t)-\theta_1(t_0))}\right) \\ & - \arctan\left(\tan(\frac{\delta\chi_3(t)-\delta\chi_2(t)}{4})\frac{\cos(\theta_2(t)+\theta_2(t_0))}{\cos(\theta_2(t)-\theta_2(t_0))}\right) \\ & - \int_{t_0}^{t} \dot{\chi}_2(s)\cos^2\theta_2(s)\sin^2\theta_1(s)\,ds - \int_{t_0}^{t} \dot{\chi}_3(s)\sin^2\theta_1(s)\sin^2\theta_2(s)\,ds \end{aligned}$$

where $\delta\chi_i(t) = \chi_i(t) - \chi_i(t_0)$. This result is difficult to interpret. We recover however the 2D case if $\theta_1(t) = \pi/2$ or $\theta_2(t) = 0$.

4.5.2 Euler angles based parametrization

The parametrization adopted by Lilly in [5] may be preferable to physically understand the different angles. Lilly's model is defined by

$$a_{\boldsymbol{x}}(t)e^{\boldsymbol{j}\varphi_{\boldsymbol{x}}(t)} J_3(\alpha_{\boldsymbol{x}}(t))J_1(\beta_{\boldsymbol{x}}(t))J_3(\theta_{\boldsymbol{x}}(t)) \begin{pmatrix} \cos\chi_{\boldsymbol{x}}(t) \\ -\boldsymbol{j}\sin\chi_{\boldsymbol{x}}(t) \\ 0 \end{pmatrix}$$

$J_1(\theta)$ (resp. J_3) is a rotation of angle θ in the plane (y, z) (resp. the plane (x, y). Thus the physical interpretation of this model is simple: Define the elliptic motion in the two dimensional plane (x, y), and then perform two rotations in space to obtain any possible case. We refer the interested readers to [5] for further comments. We make a link with the bivariate case here.

Note that $\boldsymbol{p}_{\boldsymbol{x}}(t)$ in Lilly's parametrization can be written as $J_3(\alpha_{\boldsymbol{x}}(t))J_1(\beta_{\boldsymbol{x}}(t))(\boldsymbol{p}_2^\top \; 0)^\top$, where $\boldsymbol{p}_2$ is the polarisation vector in the bivariate case, as given in equation (16).

A direct calculation (detailed in the appendix) shows that $\mathrm{Im}(\boldsymbol{p}_{\boldsymbol{x}}(t)^\dagger \dot{\boldsymbol{p}}_{\boldsymbol{x}}(t)) = \mathrm{Im}(\boldsymbol{p}_2(t)^\dagger \dot{\boldsymbol{p}}_2(t)) + 2\dot{\alpha}_{\boldsymbol{x}}(t)\cos(\beta_{\boldsymbol{x}}(t))\mathrm{Im}(p_{21}p_{22}^*)$, $p_{2i}, i = 1, 2$ being the two components of vector $\boldsymbol{p}_2$. The geometric phase is then found to be

$$\phi_{g3,\mathbf{x},\mathbf{t_0}}(t) = \phi_{g2,\mathbf{x},\mathbf{t_0}}(t) + \arg\left(\frac{\boldsymbol{p_x}(t_0)^\dagger \boldsymbol{p_x}(t)}{\boldsymbol{p}_2(t_0)^\dagger \boldsymbol{p}_2(t)}\right) - \int_{t_0}^t \dot{\alpha}_{\boldsymbol{x}}(s)\cos(\beta_{\boldsymbol{x}}(s))\sin(2\chi_{\boldsymbol{x}}(s))ds$$

which provides an interesting decomposition of the geometric phase in the trivariate case: It is the sum of the geometric phase due to the two dimensional polarization, of a term measuring the difference in Pancharatnam phases between the 3D and the 2D polarization vectors, and of a term taking into account the zenith and azimuth angles.

If $\alpha_{\boldsymbol{x}}$ and $\beta_{\boldsymbol{x}}$ are constant, then $\boldsymbol{p}_{\boldsymbol{x}}^\dagger(t_0)\boldsymbol{p_x} = \boldsymbol{p}_2^\dagger(t_0)\boldsymbol{p}_2$. The extra terms vanish and we obtain the fact that $\phi_{g3,\mathbf{x},\mathbf{t_0}}(t) = \phi_{g2,\mathbf{x},\mathbf{t_0}}(t)$ showing that the problem is indeed two dimensional, and showing that the calculation does not depend on the particular parametrization adopted for $\boldsymbol{p_x}$. Note however that this does not allow to detect that the problem is indeed two dimensional.

4.5.3 Intrinsic interpretation

A way to couple the two parametrization is to provide an intinsinc view of the dynamics. Recall that the signal reads,

$$\boldsymbol{x}(t) = a_{\boldsymbol{x}}(t)e^{\boldsymbol{j}\varphi_{\boldsymbol{x}}(t)}\boldsymbol{p_x}(t) \tag{21}$$

Write $\boldsymbol{p_x}(t) = \boldsymbol{p}_r(t) + \boldsymbol{j}\boldsymbol{p}_i(t)$ where $\boldsymbol{p}_r$ and $\boldsymbol{p}_i$ are real vectors. The real part of $\boldsymbol{x}(t)$ then describes an ellipse in the plane spanned by $\boldsymbol{p}_r$ and $\boldsymbol{p}_i$. A normal vector to this plane is $\boldsymbol{p}_r(t) \wedge \boldsymbol{p}_i(t)$ where $\wedge$ is the wedge product. The unit normal vector is then defined as

$$\boldsymbol{n_x}(t) = \frac{\boldsymbol{p}_r(t) \wedge \boldsymbol{p}_i(t)}{\|\boldsymbol{p}_r(t) \wedge \boldsymbol{p}_i(t)\|} \tag{22}$$

This can be written in term of the complex vector $\boldsymbol{p_x}(t)$ as

$$\boldsymbol{n_x}(t) = \frac{\mathrm{Im}(\boldsymbol{p}_{\boldsymbol{x}}^*(t) \wedge \boldsymbol{p_x}(t))}{\|\mathrm{Im}(\boldsymbol{p}_{\boldsymbol{x}}^*(t) \wedge \boldsymbol{p_x}(t))\|} \tag{23}$$

Since the plane is constantly rotated, the normal follows the equation

$$\dot{\boldsymbol{n}}_{\boldsymbol{x}}(t) = \boldsymbol{\Omega_x}(t) \wedge \boldsymbol{n_x}(t) \tag{24}$$

where $\boldsymbol{\Omega_x}(t)$ is the instantaneous rotation vector of the plane. We can invert this equation by taking the wedge product with $\boldsymbol{n_x}(t)$ leading to

$$\boldsymbol{n_x}(t) \wedge \dot{\boldsymbol{n}}_{\boldsymbol{x}}(t) = \boldsymbol{n_x}(t) \wedge \boldsymbol{\Omega_x}(t) \wedge \boldsymbol{n_x}(t) \tag{25}$$

$$= \boldsymbol{\Omega_x}(t) - (\boldsymbol{\Omega_x}(t)^\top \boldsymbol{n_x}(t))\boldsymbol{n_x}(t) \tag{26}$$

since $\boldsymbol{n}_{\boldsymbol{x}}(t)$ is of unit norm and thanks to the identity $\boldsymbol{a} \wedge \boldsymbol{b} \wedge \boldsymbol{c} = (\boldsymbol{a}^\top \boldsymbol{b})\boldsymbol{c} - (\boldsymbol{b}^\top \boldsymbol{c})\boldsymbol{a}$ for any vectors $\boldsymbol{a}, \boldsymbol{b}$ and $\boldsymbol{c}$. The rotation vector thus writes

$$\boldsymbol{\Omega}_{\boldsymbol{x}}(t) = \boldsymbol{n}_{\boldsymbol{x}}(t) \wedge \dot{\boldsymbol{n}}_{\boldsymbol{x}}(t) + (\boldsymbol{\Omega}_{\boldsymbol{x}}(t)^\top \boldsymbol{n}_{\boldsymbol{x}}(t))\boldsymbol{n}_{\boldsymbol{x}}(t) \tag{27}$$

The velocity of $\boldsymbol{p}_{\boldsymbol{x}}(t)$ due to this rotation thus write

$$\dot{\boldsymbol{p}}_{\boldsymbol{x}}(t) = \boldsymbol{\Omega}_{\boldsymbol{x}}(t) \wedge \boldsymbol{p}_{\boldsymbol{x}}(t) \tag{28}$$

Using the fact that $\boldsymbol{p}_{\boldsymbol{x}}(t)$ is orthogonal to $\boldsymbol{n}_{\boldsymbol{x}}(t)$ implying $\dot{\boldsymbol{n}}_{\boldsymbol{x}} \boldsymbol{p}_{\boldsymbol{x}} = -\boldsymbol{n}_{\boldsymbol{x}} \dot{\boldsymbol{p}}_{\boldsymbol{x}}$, using the above identity, the velocity reduces to

$$\dot{\boldsymbol{p}}_{\boldsymbol{x}}(t) = (\boldsymbol{n}_{\boldsymbol{x}}^\top \dot{\boldsymbol{p}}_{\boldsymbol{x}}(t))\boldsymbol{n}_{\boldsymbol{x}}(t) + \frac{\boldsymbol{j}(\boldsymbol{\Omega}_{\boldsymbol{x}}(t)^\top \boldsymbol{n}_{\boldsymbol{x}}(t))}{2e} \left(\boldsymbol{p}_{\boldsymbol{x}}^\top \dot{\boldsymbol{p}}_{\boldsymbol{x}}(t) \dot{\boldsymbol{p}}_{\boldsymbol{x}}(t)^* - \dot{\boldsymbol{p}}_{\boldsymbol{x}}(t)\right) \tag{29}$$

where $e = \|\boldsymbol{p}_{\boldsymbol{x}}(t) \wedge \boldsymbol{p}_{\boldsymbol{x}}^*(t)\|$ can be called the ellipticity of the ellipse.

The Berry connection then reads

$$\mathrm{Im}(\boldsymbol{p}_{\boldsymbol{x}}^\dagger(t)\dot{\boldsymbol{p}}_{\boldsymbol{x}}(t)) = e \cdot \boldsymbol{\Omega}_{\boldsymbol{x}}(t)^\top \boldsymbol{n}_{\boldsymbol{x}}(t) \tag{30}$$

which provide an complete intrinsic interpretation of the geometric phase: It is the integral over time of the projection of the rotation vector of the plane of the ellipse onto the normal vector to this plane, weighted by the ellipticity.

In the case of Lilly's parametrization, the normal vector is $\boldsymbol{n}_{\boldsymbol{x}}(t) = \sin 2\chi_{\boldsymbol{x}}(t) \cdot (\sin \beta_{\boldsymbol{x}}(t) \sin \alpha_{\boldsymbol{x}}(t), -\sin \beta_{\boldsymbol{x}}(t) \cos \alpha_{\boldsymbol{x}}(t), \cos \beta_{\boldsymbol{x}}(t))^\top$. The rotation vector is the sum of the rotation vector associated to the precession and nutation angles $\boldsymbol{\Omega}_{\alpha,\beta}(t) = (\dot{\beta}_{\boldsymbol{x}}(t) \cos \alpha_{\boldsymbol{x}}(t), \dot{\beta}_{\boldsymbol{x}}(t) \sin \alpha_{\boldsymbol{x}}(t), \dot{\alpha}_{\boldsymbol{x}}(t))^\top$ with the rotation vector $\dot{\theta}(t)\boldsymbol{n}_{\boldsymbol{x}}$ of the rotation of the ellipse in its plane. The ellipticity is $e = \sin(2\chi_{\boldsymbol{x}}(t))$. Finally we obtain in this parametrization

$$\mathrm{Im}(\boldsymbol{p}_{\boldsymbol{x}}^\dagger(t)\dot{\boldsymbol{p}}_{\boldsymbol{x}}(t)) = \sin(2\chi_{\boldsymbol{x}}(t)) \left(\dot{\theta}_{\boldsymbol{x}}(t) + \dot{\alpha}_{\boldsymbol{x}}(t) \cos \beta_{\boldsymbol{x}}(t)\right) \tag{31}$$

which allow to recover the result of the previous section.

4.5.4 Majorana's stellar representation

Here, we prefer to introduce for the signal processing literature an alternate description developed in 1932 by Ettore Majorana [22]. This representation identifies a J spin as the normalized and symmetrized sum of pairwise tensorial products of $J-1$ 1/2 spins. This representation has been introduced for the study of Berry phases by Hannay [23], and further developed by many authors, among whom Kam&Liu [24] has inspired the development here. For generalities and mathematical details, we refer to books of Penrose [25, 26].

We concentrate here on $n = 3$. The idea of Majorana is to use the identity between $\mathbb{C}^3$ and the symmetrized tensorial product $Sym^2(\mathbb{C}^2)$. Let $\boldsymbol{e}_1, \boldsymbol{e}_2$ be the canonical basis of $\mathbb{C}^2$. We construct a basis of $Sym^2(\mathbb{C}^2)$ by setting $\boldsymbol{S} = (\boldsymbol{e}_1 \otimes \boldsymbol{e}_1, (\boldsymbol{e}_1 \otimes \boldsymbol{e}_2 + \boldsymbol{e}_2 \otimes \boldsymbol{e}_1)/\sqrt{2}, \boldsymbol{e}_2 \otimes \boldsymbol{e}_2)$ which spans $Sym^2(\mathbb{C}^2)$.

Let $(\boldsymbol{f}_1, \boldsymbol{f}_2, \boldsymbol{f}_3)$ the canonical basis of $\mathbb{C}^3$. The so-called spin basis or spherical basis in $\mathbb{C}^3$ is given by $\boldsymbol{h}_0 = -(\boldsymbol{f}_1 - \boldsymbol{j}\boldsymbol{f}_2)/\sqrt{2}, \boldsymbol{h}_1 = \boldsymbol{f}_3, \boldsymbol{h}_2(\boldsymbol{f}_1 + \boldsymbol{j}\boldsymbol{f}_2)/\sqrt{2})$. The isomorphism between $\mathbb{C}^3$ and $Sym^2(\mathbb{C}^2)$ can be defined by identifying this last basis with the basis $\boldsymbol{S}$.

Any vector $\boldsymbol{y}$ of $\mathbb{C}^3$ can then be written as $\boldsymbol{x}_1 \otimes \boldsymbol{x}_2 + \boldsymbol{x}_2 \otimes \boldsymbol{x}_1$, where $\boldsymbol{x}_1$ and $\boldsymbol{x}_2$ are elements of $\mathbb{C}^2$:

Obviously $\boldsymbol{x}_1 \otimes \boldsymbol{x}_2 + \boldsymbol{x}_2 \otimes \boldsymbol{x}_1$ is in $\mathbb{C}^3$. If $\boldsymbol{x}_i = \alpha_i \boldsymbol{e}_1 + \beta_i \boldsymbol{e}_2$, then the coordinates of $\boldsymbol{y}$ in basis $\boldsymbol{S}$ are $(y_0, y_1, y_2) = (2\alpha_1\alpha_2, \sqrt{2}(\alpha_1\beta_2 + \alpha_2\beta_1), 2\beta_1\beta_2)$, which can be expressed as $\sqrt{2}(\beta_1\beta_2 - \alpha_1\alpha_2, \boldsymbol{j}(\alpha_1\alpha_2 + \beta_1\beta_2), \alpha_1\beta_2 + \alpha_2\beta_1)$ in the canonical basis of $\mathbb{C}^3$.

The converse is more tedious to show, but the fundamental reason is the fact that $\mathbb{C}$ is algebraically closed! Indeed, associate to $\boldsymbol{y}$ the second order polynomial $P(z) = y_0 z^2 - \sqrt{2} y_1 z + y_2$ where (y_0, y_1, y_2) are the coordinates of $\boldsymbol{y}$ in the spin basis or equivalently in basis $\boldsymbol{S}$. This polynomial can be be factorized[6] as $P(z) = 2(\alpha_1 z - \beta_1)(\alpha_2 z - \beta_2)$. The roots $z_i = \beta_i/\alpha_i$ can be projected back on $\mathcal{S}^2$ by the inverse stereographic projection. The two vectors pointing on the sphere can then be lifted to two vectors of $\mathbb{C}^2$! Polynomial $P(z)$ is called Majorana's polynomial.

If we use the stereographic projection with respect to the North pole down to the equatorial plane, a Jones vector $(\cos(\theta), \sin(\theta)e^{\boldsymbol{j}\chi})$ is projected to the complex number $\tan(\theta)e^{\boldsymbol{j}\chi}$, and conversely, any complex number z is projected back to the Jones vector $(\cos(\arctan|z|), \sin(\arctan|z|)e^{\boldsymbol{j}\arg(z)})$. Thus complex numbers such that $|z| < 1$ are projected onto the south hemisphere, whereas complex numbers with $|z| > 1$ go back to the northern hemisphere. As usual in stereographic projection the North pole is projected to infinity. The vector can also be written $(1 + |z|^2)^{-1/2}(1, -z)$.

The particular choice of the basis above ensures that the Majorana constellation is covariant to to a rotation of the three dimensional space, and thus reflects the internal parameters of the polarization.

4.5.5 The geometry of Majorana's stellar representation

Let $\boldsymbol{p} \in \mathbb{C}^3$ and $\boldsymbol{p}_1$ and $\boldsymbol{p}_2$ in $\mathbb{C}^2$, all with unit norm. Note that $\boldsymbol{p}_i$ projects by π on the two dimensional sphere on a point which can be represented by a euclidean vector $\boldsymbol{u}_i$ which satisfies $\pi(\boldsymbol{p}_i) = \boldsymbol{p}_i\boldsymbol{p}_i^\dagger = (\boldsymbol{I} + \boldsymbol{u}_i \cdot \sigma)/2$, where σ is the basis of Pauli matrices. Then

$$\boldsymbol{p} = \frac{1}{\sqrt{2A_2}}\tilde{\boldsymbol{p}} \quad \text{where} \quad \tilde{\boldsymbol{p}} = \boldsymbol{p}_1 \otimes \boldsymbol{p}_2 + \boldsymbol{p}_2 \otimes \boldsymbol{p}_1 \tag{32}$$

and the normalizing constant reads

$$2A_2 = \boldsymbol{p}^\dagger \boldsymbol{p} = 1 + \frac{1 + \boldsymbol{u}_1 \cdot \boldsymbol{u}_2}{2} = \frac{3 + \boldsymbol{u}_1 \cdot \boldsymbol{u}_2}{2} \tag{33}$$

This is shown using Pauli representation that proves $|\boldsymbol{p}_1^\dagger \boldsymbol{p}_2|^2 = (1 + \boldsymbol{u}_1 \cdot \boldsymbol{u}_2)/2$. Now as shown in the appendix, it is possible to relate the Berry connection (or equivalently

[6] The cases $\alpha_1 = 0$ and/or $\alpha_2 = 0$ are excluded from the discussion, but really mean that the problem is not three dimensional but rather two or one.

linear form $A_{\boldsymbol{p}}(\dot{\boldsymbol{p}}))$ in $\mathbb{C}^3$ to the Berry connections $A_{\boldsymbol{p}_i}(\dot{\boldsymbol{p}}_i)$ in $\mathbb{C}^2$ and the vector $\boldsymbol{u}_i$ pointing on the Poincaré sphere. This link is provided by

$$\operatorname{Im}\left(\boldsymbol{p}^\dagger \dot{\boldsymbol{p}}\right) = \operatorname{Im}\left(\boldsymbol{p}_1^\dagger \dot{\boldsymbol{p}}_1\right) + \operatorname{Im}\left(\boldsymbol{p}_2^\dagger \dot{\boldsymbol{p}}_2\right) + \frac{1}{2}\frac{(\dot{\boldsymbol{u}}_1 - \dot{\boldsymbol{u}}_2)\cdot(\boldsymbol{u}_1 \wedge \boldsymbol{u}_2)}{3 + \boldsymbol{u}_1 \cdot \boldsymbol{u}_2}$$

If $\boldsymbol{p}_i$ is parametrized as a Jones vector as $(\cos(\alpha_i), \sin(\alpha_i)e^{\boldsymbol{j}\gamma_i})$, the Berry connection of each vectors writes $\operatorname{Im}\left(\boldsymbol{p}_i^\dagger \dot{\boldsymbol{p}}_i\right) = \dot{\gamma}_i \sin^2(\alpha_i)$, and vectors $\boldsymbol{u}_i$ have spherical coordinates $(\sin(2\alpha_i)\cos(\gamma_i), \sin(2\alpha_i)\sin(\gamma_i), \cos(2\alpha_i))$. The previous equation along with the calculation of $\arg(\boldsymbol{p}^\dagger(t_0)\boldsymbol{p}(t))$ allows to evaluate the geometric phase.

A beautiful result appears nicely if $\boldsymbol{p}(t)$ is T-periodic. In this case, the roots of Majorana's polynomial are T-periodic and the vectors $\boldsymbol{p}_i$ as well. Thus after an interval on length T, the different vectors are in phase with the origins, meaning $\arg(\boldsymbol{p}^\dagger(t_0)\boldsymbol{p}(t_0+T)) = 0 = \arg(\boldsymbol{p}_i^\dagger(t_0)\boldsymbol{p}_i(t_0+T)), i = 1, 2$. Therefore for a closed loop, we obtain

$$\varphi_{g,\boldsymbol{x},t_0} = \varphi_{g,\boldsymbol{x}_1,t_0} + \varphi_{g,\boldsymbol{x}_2,t_0} - \frac{1}{2}\int_{t_0}^{t_0+T} \frac{(\dot{\boldsymbol{u}}_1 - \dot{\boldsymbol{u}}_2)\cdot(\boldsymbol{u}_1 \wedge \boldsymbol{u}_2)}{3 + \boldsymbol{u}_1 \cdot \boldsymbol{u}_2} dt$$

where we see the contribution from each star is completed by a contribution linked to the interaction between these two stars.

We can study the conditions under which the interaction contribution is zero:

- if $\boldsymbol{u}_1 = \boldsymbol{u}_2$: We end up with a two dimensional vector $\boldsymbol{p}$, and we are back to the bivariate case. This ocurrs if $\theta_1(t) = \pi/2$ or $\theta_2(t) = 0$. In this case, Majorana's polynomial is of degree 1, and there is only one star;
- if $\dot{\boldsymbol{u}}_1 = \dot{\boldsymbol{u}}_2$: This situation occurs if the great circle connecting the two stars experiences a rotation and the stars do not move along this circle;
- if $\dot{\boldsymbol{u}}_1 - \dot{\boldsymbol{u}}_2 \perp \boldsymbol{u}_1 \wedge \boldsymbol{u}_2$: The two stars slide along the great circle connecting them.

In all others cases, the interaction term is present. Kam&Liu in [24] explicit the following beautifull form shown in the appendix as well. Let $\boldsymbol{R} = (\boldsymbol{u}_1 + \boldsymbol{u}_2)/2$ and $\boldsymbol{r} = \boldsymbol{u}_1 - \boldsymbol{u}_2$, and $\tilde{\boldsymbol{R}}$, $\tilde{\boldsymbol{r}}$ their normalized counterparts. Then $\boldsymbol{u}_1 = \boldsymbol{R} + \boldsymbol{r}/2$ and $\boldsymbol{u}_2 = \boldsymbol{R} - \boldsymbol{r}/2$.Then the Berry connection writes

$$\operatorname{Im}\left(\boldsymbol{p}^\dagger \dot{\boldsymbol{p}}\right) = \operatorname{Im}\left(\boldsymbol{p}_1^\dagger \dot{\boldsymbol{p}}_1\right) + \operatorname{Im}\left(\boldsymbol{p}_2^\dagger \dot{\boldsymbol{p}}_2\right) - \frac{1}{2} f(\psi)\cos(\psi)\,(\tilde{\boldsymbol{r}} \wedge \dot{\tilde{\boldsymbol{r}}}) \cdot \tilde{\boldsymbol{R}}$$

where $2\psi = \arccos(\boldsymbol{u}_1 \cdot \boldsymbol{u}_2)$ and $f(\psi) = \sin^2(\psi)/(1 + \cos^2(\psi))$. Thus ψ is the angle between the $\boldsymbol{u}_i$'s and $\boldsymbol{R}$, or equivalently the half the angle between the the two stars, and $(\tilde{\boldsymbol{r}} \wedge \dot{\tilde{\boldsymbol{r}}}) \cdot \tilde{\boldsymbol{R}}$ is the infinitesimal angle between $\boldsymbol{r}$ and $\dot{\boldsymbol{r}}$.

5 Estimation of the geometric phase

In this section, we consider the problem of estimating the geometric phase of a discretized multivariate signal $\{\mathbf{x}[t_k]\}_{k=1}^K$ for a set of time instants $\{t_k\}_{k=1}^K$, with $\mathbf{x}[t_k] \in \mathbb{C}^n$. The goal is to estimate the geometric phase accumulated by this signal during the (discrete) time interval $\Delta_K t = t_K - t_1$. Obviously, the measurement is always dependent of the initial instant of measure, and we always consider that it is t_0 in the sequel. For this purpose, we introduce the Bargmann invariant which is well known in quantum mechanics [14] and propose a recursive estimator inspired by the work in [27]. We also describe the influence of the sampling frequenc rate on the geometric phase estimation accuracy. Recall that we consider the case of normalized signals, *i.e.* $\mathbf{x}^{\mathsf{H}}[t_k]\mathbf{x}[t_k] = 1 \quad \forall t_k$.

Note that in hat follows, we consider signals sampled uniformly sampling. However, the theory presented works with non-uniform sampling as well.

5.1 Bargmann invariant

A way to estimate the geometric phase is to use its connexion to the so-called *Bargmann invariant*, which was initially introduced in the context of quatum mechanics, while looking for invariants for the Wigner theorem [28]. The use of Bargmann invariant to compute the geometric phase was detailed in [14]. In our contect of signal processing it reads as follows: the geometric phase accumulated over the time interval $\Delta_K t$ by the multivariate signal $\{\mathbf{x}[t_k]\}_{k=1}^{K}$ is given as:

$$\Phi_{\mathsf{geo},\mathbf{x}}[\Delta_K t] = -\arg \mathsf{B}_{\mathbf{x}}^{1:K} \tag{34}$$

where the Bargmann invariant $\mathsf{B}_{\mathbf{x}}^{1:K}$ reads:

$$\mathsf{B}_{\mathbf{x}}^{1:K} = \mathbf{x}^{\dagger}[t_1]\mathsf{x}[t_2]\left[\prod_{k=2}^{K-1}\mathbf{x}^{\dagger}[t_k]\mathbf{x}[t_{k+1}]\right]\mathbf{x}^{\dagger}[t_K]\mathbf{x}[t_1] \tag{35}$$

A demonstration (directly inspired from [14]) of the relation given in equation (34) can be found in E.

The Bargmann invariant is a $\mathsf{U}(1)^{\otimes K}$-invariant, which means that, given a smooth function $\alpha(\mathsf{t})$, the discrete multivariate time series $\tilde{\mathbf{x}}$ given as:

$$\tilde{\mathsf{x}}[\mathsf{t}_k] = e^{\boldsymbol{j}\alpha(\mathsf{t}_k)}\mathsf{x}[\mathsf{t}_k] \qquad \forall \mathsf{t}_k \tag{36}$$

and for $k = 1, \ldots, K$ has the same Bargmann invariant as $\mathbf{x}$. This reads:

$$\mathsf{B}_{\mathbf{x}}^{1:K} = \mathsf{B}_{\tilde{\mathbf{x}}}^{1:K} \tag{37}$$

This actually means that two multivariate signals in being two sections of the fiber bundle with the same projection in the projective space have the same Bargmann invariant. This is a gauge invariant property that is shared by the geometric phase, which makes somehow the relation between the Bargmann invariant and the geometric phase a rather geometrically intuitive one.

Note also that, when considering unit norm multivariate signals (*i.e.* signals with $||\mathbf{x}[\mathsf{t}_\mathsf{k}]||^2 = 1 \ \forall \mathsf{t}_k$), the Bargmann invariant takes the form

$$\mathsf{B}_{\mathbf{x}}^{1:K} = \mathrm{Tr}\left(\rho(\mathsf{t}_1)\rho(\mathsf{t}_2)\ldots\rho(\mathsf{t}_K)\right) \tag{38}$$

with $\rho(\mathsf{t}_k) = \mathbf{x}[\mathsf{t}_k]\mathbf{x}^\dagger[\mathsf{t}_k]$. This way to express the Bargmann invariant shows its algebraic nature (the trace of a product of Hermitian matrices), and is used in **??** to relate its algebraic properties to the geometric ones of the geometric phase.

Note also that taking $\boldsymbol{j}\log \mathsf{B}_{\mathbf{x}}^{1:K}$, and using the expression on the right hand side in equation (35), one gets a direct expression of the geometric phase as ths sum of pairwise angles between (normalized) signal values, *i.e.*:

$$\Phi_{\mathsf{geo},\mathbf{x}}[\Delta_K \mathsf{t}] = -\arg \mathsf{B_x}^{1:K} = -\sum_{i=1}^{K-1} \arg \mathbf{x}[\mathsf{t_i}]^{\mathsf{H}}\mathbf{x}[\mathsf{t_{i+1}}] - \arg \mathbf{x}[\mathsf{t_K}]^{\mathsf{H}}\mathbf{x}[\mathsf{t_1}] \tag{39}$$

which is actually the expression used in [27] for the computation of the geometric phase (in optics). Thus, expression (35) and (39) provide effective, and equivalent, ways to compute the geometric phase of a discretized multivariate signal.

5.2 Recursive estimator

It is also possible to define a recursive estimator for the geometric phase, which allows to obtain its estimation at time instant t_k given the one at t_{k-1}. The expression of this estimator, denoted $\Phi^{\mathsf{r}}_{\mathsf{geo},\mathbf{x}}[\Delta_{K+1}\mathsf{t}]$, reads:

$$\Phi^{\mathsf{r}}_{\mathsf{geo},\mathbf{x}}[\Delta_{K+1}\mathsf{t}] = \Phi^{\mathsf{r}}_{\mathsf{geo},\mathbf{x}}[\Delta_K \mathsf{t}] + \arg\left(\frac{\mathbf{x}^\dagger[\mathsf{t}_{K+1}]\mathbf{x}[\mathsf{t}_K]\mathbf{x}^\dagger[\mathsf{t}_K]\mathbf{x}[\mathsf{t}_1]}{\mathbf{x}[\mathsf{t}_{K+1}]^H\mathbf{x}[\mathsf{t}_1]}\right) \tag{40}$$

where a natural choice for the initialization is to set $\Phi^{\mathsf{r}}_{\mathsf{geo},\mathbf{x}}[\Delta_0 \mathsf{t}] = 0$. The derivation to obtain the expression (40) are detailed in Appendix F. In the sequel, we will make use of both estimators indistinctively as we illustrate the geometric phase concept in noise-free scenarios, leaving detailed study of the estimators for future work.

5.3 Estimation of the geometric phase: influence of sampling frequency

Consider a continuous-time multivariate signal $\mathbf{x}(t)$ and its sampled version $\{\mathbf{x}(t_k)\}_{k\in\mathbb{Z}}$ with $t_k = kT_e$, where $1/T_e = F_e$ is the sampling frequency. Let us fix a finite time horizon, say $\Delta = [0, 1]$, and assume that the *continuous* time signal $\mathbf{x}(t)$ describes a continuous path $\mathcal{C}$ in projective space $\mathbb{PC}^{n-1}$. A natural question is whether the geometric phase $\Phi_{\mathrm{geo}}(\mathcal{C})$ can be related to the geometric phase of the *discrete* signal $\{\mathbf{x}(t_k)\}$ over the the same horizon Δ.

It is well known [14, 29] that the expression of the Bargmann invariant associated to the discrete signal $\{\mathbf{x}(t_k)\}$ defines a geometric phase corresponding to a polygon in the projective space $\mathbb{PC}^{n-1}$. More precisely, let $\rho_k = \mathbf{x}(t_k)\mathbf{x}(t_k)^H$. Then the geometric phase associated to $\rho_1, \ldots, \rho_n$ corresponds to that of a n-vertex polygon defining a path $\mathcal{C}_{T_e}$ in $\mathbb{PC}^{n-1}$, such that successives ρ_k (modulo n) are connected by (free) geodesics [29, Eq. (2.9)]. As the sampling period T_e decreases (i.e., sampling frequency increases), the polygon vertices become denser, and the discrete path approaches the continuous curve $\mathcal{C}$.

Fig. 3 illustrates this phenomenon for a bivariate signal $\mathbf{x}(t)$ such that $\rho(0) = \rho(1)$ and $\rho(t)$ describes an iso-latitude trajectory in $\mathbb{CP}^1 \equiv \mathcal{S}^2$. In this case, the geometric

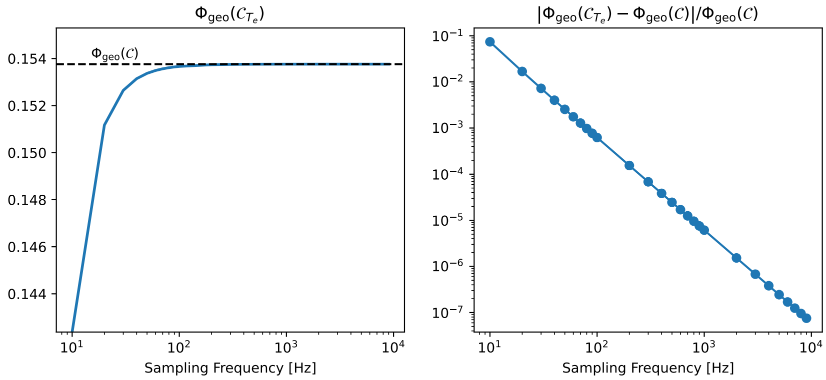


Fig. 3 Convergence of the geometric phase computed from discrete samples its continuous counterpart as a function of the sampling frequency F_e.

phase of the continuous signal is available in closed form, i.e., $\Phi_{\text{geo}}(\mathcal{C}) = \pi(1 - \cos\alpha)$ where α is the co-latitude. We evaluate for increasing values of sampling frequencies the geometric phase of the discrete signal using (34); we observe rapid convergence of the discrete geometric phase towards its continuous part, which falls below 10^{-3} relative error at $F_e \simeq 100$ Hz and below 10^{-4} relative error at $F_2 \simeq 10^3$ Hz – convergence is quadratic.

6 Illustration

In this section, several examples of the manifestation/appearance/existence of a geometric phase for multivariate signals are provided. First, we provide an illustration of the behaviour of the geometric phase, in the bivariate case, for a specific polarization dynamics (*i.e.* a specific path in the projective space) of the signal. Then, a second example highlights how the geometric phase affects the instantaneous frequency interpretation/concept. This phenomenon is also illustrated in thecase of a bivariate signal. The third example concerns multivariate signals of higher dimension and relates to an array processing scenario: we introduce a case where the geometric phase reveals variations of inter-sensor distance during an array measurement.

6.1 Geodesic triangle and spherical cap on the Poincaré sphere

The path in the projective space is the key ingredient to the values that the geometric phase can take. The followings exmaples illustrate some of the properties of the geometric phase.

6.1.1 Geodesic triangle path

As a first example, we consider the case of a monochromatic bivariate signal $\mathbf{x}(t)$ with constant amplitude and which coherence matrix $\rho_{\mathbf{x}}(t)$ follows a path made of two geodesic triangles. Figure 4 shows the trajectory followed by $\rho_{\mathbf{x}}(t)$ in the projective space (which is $\mathcal{S}^2$ in the bivariate case). The initial point (at time $t = 0s$) is 'A' and arrows show the order with which the differents points/locations are visited: 'A' $\rightarrow$ 'B' $\rightarrow$ 'C' $\rightarrow$ 'A' $\rightarrow$ 'D' $\rightarrow$ 'C' $\rightarrow$ 'A'. The blue trajectory ('A' $\rightarrow$ 'B' $\rightarrow$ 'C' $\rightarrow$ 'A') is followed firstly, from time $t = 0s$ to $t = 3s$, while the yellow trajectory ('A' $\rightarrow$ 'D' $\rightarrow$ 'C' $\rightarrow$ 'A') is followed from $t = 3s$ to $t = 6s$. The blue path is traversed counterclockwise while the yellow one is traversed clockwise. The full trajectory thus consists in two consecutive geodesic triangle paths traveled in opposite directions.

This trajectory followed in the projective space induces a geometric phase for the bivariate signal $\mathbf{x}(t)$. In Figure 5, the two components $x_1(t)$ and $x_2(t)$ of the bivariate signal $\mathbf{x}(t)$ are displayed in the two upper rows, while the geometric phase accumulated along the path is displayed in the lower row. Along the time axis, the various points traversed along the path are given. Remembering that the geometric phase is computed with respect to the initial point 'A' at time $t = 0s$, its variations along time can be understood as follows:

- From 'A' to 'B': the path is a geodesic (a great circle) from point 'A' on the equator to point 'B' also on the equator. As a consequence, the geometric phase accumulated is zero from $t = 0s$ to $t = 1s$.
- From 'B' to 'C': from position 'B', the complete path (starting from 'A' the initial point) is no longer a geodesic as there is a right angle turn. As a consequence, the geometric phase acculumated from $t = 0s$ to any point between 'B' and 'C' (*i.e.* for $1s < t \leq 2s$) is no longer null. As the area surrounded by the path is not closed, the computation of the geometric phase consists in closing it with a geodesic. The area is thus a geodesic triangle with non-equal edges which values is given by the Girard theorem Est-ce qu'on fait une annexe pour justifier le comportement en artan ? which fits the geometric phase curve in Figure 5 for $1s < t \leq 2s$. When reaching position 'C', the geodesic triangle covers a surface of a $1/16$ of the sphere surface, *i.e.* $4\pi/16$). This corresponds to a geometric phase of half this area, which is $\pi/8 \simeq -0.39$ as can be seen on the lower row of Figure 5.
- From 'C' to 'A': as the edge between 'C' and 'A' is a geodesic, the geometric phase does not accumulate along this portion of the path, so that it remains identical to what it reachs at position 'C', *i.e.* $\pi/8$. This causes the plateau in lower row of Figure 5 between $t = 2s$ and $t = 3s$. At this point, half of the path as been traversed.
- From 'A' to 'D', then to 'C' and back to 'A': from $t = 3s$, the first geodesic triangle has been traversed and the second one is traversed during the last $3s$. This second part of the path consists in the same shape, but traveled in opposite direction. The consequence is that the variations of the geometric phase in the interval starts with a plateau, then an increse with the same Girard theorem behaviour and finally a plateau during the last second, with a geometric phase having reached the zero value.

The behaviour of the geometric phase on this path highlights several of its properties listed in Section 3.8. For example, the fact that following a geodesic in the projective space does accumulate zero geometric phase can be seen on several parts of this. One needs to understand that when saying that, it is always relative to the original/reference point, so that a path made of several geodesic arcs (just like here as can be seen in Figure 4) may not have a null geometric phase. It is the total path, relative to the reference point that should be a geodesic, for the geometric phase to stay null. Then, the direction on the path is important as it controls the sign of the geometric phase accumulated, allowing potentially (as shown is this example) the geometric phase to vanish at some point after having reached high values. This property is important as it means that the interpretation of the geometric phase should be made on a time interval (and initial point) and not for its value at a given time instant.

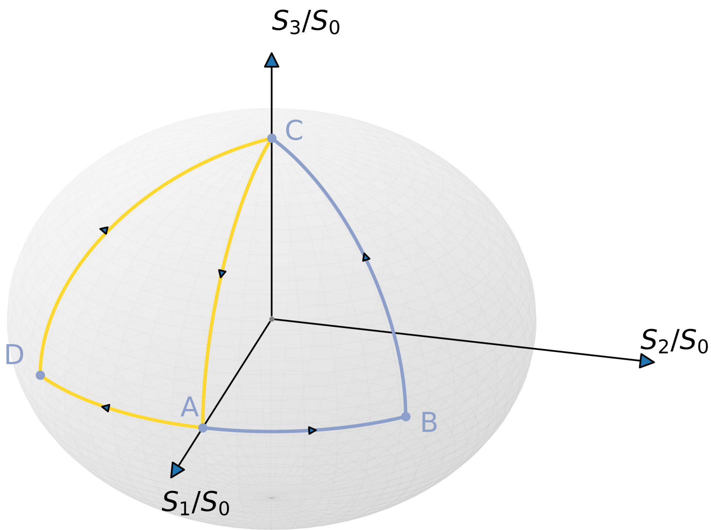


Fig. 4 Path followed by $\rho_{\mathbf{x}}(t)$ on the Poincaré Sphere (*i.e.* $\mathcal{S}^2$) between time $t = 0s$ at initial location 'A' and time $t = 6s$ at point 'A'. The blue path is traversed counterclockwise while the yellow path is traversed clockwise.

NLB: Changer la notation dans les figures pour les phases.

This first example gives some indications on the cases where a geometric phase can be accumulated over time for a bivariate signal. The computation of this phase can be

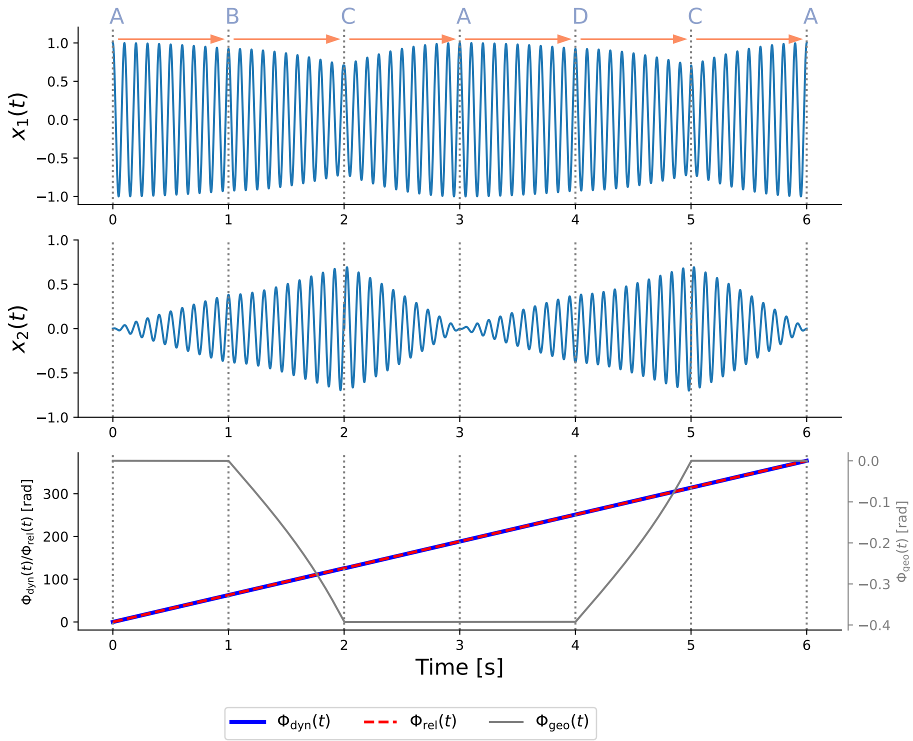


Fig. 5 Components $x_1(t)$ (top) and $x_2(t)$ (middle) of the bivariate signal and the accumulated geometric phase (bottom) during the time interval taken to traverse the path in the projective space depicted in Figure 4. Positions on the projective space are label at time they are reached.

made using an estimator as given in 5. The effect of the phase on the signal may not be obvious at first sight, for example on $x_1(t)$ and $x_2(t)$ as displayed in the first two rows of Figure 5. The next section illustrates how some important observables/parameters of a bivariate modulated signal can be affected by the existence of a geometric phase.

6.1.2 Spherical cap path

Still in the bivariate case, we now consider a trajectory for the coherence matrix which is an iso-lattitude closed loop on the Poincaré sphere, as depicted on the right-hand side of Figure 6. The starting and ending points of the trajectory are actually identical and indicated by a light blue dot. During the time interval $[0, 1]$, the bivariate signal has thus its coherence matrix $\rho_{\boldsymbol{x}}(t)$ that follows a path with constant co-elevation parameter $\chi_{\boldsymbol{x}}(t) = \pi/6$, and attitude parameter $\theta_{\boldsymbol{x}}(t)$ that takes various expressions, namely πt, $\pi(4(t-\frac{1}{2})^3+\frac{1}{2})$, $\pi((\frac{t}{4}-\frac{1}{8})^{1/3}+\frac{1}{2})$, πt^3 and $\pi t^{1/3}$. For each of one of these evolutions, the geometric phase is computed and displayed on the left-hand side of Figure 6. One can see that in all the cases, the initial value of the geometric phase is 0, and its final value, accumulated along the paths during the time interval $[0, 1]$ is $\pi/2$. As explained in 4.4, this value of the geometric phase corresponds to half the area of the cap delimited by the path. Here, the cap has a surface of π so that $\Phi_{g,\boldsymbol{x},0}(1) = \pi/2$. The cap with such an area (of π) corresponds to a co-elevation parameter $\chi_{\boldsymbol{x}}(t) = \pi/6$ with the classical parametrization of the Poincaré sphere [9].

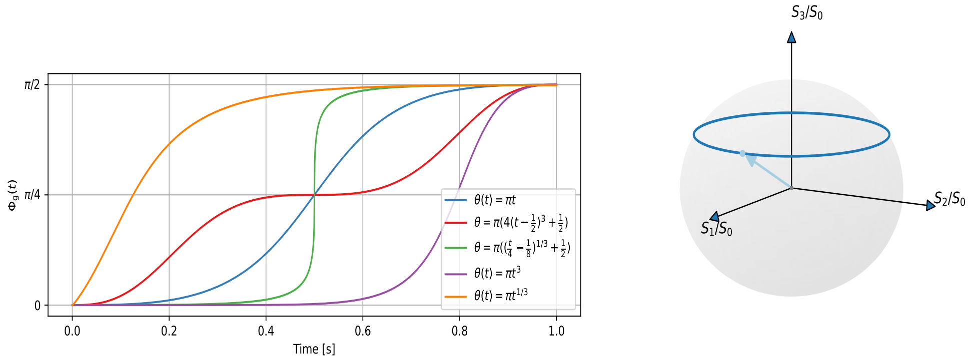


Fig. 6 Geometric phase (left hand-side) for an iso-lattitude loop path on the Poincaré sphere (right hand side) followed at different velocities. The closed loop on the Poincaré sphere is at constant co-elevation $\chi_{\boldsymbol{x}} = \pi/4$ and with various $\theta_{\boldsymbol{x}}(t)$. The initial and final values of the geometric phase are identical for all $\theta_{\boldsymbol{x}}(t)$, but their evolution depend on the dynamics of the azimuth angle $\theta_{\boldsymbol{x}}(t)$.

The various behaviour of the geometric phase accumulated during the time interval are displayed in Figure 6. The way $\theta_{\boldsymbol{x}}(t)$ varies with time dictates the geometric phase accumulation along time. As the same loop (starting and arrival point) is followed for each case, the total accumulated phase is the same once the path is entirely parcoured. This illustrates two properties of the geometric phase: first it is relative to the initial point, and second the velocity at which the path is followed does not influence the result, as long as the final point is the same. The differences in velocities displayed for $\theta_{\boldsymbol{x}}(t)$ show that for example at time $0.5s$, three curves have a value of $\pi/4$, so that they are at the location on the path on the sphere. The behaviour are different thanks to the dynamics of $\theta_{\boldsymbol{x}}(t)$. One can thus say that the value of the geometric phase is really dependent on the path length in the projective space and does not depend on the velociy used to go from the intial point to the final point.

We now turn to an example that demonstrate the necessity to be carefull when dealing with the concept of instantaneous phase for multivariate signals.

6.2 Geometric phase effect on instantaneous frequency

We consider a type of bivariate signal $\mathbf{x}(t)$ that is narrow-banded and with polarization parameters fluctuating (with predefined law) over time. This is a modulated bivariate signal. As was detailed previously, in the bivariate case, the projective space $\mathbb{C}\mathrm{P}^{n-1}$ is the 2-sphere $\mathcal{S}^2$ in $\mathbb{R}^3$ which can be parametrized with two angles: $\theta_{\mathbf{x}}(t)$ and $\chi_{\mathbf{x}}(t)$. These parameters are the one used in Eq. 16 for a modulated bivariate signal. In this example, we set the following time dependency for the parameters of $\mathbf{x}(t)$:

$$\begin{cases} a_{\mathbf{x}}(t) &= 1 \\ \phi_{\mathbf{x}}(t) &= 2\pi\dfrac{(\nu_1-\nu_0)}{2}t^2 + 2\pi\nu_0 t \\ \theta_{\mathbf{x}}(t) &= 2\pi\nu_\theta t^\alpha - \dfrac{\pi}{4} \\ \chi_{\mathbf{x}}(t) &= 2\pi\nu_\chi t^\beta + \dfrac{\pi}{16} \end{cases} \tag{41}$$

From the expression of $\phi_{\mathbf{x}}(t)$ in (41) one can see that $\mathbf{x}(t)$ is a linear chirp in frequency, while its parameters $\theta_{\mathbf{x}}(t)$ and $\chi_{\mathbf{x}}(t)$ have power-law variations with respect to time. In the simulations presented below, the frequencies have been set to $\nu_0 = 100$, $\nu_1 = 120$, $\nu_\theta = 2$ and $\nu_\chi = 1$.

In Figures 8, 9 and 7, the time frequency representation of $\mathbf{x}(t)$ is given for three sets of values, namely $\alpha = 1$ and $\beta = 0.5$ for Figure 8, while $\alpha = 2$ and $\beta = 0.5$ for Figure 9, and $\alpha = 0$ and $\beta = 0$ for Figure 7. The time-frequency representations are computed using the Bispy toolbox [30] and are composed of four images corresponding to the Stokes time-frequency representation introduced in [9] to analyze non-stationnary bivariate signals. The S_0 component encodes the energy (just as standard time-frequency representation), while S_1, S_2 and S_3 solely encode the ploarization information related to the parameters $\theta_{\mathbf{x}}(t)$ and $\chi_{\mathbf{x}}(t)$. Details and exact expressions can be found in [9]. In Figure 7, one can see that the polarization information is constant in the bandwidth of the signal and during the whole 20 seconds duration of the time interval. As $\theta_{\mathbf{x}}(t)$ and $\chi_{\mathbf{x}}(t)$ are both constants, there is no geometric phase acquired over time for this case and the linear frequency can be seen unaltered (*i.e.* it stays linear on the whole time interval) in the time frequency plane (S_0 components in the upper right plot of Figure 7).

On the opposite, in Figures 8 and 9, parameters $\theta_{\mathbf{x}}(t)$ and $\chi_{\mathbf{x}}(t)$ are fluctuating with time, according to (41), with two different sets of values (α, β). The fluctuations of these parameters can be seen on S_1, S_2 and S_3 as they are functions of them (see [9] for detailed expressions). In addition, an extra phenomenon happens in the S_0 component in Figures 8 and 9 (upper left corners): the instantaneous frequency, which should be vary linearly with time as in the model (41), exhibits fluctuations around the value it has when $\theta_{\mathbf{x}}(t)$ and $\chi_{\mathbf{x}}(t)$ are null/constant. This effect is actually due to the exitence of a geometric phase, and this effects is apmlified as time grows. More precisely, the deviations from the straight slope (from 100Hz to 250Hz in the time-frequency plane) that can be seen in Figures 8 and 9 is due to the $2\pi^{-1}\mathrm{Im}\left(\mathbf{p}^\dagger(t)\dot{\mathbf{p}}(t)\right)$ term as described in (**??**). As a consequence, denoting $\nu_{\mathbf{x}_{(\alpha,\beta)}}(t)$ the instantaneous frequency of the signal from the model (41) one get the following expression:

$$\nu_{\mathbf{x}}(t) = \nu_{\mathbf{x}_{(0,0)}}(t) + \nu_\theta \alpha t^{\alpha-1} \sin\left(4\pi\nu_\chi t^\beta + \frac{\pi}{8}\right) \tag{42}$$

where $\nu_{\mathbf{x}_{(0,0)}}(t)$ is the instantaneous frequency of the signal with constant $\theta_{\mathbf{x}}(t)$ and $\chi_{\mathbf{x}}(t)$ parameters (the type of signal displayed in Figure 7). In the simple case of a linear chirp, we have that $\nu_{\mathbf{x}_{(0,0)}}(t) = (\nu_0 - \nu_1)t + \nu_0$. The perturbations due to the geometric phase are not dependent of the actual "true" instantaneous frequency of the signal $\nu_{\mathbf{x}_{(0,0)}}(t)$, but only depend on trajectory associated to $\rho_{\mathbf{x}}(t)$, the projection of $\mathbf{x}(t)$, in the projective space $\mathbb{C}\mathsf{P}^{n-1}$.

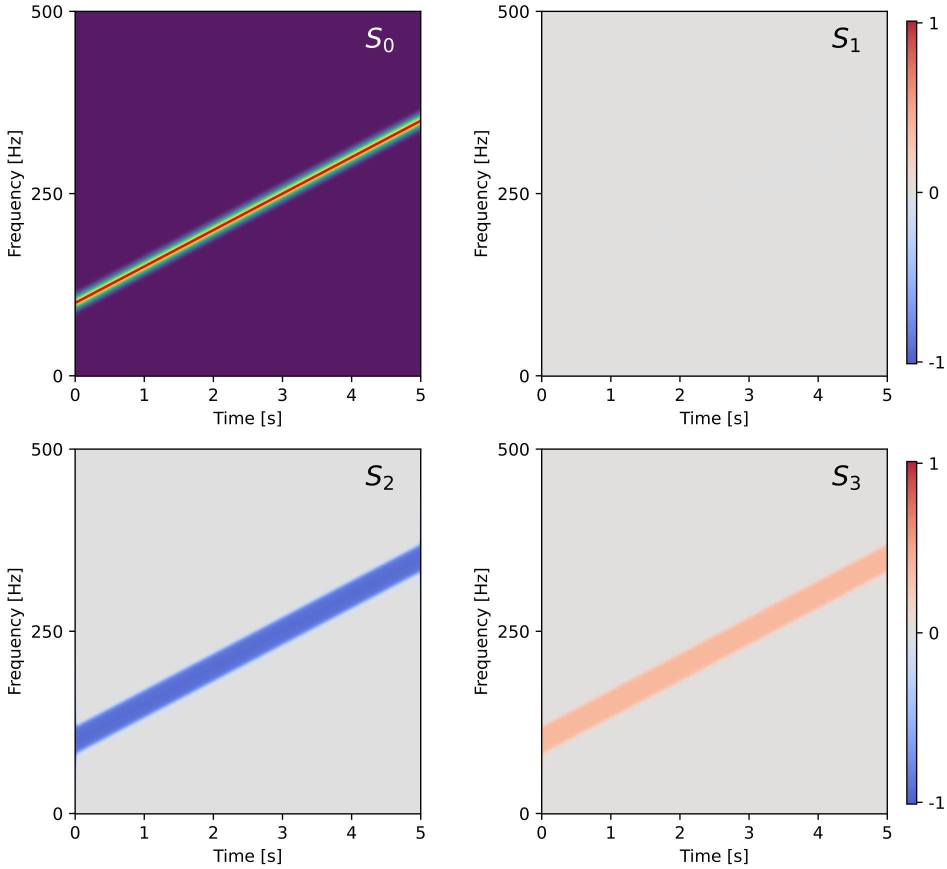


Fig. 7 Time frequency representation of $\boldsymbol{x}(t)$ as in (41) for $\alpha = 0$ and $\beta = 0$.

The results displayed in this section illustrate the way a geometric phase can affect a central parameter/observable of modulated bivariate (and more generally multivariate) signals: its instantaneous frequency. Such a phenomenon was not forseen by authors in [4–6] Changer pour dire qu'ils n'avaient pas vu que c'est une PG, mais qu'ils avaient vu des trucs dans le papier trivarié as they only considered constant values $\mathbf{p}(t) = \mathbf{p}_0$ in the model depicted in (2). Letting this parameter change with time allows to model a much larger classe of multivariate signals, but the time dependencies of $\mathbf{p}(t)$ comes with the possible appearance of a geometric phase that can perturbate the instantaneous frequency evaluation (in a scenario where one wants to estimate this

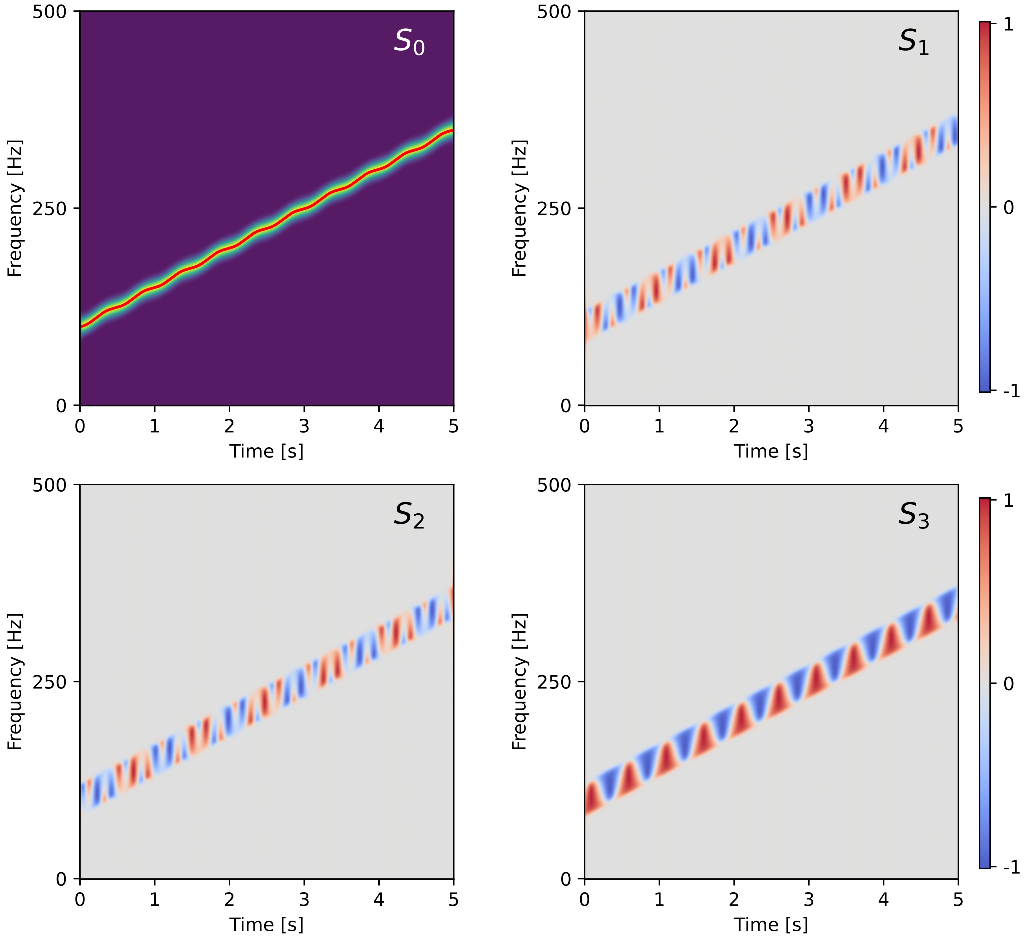


Fig. 8 Time frequency representation of $\boldsymbol{x}(t)$ as in (41) for $\alpha = 1$ and $\beta = 1.5$

frequency from an observed signal). Monitoring (through its evaluation/estimation) the geometric phase allows potentially evaluate its contribution to the instantaneous frequency estimation. The geometric phase effect should thus be taken into account systematically to avoid any mismatch with the "true" instantaneous frequency. Note also that the fluctuations of the instantaneous frequency due to the geometric phase carry some informations on the $\mathbf{p}(t)$ vector. In the bivariate case, this can be understood as polarization fluctuations and could be of interest to estimate in applications where polarization fluctuations carry information on the souce that generates the bivariate signal (see [21] for an exmaple on seismology data).

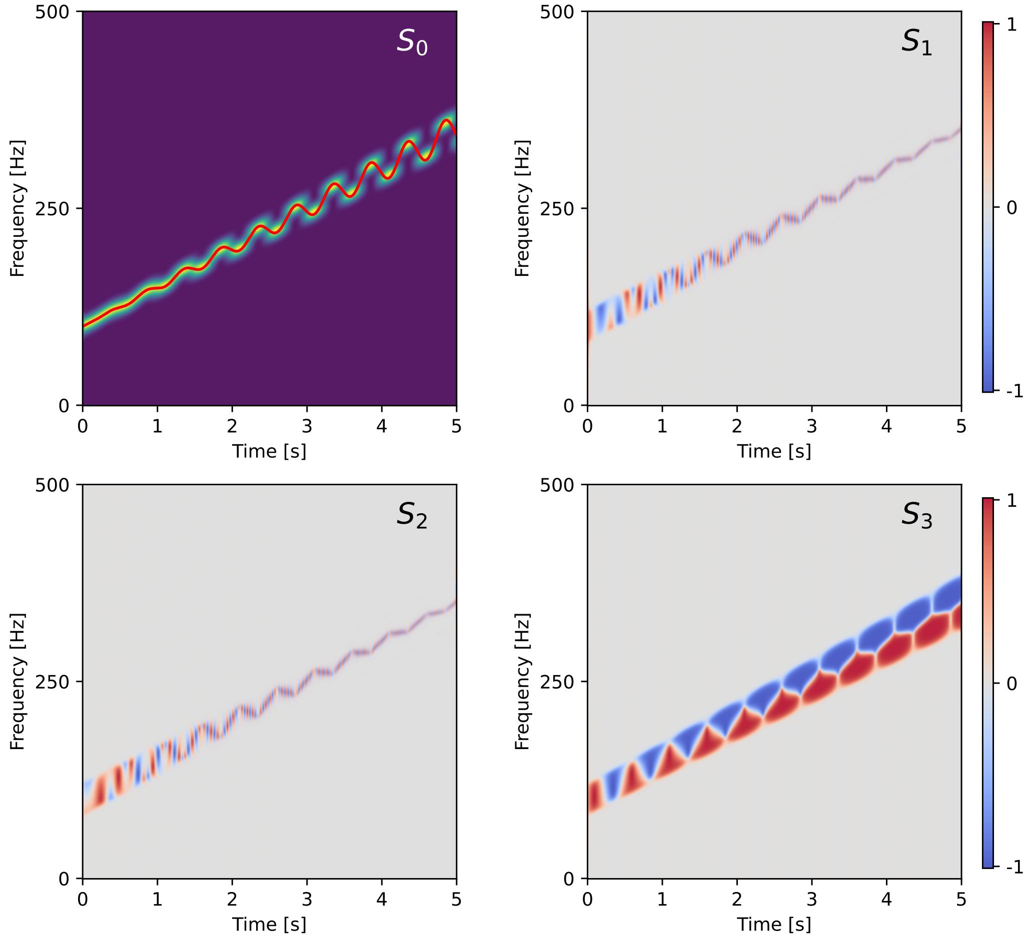


Fig. 9 Time frequency representation of $\boldsymbol{x}(t)$ as in (41) for $\alpha = 2$ and $\beta = 1.5$.

This example shows that the geometric phase has potential influence on some central quantities of multivariate modulated signals, and thus should be evaluated in the process of analysing time-varying properties of such signals.

6.3 Geometric phase effects in array processing

In this section, we consider the case where the multivariate nature of the signal comes from the multiple measurements of an array of sensors. The multivariate signal $\boldsymbol{x}(t) \in \mathbb{C}^{N_s}$ is made of the recordings of the N_s sensors of the array. In the sequel, we only detail cases including linear arrays, but a generalization to arbitrarily shaped arrays could be made. In the examples below, it is shown how geometric phases can appear when Uniform Linear Arrays (ULA) of sensors suffer from imperfections or are subject to perturbations. ULAs are used on a daily basis in a large range applications such as seismology, communications, acoustics (from medical ultrasound imaging to oceanography) or astrophysics to name just a few.

Consider a set of N_s sensors dispatched along a line and equally spaced with an inter-sensor distance d. The multivariate signal received on the antenna is $\boldsymbol{x}(t) = [x_1(t), x_2(t), \ldots, x_{N_s}(t)]^T \in \mathbb{C}$, where $x_k(t)$ is the signal recorded by the k^{th} sensor of the array. The signal is supposed assumed narrow-banded (a classical assumption in array processing) with central frequency ν_0, so that it can be written like:

$$\boldsymbol{x}(t) = a_{\boldsymbol{x}}(t) e^{-\boldsymbol{j}2\pi\nu_0 t} \mathbf{s}(\tau) \tag{43}$$

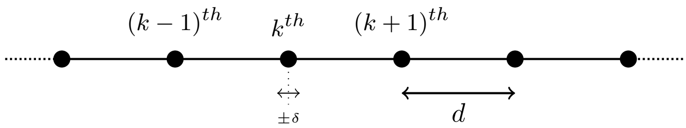
where the normalized *steering vector* $\mathbf{s}(\tau) \in \mathbb{C}^{N_s}$ associated to a source located in the far-field of the array and impinging on it with an angle θ is given by:

$$\mathbf{s}(\tau) = \frac{1}{\sqrt{N_s}} \begin{bmatrix} 1 \\ e^{\boldsymbol{j}2\pi\nu_0\tau} \\ \vdots \\ e^{\boldsymbol{j}2\pi\nu_0(N_s-1)\tau} \end{bmatrix} \tag{44}$$

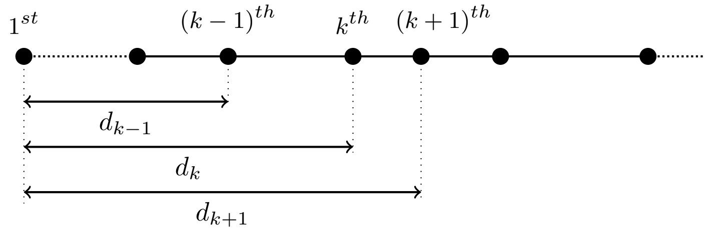
where $\tau = \frac{d\cos\theta}{c}$ is the time delay between two adjacent sensors for a source located at angle θ and c is the wave velocity in the propagation medium. The steering vector is a central quantity in array processing as it encodes the spatial signature of the source on the array. It is used in many applications such as beamforming, direction of arrival estimation, source separation, etc. Here again, we will consider cases with no amplitude variations, *i.e.* $a_{\boldsymbol{x}}(t) = 1 \, \forall t$. The steering vector in (44) is for the ULA case where the inter-sensor distance d is constant aross the array.

In the sequel, the following two scenarios are detailed:

- *i)* one sensor is perturbated during the measurement, so that the inter-sensor distance d is fluctuating for the k^{th} sensor with regards to its neighbors ($(k-1)^{th}$ and $(k+1)^{th}$ sensors) as depicted below:



- *ii)* the angle of arrival $\theta(t)$ is varying with time, while the inter-sensor distance is constant during the measurement, but this inter-sensor distance may not be equal to d (a case of mis-placement of the antenna sensor leading to a non-uniform linear array, *i.e.* NULA, case). The distance between the first sensor of the array and the $(k+1)^{th}$ sensor is denoted d_k as illustrated below[7]:



Both scenarios are relevant in practice. For example, the first one can be encountered in seismology when the sensors are deployed on the ground and thus are subject to potential local ground motion, while the second one can be encountered in communications or tracking when the source is moving with respect to the array while imperfections affect the array shape such that it is no longer an ULA. In both scenarios, the steering vector can fluctuate with time, meaning that the time delay is is a function of time, *i.e.* $\tau(t)$. As a consequence, the antenna signal $\boldsymbol{x}(t)$ now takes the form:

$$\boldsymbol{x}(t) = e^{-\boldsymbol{j}2\pi\nu_0 t}\mathbf{s}(\tau(t)) \tag{45}$$

where a said previously, $a(t)$ is set to 1 $\forall t$. The multivariate complex-valued signal $\boldsymbol{x}(t)$ of dimension N_s has a projection in $\mathbb{C}\mathsf{P}^{N_s-1}$ given as

$$\rho_{\boldsymbol{x}}(t) = \frac{\mathbf{s}(\tau(t))\mathbf{s}^{\dagger}(\tau(t))}{||\mathbf{s}(\tau(t))||} = N_s\,\mathbf{s}(\tau(t))\mathbf{s}^{\dagger}(\tau(t)) \tag{46}$$

As $\tau(t)$ is fluctuating with time, $\rho_{\boldsymbol{x}}(t)$ follows a path in $\mathbb{C}\mathsf{P}^{N_s-1}$ which induces a geometric phase for the steering vector. We now consider the two different scenarios and show how geometric phases can appear in these scenarios.

Scenario i)

Here, the inter-sensor distance d is fluctuating for the k^{th} sensor with regards to his neighbors. This can be modeled by letting $\tau(t)$ fluctuate with time as $\tau(t) = \tau_0 + \delta\tau(t)$ where $\tau_0 = d\sin\theta/c$ is the nominal time delay between two adjacent sensors and $\delta\tau(t)$ is a time-varying perturbation. The geometric phase accumulated during the measurement (time interval $[t_0, t]$) takes the following expression:

$$\Phi_{\boldsymbol{x}}(t) = \arg\left(\frac{\sin\left(2\pi\nu_0(k-1)\frac{\cos\theta}{c}\left(\delta\tau(t)-\delta\tau(t_0)\right)\right)}{N_s - 1 + \cos\left(2\pi\nu_0(k-1)\frac{\cos\theta}{c}\left(\delta\tau(t)-\delta\tau(t_0)\right)\right)}\right) - \frac{2\pi\nu_0}{N_s}(k-1)\frac{\cos\theta}{c}\left(\delta\tau(t)-\delta\tau(t_0)\right) \tag{47}$$

where one can see the dependency on the position k of the perturbated senso (meaning that this geometric phase could be used to monitor the location of the perturbated

[7]We make use of this notation for the NULA as it allows to see the ULA as a special case. As d_k is the distance between the first and k^{th}, in the ULA case we have that $d_k = kd$. This notation is useful for comparison of geometric phases expressions in ULA nd NULA cases.

sensor on the array). As an illustration, choosing a perturbation of type $\delta\tau(t) = \alpha d\cos(2\pi\nu_1 t)$, with $|\alpha| \ll 1$ mimics a resonance phenomenon of the sensor due to some possible malfunction or external parasitic harmonic source. In Figure 10, the geometric phase is displayed (together with total and dynamical phase) for a ULA of $N_s = 15$ sensors, a carrying frequency $\nu_0 = 40Hz$, an inter-sensor distance $d = 1m$, a wave velocity $c = 1m.s^{-1}$, an angle of arrival $\theta = 85.5°$ and a perturbation with $\alpha = 0.1$ and $\nu_1 = 10Hz$.

The geometric phase exhibits fluctuations with time due to the perturbation of the inter-sensor distance for the 5^{th} sensor. The amplitude of these fluctuations depends on the position $k = 5$ of the perturbated sensor, as can be seen in the expression (47) of $\Phi_{\boldsymbol{x}}(t)$ above. In this example, we set $k = 5$, which corresponds to the middle sensor of the array. The value of the geometric phase thus reveals the presence of a perturbation on the array, its location and frequency, and could be used to monitor/compensate this perturbation during the measurement. Interestingly, expression (47) generalizes naturally to the case of multiple sensors malfunctioning. The resulting geometric phase is simply the linear combination of the geometric phases for different velues of k, with possible pertubation frequencies ν_k. While in the case of a single pertubation, the perturbation is monochromatic (as displayed in Figure 10), the geometric phase whould be polychromatic in the case of multiple sensors affected by different perturbations.

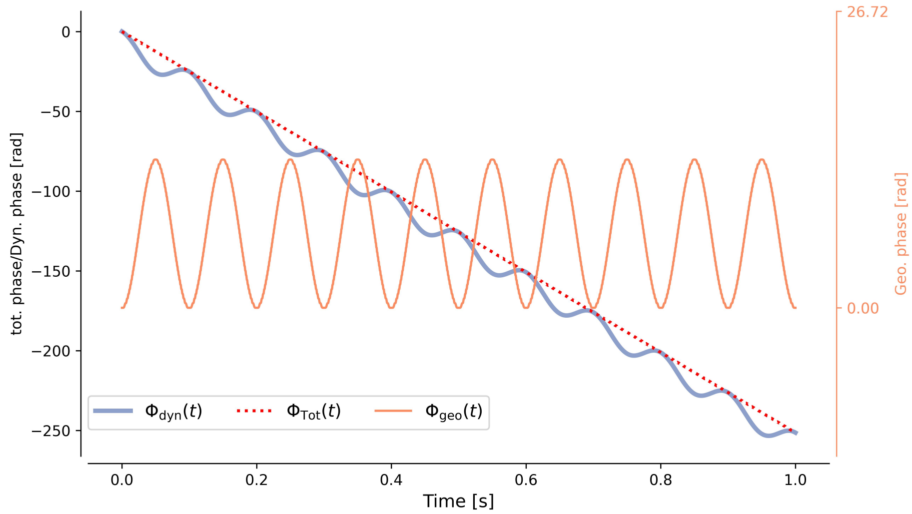


Fig. 10 Geometric phase (orange) accumulated along time for the case where the intersensor distance is fluctuating for the k^{th} sensor of the array (Scenario i)

Scenario ii)

In this second scenario, we study the effect of fluctuations over time of the direction of arrival θ in two cases. First, the inter-sensor distance d is the same between each sensor and constant with time. This is an ULA (Uniform linear array) with constant inter-sensor interval described as a). The b) case consider that the distance between two sensors is not the same for all pair of sensor (while also not varying with time). This is the NULA (Non Uniform Linear Array) case, with d_k the distance between the first sensor and the k^{th} sensor (as displayed above). For both a) and b), the distances between sensors are not varying with time, but it is the angle of arrival θ that is varying with time. This can be modeled by letting $\tau(t)$ fluctuate with time as $\tau(t) = \frac{d\cos\theta(t)}{c}$ where $\theta(t)$ is a time-varying angle of arrival.

a): **ULA case**. In the ULA case with time varying angle of arrival $\theta(t)$, the geometric phase accumulated during the measurement (time interval $[t_0, t]$) takes the following expression:

$$\Phi_{\boldsymbol{x}}^{\mathrm{ULA}}(t) = \arg\left(\frac{1}{N_s}\frac{1-q^{N_s}}{1-q}\right) - 2\pi\nu_0\frac{d}{c}\frac{N_s-1}{2}\left(\cos\theta(t) - \cos\theta(t_0)\right) \tag{48}$$

where $q = e^{\boldsymbol{j}2\pi\nu_0\frac{d}{c}(\cos\theta(t)-\cos\theta(t_0))}$. Using the fact that for a real number η one has that $\arg(\frac{1-e^{\boldsymbol{j}\eta n}}{1-e^{\boldsymbol{j}\eta}}) = \eta(n-1)/2$, then one gets directly that:

$$\Phi_{\boldsymbol{x}}^{\mathrm{ULA}}(t) = 0 \tag{49}$$

This means that for an ULA, the geometric phase $\Phi_{\boldsymbol{x}}^{\mathrm{ULA}}(t)$ is insensitive to a variation of the direction of arrival $\theta(t)$. This is a rather interesting property, as it means that a perfectly designed antenna with identical distance between all its sensors will have a null geometric phase, whatever the potentialy time varying position of the source.

b): **NULA case**. Consider a Non-Uniform Linear Array (NULA) made of N_s sensors, where the distance between the first and the k^{th} sensor is denoted d_k, as depicted previously (NLB:Mettre des numéros aux schemas d'antennes ?). Consider again a time varying angle of arrival $\theta(t)$ for the plane wave impinging the array. In such case, the geometric phase accumulated during the measurement (time interval $[t_0, t]$) is no longer null and takes the following expression:

$$\Phi_{\boldsymbol{x}}^{\mathrm{NULA}}(t) = \arg\left(\sum_{k=0}^{N_s-1}\left(e^{2\pi\boldsymbol{j}\nu_0\frac{\cos\theta(t)-\cos\theta(t_0)}{c}}\right)^{d_k}\right) - 2\pi\frac{\nu_0 L}{N_s c}\left(\cos\theta(t) - \cos\theta(t_0)\right) \tag{50}$$

where we set $d_0 = 0$ and $L = \sum_{k=1}^{N_s-1} d_k$ is the sum of the distances from the first sensor to each sensor of the array. In the ULA case, *i.e.* the peculiar case where $d_k = kd$, then $L = N_s(N_s-1)/2$ and the first term on the right hand side of (50) is equal to the first term of the right hand side of (48), leading the vanishing of the geometric phase, as mentioned in a). This means that the expression for the NULA case in (50) vanishes for the special case $d_k = kd$, which is the ULA configuration.

From equations (50) and (49), one can see that an imperfection in the design of a uniform linear antenna (a non uniformity in the inter-sensor distance) can be detected through the estimation of the geometric phase, even if the calibration source has a fluctuating direction of arrival $\theta(t)$.

7 Conclusion and future work

Appendix A Expression of the geometric phase for a bivariate signal

In the case of a bivariate signal $\mathbf{x}(t)$ expressed as in (16) and with unitary assumption ($i.e.$ $||\mathbf{x}(t)||^2 = 1$), the expression of the geometric phase given, as in equation (18), takes the form:

$$\int_{t_0}^{t_0+\Delta t} \frac{\mathrm{Im}\left(\mathbf{x}^\dagger(\mathsf{s})\dot{\mathbf{x}}(\mathsf{s})\right)}{||\mathbf{x}(\mathsf{s})||^2}\,\mathsf{ds} = \int_{t_0}^{t_0+\Delta t} \dot{\phi}_{\mathsf{x}}(\mathsf{s})\,\mathsf{ds} + \int_{t_0}^{t_0+\Delta t} \mathrm{Im}\left(\mathbf{p}^{\mathsf{H}}(\mathsf{s})\dot{\mathbf{p}}(\mathsf{s})\right)\,\mathsf{ds} \tag{A1}$$

It actually consists in the integration between t_0 and $\mathsf{t}_0 + \Delta\mathsf{t}$ of the integrand:

$$\mathrm{Im}\left(\mathbf{x}^\dagger(\mathsf{t})\dot{\mathbf{x}}(\mathsf{t})\right) = \dot{\phi}_{\mathbf{x}}(\mathsf{t}) + \mathrm{Im}\left(\mathbf{p}^{\mathsf{H}}(\mathsf{t})\dot{\mathbf{p}}(\mathsf{t})\right) \tag{A2}$$

We detail below the calculation to obtain the expression of this integrand. Recall that $\mathbf{x}(\mathsf{t}) = e^{\phi_{\mathsf{x}}(\mathsf{t})}\mathbf{p}(\mathsf{t})$, so that:

$$\begin{aligned}\mathbf{x}^{\mathsf{H}}(\mathsf{t})\dot{\mathbf{x}}(\mathsf{t}) &= \left(e^{-\boldsymbol{j}\phi_{\mathsf{x}}(\mathsf{t})}\mathbf{p}^{\mathsf{H}}(\mathsf{t})\right)\left(\boldsymbol{j}\dot{\phi}_{\mathbf{x}}(\mathsf{t})e^{\phi_{\mathsf{x}}(\mathsf{t})}\mathbf{p}(\mathsf{t}) + e^{\phi_{\mathsf{x}}(\mathsf{t})}\dot{\mathbf{p}}(\mathsf{t})\right)\\ &= \boldsymbol{j}\dot{\phi}_{\mathbf{x}}(\mathsf{t})\mathbf{p}^{\mathsf{H}}(\mathsf{t})\mathbf{p}(\mathsf{t}) + \mathbf{p}^{\mathsf{H}}(\mathsf{t})\dot{\mathbf{p}}(\mathsf{t})\end{aligned} \tag{A3}$$

Remembering that $\mathbf{p}(\mathsf{t})$ is unitary, such that $\mathbf{p}^{\mathsf{H}}(\mathsf{t})\mathbf{p}(\mathsf{t}) = 1$, as a consequence $\dot{\mathbf{p}}^{\mathsf{H}}(\mathsf{t})\mathbf{p}(\mathsf{t}) = -\mathbf{p}^{\mathsf{H}}(\mathsf{t})\dot{\mathbf{p}}(\mathsf{t}) = -\left(\dot{\mathbf{p}}^{\mathsf{H}}(\mathsf{t})\mathbf{p}(t)\right)^\dagger$. This means that the quantity $\dot{\mathbf{p}}^{\mathsf{H}}(\mathsf{t})\mathbf{p}(t)$ is purely imaginery, and one has:

$$\mathbf{p}^{\mathsf{H}}(\mathsf{t})\dot{\mathbf{p}}(\mathsf{t}) = \boldsymbol{j}\,\mathrm{Im}\left(\mathbf{p}^{\mathsf{H}}(\mathsf{t})\dot{\mathbf{p}}(\mathsf{t})\right) \tag{A4}$$

which is equivalent to say that $\mathrm{Re}\left(\mathbf{p}^{\mathsf{H}}(\mathsf{t})\dot{\mathbf{p}}(\mathsf{t})\right) = 0$. The same property holds for $\mathbf{x}(\mathsf{t})$, so that:

$$\mathbf{x}^{\mathsf{H}}(\mathsf{t})\dot{\mathbf{x}}(\mathsf{t}) = \boldsymbol{j}\,\mathrm{Im}\left(\mathbf{x}^{\mathsf{H}}(\mathsf{t})\dot{\mathbf{x}}(\mathsf{t})\right) \tag{A5}$$

which by substitution in (A3) and simplification by $\boldsymbol{j}$ on both sides, gives the final expression (A2). In this expression, the geometric phase corresponds to the term $\mathrm{Im}\left(\mathbf{p}^{\mathsf{H}}(\mathsf{t})\dot{\mathbf{p}}(\mathsf{t})\right)$. We know give the details to obtain the result given in (18). Recalling that the expression of $\mathsf{p}(\mathsf{t})$ is given in (16), then we have that:

$$\begin{aligned}\mathsf{p}^{\mathsf{H}}(\mathsf{t})\dot{\mathsf{p}}(\mathsf{t}) = {} & [\cos\theta(t)\cos\chi(t) - \boldsymbol{j}\sin\theta(t)\sin\chi(t)\,,\ \sin\theta(t)\cos\chi(t) + \boldsymbol{j}\cos\theta(t)\sin\chi(t)]\\ & \cdot \begin{bmatrix} -\dot{\theta}(t)\sin\theta(t)\cos\chi(t) - \dot{\chi}(t)\cos\theta(t)\sin\chi(t) + \boldsymbol{j}\dot{\theta}(t)\sin\chi(t) + \boldsymbol{j}\dot{\chi}(t)\sin\theta(t)\cos\chi(t) \\ \dot{\theta}(t)\cos\theta(t)\cos\chi(t) - \dot{\chi}(t)\sin\theta(t)\sin\chi(t) + \boldsymbol{j}\dot{\theta}(t)\sin\theta(t)\sin\chi(t) - \boldsymbol{j}\dot{\chi}(t)\cos\theta(t)\cos\chi(t)\end{bmatrix}\end{aligned} \tag{A6}$$

so that the imaginary part takes the form:

$$\begin{aligned}\mathrm{Im}\left(\mathsf{p}^{\mathsf{H}}(\mathsf{t})\dot{\mathsf{p}}(\mathsf{t})\right) = {} & \dot{\theta}(t)\cos^2\theta(t)\cos\chi(t)\sin\chi(t) + \dot{\chi}(t)\cos\theta(t)\sin\theta(t)\cos^2\chi(t)\\ & +\dot{\theta}(t)\sin^2\theta(t)\cos\chi(t)\sin\chi(t) + \dot{\chi}(t)\cos\theta(t)\sin\theta(t)\sin^2\chi(t)\\ & +\dot{\theta}(t)\sin^2\theta(t)\cos\chi(t)\sin\chi(t) - \dot{\chi}(t)\sin\theta(t)\cos\theta(t)\cos^2\chi(t)\\ & +\dot{\theta}(t)\cos^2\theta(t)\cos\chi(t)\sin\chi(t) - \dot{\chi}(t)\cos\theta(t)\sin\theta(t)\sin^2\chi(t)\end{aligned} \tag{A7}$$

which simplifies into:

$$\mathrm{Im}\left(\mathsf{p}^{\mathsf{H}}(\mathsf{t})\dot{\mathsf{p}}(\mathsf{t})\right) = \dot{\theta}(t)\sin 2\chi(t) \tag{A8}$$

and gives the expression of the geometric phase for bivariate signals as in (18).

Appendix B Expression of the geometric phase for a bivariate signal

The geometric phase acquired during the interval $[t, t_0]$ is

$$\phi_{g,\mathbf{x},\mathbf{t_0}} = \arg p_x^\dagger(t_0)p_x^\dagger(t) - \int_{t_0}^{t} \dot{\theta}_{\boldsymbol{x}}(s)\sin 2\chi_{\boldsymbol{x}}(s)\,ds$$

More over, we can evaluate $p_x^\dagger(t_0)p_x(t)$ to find

$$\begin{aligned}
p_x^\dagger(t_0)p_x(t) &= \cos\theta_0\cos\chi_0\cos\theta\cos\chi + \sin\theta_0\sin\chi_0\sin\theta\sin\chi \\
&+ \boldsymbol{j}(\cos\theta_0\cos\chi_0\sin\theta\sin\chi - \sin\theta_0\sin\chi_0\cos\theta\cos\chi) \\
&+ \sin\theta_0\cos\chi_0\sin\theta\cos\chi + \cos\theta_0\sin\chi_0\cos\theta\sin\chi \\
&+ \boldsymbol{j}(-\sin\theta_0\cos\chi_0\cos\theta\sin\chi + \cos\theta_0\sin\chi_0\sin\theta\cos\chi) \\
&= \cos\theta_0\cos\theta\cos(\chi_0-\chi) + \sin\theta_0\sin\theta\cos(\chi_0-\chi) \\
&+ \boldsymbol{j}(-\sin\theta_0\cos\theta\sin(\chi+\chi_0) + \cos\theta_0\sin\theta\sin(\chi+\chi_0)) \\
&= \cos(\theta_0-\theta)\cos(\chi_0-\chi) + \boldsymbol{j}\sin(\theta_0-\theta)\sin(\chi+\chi_0)
\end{aligned}$$

and conclude that

$$\phi_{g,\mathbf{x},\mathbf{t_0}}(t) = \arctan\left(\frac{\sin(\theta-\theta_0)\sin(\chi+\chi_0)}{\cos(\theta-\theta_0)\cos(\chi+\chi_0)}\right) - \int_{t_0}^{t}\dot{\theta}_{\boldsymbol{x}}(s)\sin 2\chi_{\boldsymbol{x}}(s)\,ds$$

Appendix C Instantenous frequency of a bivariate signal as the first instantaneous moment of its energy density

Similarly to the univariate case, it is possible to show that the instantaneous frequency of a bivariate signal is the first moment of the energy density. This property is well known in time-frequency analysis [1, 2]. Here, we derive the calculation to show that the first moment of the energy density of a bivariate signal exhibits an extra term (in addition to the instantaneous phase) which is precisley the geometric phase. The derivation we use is based on complex vector valued signal. In [18, Section 1.4 & Appendix], similar results were presented using the quaternion formalism.

First, consider a bivariate signal $\mathbf{x}(\mathsf{t}) = [\mathsf{x}_1(\mathsf{t}), \mathsf{x}_2(\mathsf{t})]^\mathsf{T} \in \mathbb{C}^2$, and its *vector* Fourier transform[8] $\mathbf{X}(\omega) = [\mathsf{X}_1(\omega), \mathsf{X}_2(\omega)]^\mathsf{T} \in \mathbb{C}^2$. For such signals, its *first instantaneous*

[8] For a signal $\mathsf{x}_\beta(\mathsf{t})$, with $\beta = \{1, 2\}$, we use the definition $\mathsf{X}_\beta(\omega) = \int_{-\infty}^{+\infty}\mathsf{x}_\beta(\mathsf{t})e^{-\boldsymbol{j}\omega\mathsf{t}}\mathsf{d}\omega$ for its Fourier transform.

moment, denoted $\omega_{\mathbf{x}}(\mathsf{t})$, is defined as:

$$\int \omega \left|\mathbf{X}(\omega)\right|^2 \mathsf{d}\omega = \int \omega_{\mathbf{x}}(\mathsf{t}) \left|\mathbf{x}(\mathsf{t})\right|^2 \mathsf{dt} \tag{C9}$$

The left-hand side of this equation can be re-expressed as:

$$\begin{aligned} \int \omega \left|\mathbf{X}(\omega)\right|^2 \mathsf{d}\omega &= \int \left[\bar{\mathsf{X}}_1(\omega), \bar{\mathsf{X}}_2(\omega)\right] . \begin{bmatrix} \omega \mathsf{X}_1(\omega) \\ \omega \mathsf{X}_2(\omega) \end{bmatrix} \mathsf{d}\omega \\ &= -\boldsymbol{j} \int \left[\bar{\mathsf{X}}_1(\omega), \bar{\mathsf{X}}_2(\omega)\right] . \begin{bmatrix} \mathcal{F}\left[\dot{\mathsf{X}}_1\right](\omega) \\ \\ \mathcal{F}\left[\dot{\mathsf{X}}_2\right](\omega) \end{bmatrix} \mathsf{d}\omega \end{aligned} \tag{C10}$$

where we used the basic property Fourier transform for derivates. Now, making use of the Parseval-Plancherel theorem, it comes that:

$$\begin{aligned} \int \omega \left|\mathbf{X}(\omega)\right|^2 \mathsf{d}\omega &= -\boldsymbol{j} \int \left[\bar{\mathsf{x}}_1(\mathsf{t}), \bar{\mathsf{x}}_2(\mathsf{t})\right] . \begin{bmatrix} \dot{\mathsf{x}}_1(\mathsf{t}) \\ \\ \dot{\mathsf{x}}_2(\mathsf{t}) \end{bmatrix} \mathsf{dt} \\ &= -\boldsymbol{j} \int \frac{\bar{\mathsf{x}}_1(\mathsf{t})\dot{\mathsf{x}}_1(\mathsf{t}) + \bar{\mathsf{x}}_2(\mathsf{t})\dot{\mathsf{x}}_1(\mathsf{t})}{\left|\mathbf{x}_1(\mathsf{t})\right|^2 + \left|\mathbf{x}_2(\mathsf{t})\right|^2} \left|\mathbf{x}(\mathsf{t})\right|^2 \mathsf{dt} \end{aligned} \tag{C11}$$

where we used the fact that $\left|\mathbf{x}(\mathsf{t})\right|^2 = \left|\mathbf{x}_1(\mathsf{t})\right|^2 + \left|\mathbf{x}_2(\mathsf{t})\right|^2$. This means that the *first instantaneous moment*, denoted $\omega_{\mathbf{x}}(\mathsf{t})$, is given as for a bivariate signal $\mathbf{x}(\mathsf{t})$:

$$\omega_{\mathbf{x}}(\mathsf{t}) = -\boldsymbol{j} \, \frac{\bar{\mathsf{x}}_1(\mathsf{t})\dot{\mathsf{x}}_1(\mathsf{t}) + \bar{\mathsf{x}}_2(\mathsf{t})\dot{\mathsf{x}}_2(\mathsf{t})}{\left|\mathbf{x}_1(\mathsf{t})\right|^2 + \left|\mathbf{x}_2(\mathsf{t})\right|^2} \tag{C12}$$

Now, using the parametrization of a bivariate signal as given in (16) and remembering that we consider normalized signals (*i.e.* $\mathsf{a}_{\mathbf{x}}(\mathsf{t}) = \sqrt{\left|\mathbf{x}_1(\mathsf{t})\right|^2 + \left|\mathbf{x}_2(\mathsf{t})\right|^2} = 1 \quad \forall \mathsf{t}$) one can get the expression of the two terms in expression (C12):

$$\begin{aligned}
\bar{\mathsf{x}}_1(\mathsf{t})\dot{\mathsf{x}}_1(\mathsf{t}) &= \left(\cos\theta(t)\cos\chi(t) - \boldsymbol{j}\sin\theta(t)\sin\chi(t)\right)e^{-\boldsymbol{j}\phi(t)} \\
&\quad .e^{\boldsymbol{j}\phi(t)}\Big(\boldsymbol{j}\dot{\phi}(t)\left(\cos\theta(t)\cos\chi(t) + \boldsymbol{j}\sin\theta(t)\sin\chi(t)\right) \\
&\quad + -\dot{\theta}(t)\sin\theta(t)\cos\chi(t) - \dot{\chi}(t)\cos\theta(t)\sin\chi(t) \\
&\quad +\boldsymbol{j}\dot{\theta}(t)\cos\theta(t)\sin\chi(t) + \boldsymbol{j}\dot{\chi}(t)\sin\theta(t)\cos\chi(t)\Big) \\
&= \boldsymbol{j}\dot{\phi}(t)\left(\cos^2\theta(t)\cos^2\chi(t) + \sin^2\theta(t)\sin^2\chi(t)\right) \\
&\quad +\dot{\theta}(t)\left(\cos\theta(t)\sin\theta(t)\left(\sin^2\chi(t) - \cos^2\chi(t)\right)\right. \\
&\quad \left.+\boldsymbol{j}\cos\chi(t)\sin\chi(t)\left(\cos^2\theta(t) + \sin^2\theta(t)\right)\right) \\
&\quad +\dot{\chi}(t)\left(\cos\chi(t)\sin\chi(t)\left(\sin^2\theta(t) - \cos^2\theta(t)\right)\right. \\
&\quad \left.+\boldsymbol{j}\cos\theta(t)\sin\theta(t)\left(\cos^2\chi(t) + \sin^2\chi(t)\right)\right)
\end{aligned} \tag{C13}$$

Using similar calculation with $\mathsf{x}_2(\mathsf{t})$, one gets:

$$\begin{aligned}
\bar{\mathsf{x}}_2(\mathsf{t})\dot{\mathsf{x}}_2(\mathsf{t}) &= \boldsymbol{j}\dot{\phi}(t)\left(\sin^2\theta(t)\cos^2\chi(t) + \cos^2\theta(t)\sin^2\chi(t)\right) \\
&\quad +\dot{\theta}(t)\left(\cos\theta(t)\sin\theta(t)\left(\cos^2\chi(t) - \sin^2\chi(t)\right)\right. \\
&\quad \left.+\boldsymbol{j}\cos\chi(t)\sin\chi(t)\left(\cos^2\theta(t) + \sin^2\theta(t)\right)\right) \\
&\quad +\dot{\chi}(t)\left(\cos\chi(t)\sin\chi(t)\left(\cos^2\theta(t) - \sin^2\theta(t)\right)\right. \\
&\quad \left.-\boldsymbol{j}\cos\theta(t)\sin\theta(t)\left(\cos^2\chi(t) + \sin^2\chi(t)\right)\right)
\end{aligned} \tag{C14}$$

By combination of (C13) and (C14) and after simplification, one gets that:

$$\begin{aligned}
\bar{\mathsf{x}}_1(\mathsf{t})\dot{\mathsf{x}}_1(\mathsf{t}) + \bar{\mathsf{x}}_2(\mathsf{t})\dot{\mathsf{x}}_2(\mathsf{t}) &= \boldsymbol{j}\dot{\phi}(t)\left(\cos^2\chi(t)\left(\cos^2\theta(t) + \sin^2\theta(t)\right)\right. \\
&\quad \left.+\sin^2\chi(t)\left(\cos^2\theta(t) + \sin^2\theta(t)\right)\right) + \boldsymbol{j}\dot{\theta}(t)2\sin\chi(t)\cos\chi(t)
\end{aligned} \tag{C15}$$

Finally, using (C12) and (C15), and remembering using the fact that $|\mathbf{x}_1(\mathsf{t})|^2 + |\mathbf{x}_2(\mathsf{t})|^2 = 1$, we get that:

$$\omega_{\mathbf{x}}(\mathsf{t}) = \dot{\phi}(t) + \dot{\theta}(t)\sin 2\chi(t) \tag{C16}$$

This expression showh that for the bivariate case, the first instantaneous moment contains the instantaneous frequency (up to a 2π factor) $\dot{\phi}(t)$ as well as the geometric phase term $\dot{\theta}(t)\sin 2\chi(t)$, as given in (18) and demonstrated in A. Consequences of this extra geometric term are illustrated in Section 6.

Appendix D Geometric phase along a geodesic path

First, consider a path between $\rho(t_1)$ and $\rho(t_2)$), two elements in the projective space $\mathbb{CP}^{p-1}$. This path is a *geodesic*, denoted $\mathcal{G}_{\rho(t_1)\to\rho(t_2)}$, if ....

Given $\rho(t_1), \rho(t_2) \in \mathbb{CP}^{p-1}$, and assuming $\mathrm{Tr}(\rho(t_1)\rho(t_2)) \neq 0$ the geometric phase acquired along the geodesic $\mathcal{G}_{\rho(t_1)\to\rho(t_2)}$ in $\mathbb{CP}^{p-1}$ is null, which reads:

$$\Phi_{\mathrm{geo}}[\mathcal{G}_{\rho(t_1)\to\rho(t_2)}] = 0 \tag{D17}$$

Give the lines of the proof as in Mukunda 1993.

Appendix E Geometric phase and Bargmann invariant

Consider a path $\mathcal{C} \subset \mathbb{CP}^{p-1}$ with n times instants $t_1, t_2, \dots, t_n$ and $n-1$ geodesic arcs connecting $\rho(t_r)$ to $\rho(t_{r+1})$ for $r \in [1; n-1]$. This actually corresponds to decomposing the path $\mathcal{C}$ into a set of $n-1$ geodesic arcs[9]. NLB:Preuve qu'on peut toujours découper en ensemble d'arcs géodésiques ? en continu c'est ok, mais parler de l'approx que ça introduit dans le calcul de la phase géo.

The geometric phase associated to $\mathcal{C}$ is given as:

$$\begin{aligned}
\Phi_{\text{geo}}[\mathcal{C}] &= \Phi_{\text{rel}}[\mathcal{C}] - \Phi_{\text{dyn}}[\mathcal{C}] \\
&= \arg\left(\mathsf{x}^\dagger(t_1)\mathsf{x}(t_n)\right) - \sum_{r=1}^{n-1} \Phi_{\text{dyn}}[\mathcal{G}_{\rho(t_r)\to\rho(t_{r+1})}] \\
&= \arg\left(\mathsf{x}^\dagger(t_1)\mathsf{x}(t_n)\right) \\
&\quad - \sum_{r=1}^{n-1} \left(\Phi_{\text{geo}}[\mathcal{G}_{\rho(t_r)\to\rho(t_{r+1})}] - \arg\left(\mathsf{x}^\dagger(t_r)\mathsf{x}(t_{r+1})\right)\right) \\
&= \arg\left(\mathsf{x}^\dagger(t_1)\mathsf{x}(t_n)\right) - \sum_{r=1}^{n-1} \arg\left(\mathsf{x}^\dagger(t_r)\mathsf{x}(t_{r+1})\right) \\
&= -\arg\left(\mathsf{x}^\dagger(t_1)\mathsf{x}(t_2)\right) - \arg\left(\mathsf{x}^\dagger(t_2)\mathsf{x}(t_3)\right)\dots \\
&\quad - \arg\left(\mathsf{x}^\dagger(t_{n-1})\mathsf{x}(t_n)\right) + \arg\left(\mathsf{x}^\dagger(t_1)\mathsf{x}(t_n)\right) \\
&= -\arg\left(\mathsf{x}^\dagger(t_1)\mathsf{x}(t_2)\mathsf{x}^\dagger(t_2)\mathsf{x}(t_3)\dots\mathsf{x}^\dagger(t_{n-1})\mathsf{x}(t_n)\mathsf{x}^\dagger(t_n)\mathsf{x}(t_1)\right) \\
&= -\arg \mathsf{B}_{\mathsf{x}}^{1:n}
\end{aligned}$$

where we used the relation between geometric, total and dynamical phase (**??**) and the fact that the geometric phase acquired along a geodesic arc is zero.

Appendix F Derivation for the recursive estimator expression for the geometric phase

We present here the details of the calculation to obtain the recursive estimator for the geometric phase $\Phi^{\mathsf{r}}_{\mathsf{geo},\mathbf{x}}[\Delta_K \mathsf{t}]$ given in (40). First, using the expression of the geometric phase as in (39) one can the one for the interval $\Delta_K \mathsf{t}$ like:

$$\Phi_{\mathsf{geo},\mathbf{x}}[\Delta_{K+1}\mathsf{t}] = -\sum_{i=1}^{K} \arg \mathbf{x}[\mathsf{t_i}]^{\mathsf{H}}\mathbf{x}[\mathsf{t_{i+1}}] - \arg \mathbf{x}[\mathsf{t_{K+1}}]^{\mathsf{H}}\mathbf{x}[\mathsf{t_1}] \tag{F18}$$

[9] talk about the discretization of the path using geodesic arcs

Next, one can rewrite this expression the following way:

$$
\begin{aligned}
\Phi_{\mathsf{geo},\mathbf{x}}[\Delta_{K+1}\mathsf{t}] &= -\sum_{i=1}^{K-1} \arg \mathbf{x}[\mathsf{t_i}]^{\mathsf{H}}\mathbf{x}[\mathsf{t_{i+1}}] - \arg \mathbf{x}[\mathsf{t_{K+1}}]^{\mathsf{H}}\mathbf{x}[\mathsf{t_1}] - \arg \mathbf{x}[\mathsf{t_K}]^{\mathsf{H}}\mathbf{x}[\mathsf{t_{K+1}}] \\
&= -\arg \mathbf{x}[\mathsf{t_K}]^{\mathsf{H}}\mathbf{x}[\mathsf{t_1}] - \sum_{i=1}^{K-1} \arg \mathbf{x}[\mathsf{t_i}]^{\mathsf{H}}\mathbf{x}[\mathsf{t_{i+1}}] \\
&\quad + \arg \mathbf{x}[\mathsf{t_K}]^{\mathsf{H}}\mathbf{x}[\mathsf{t_1}] - \arg \mathbf{x}[\mathsf{t_{K+1}}]^{\mathsf{H}}\mathbf{x}[\mathsf{t_1}] \arg \mathbf{x}[\mathsf{t_K}]^{\mathsf{H}}\mathbf{x}[\mathsf{t_{K+1}}] \\
&= \Phi_{\mathsf{geo},\mathbf{x}}[\Delta_K \mathsf{t}] + \arg\left(\frac{\mathbf{x}[\mathsf{t_{K+1}}]^{\mathsf{H}}\mathbf{x}[\mathsf{t_K}]\mathbf{x}[\mathsf{t_K}]^{\mathsf{H}}\mathbf{x}[\mathsf{t_1}]}{\mathbf{x}[\mathsf{t_{K+1}}]^{\mathsf{H}}\mathbf{x}[\mathsf{t_1}]}\right)
\end{aligned}
\tag{F19}
$$

where we use the additivity of arguments and a simple rearrangement of the terms to obtain the last line expression.

Appendix G Decomposition of the 3D geometric phase into the 2D corrected by fluctuations of the elliptic plane

In the sequel we omit all uneccessary indexes variables such $_{\boldsymbol{x}}$ or (t), and t_0 will be represented by a $_0$. We have $\boldsymbol{p} = J_3(\alpha)J_1(\beta)(\boldsymbol{p}_2^\top\ 0)^\top$ so that its derivative writes

$$
\dot{\boldsymbol{p}} = \dot{J}_3(\alpha)J_1(\beta)\begin{pmatrix}\boldsymbol{p}_2\\0\end{pmatrix} + J_3(\alpha)\dot{J}_1(\beta)\begin{pmatrix}\boldsymbol{p}_2\\0\end{pmatrix} + J_3(\alpha)J_1(\beta)\begin{pmatrix}\dot{\boldsymbol{p}}_2\\0\end{pmatrix}
$$

The derivative of the matrices are given by

$$
\dot{J}_3(\alpha) = \dot{\alpha}\begin{pmatrix} -\sin(\alpha) & -\cos(\alpha) & 0\\ \cos(\alpha) & -\sin(\alpha) & 0\\ 0 & 0 & 0\end{pmatrix} = -\dot{\alpha}\begin{pmatrix} \cos(\alpha-\frac{\pi}{2}) & -\sin(\alpha-\frac{\pi}{2}) & 0\\ \sin(\alpha-\frac{\pi}{2}) & \cos(\alpha-\frac{\pi}{2}) & 0\\ 0 & 0 & 0\end{pmatrix}
$$

(and likewise for J_1). Furthermore, $J_3^\dagger(\alpha) = J_3(-\alpha)$ so that $J_3(-\alpha)J_3(\alpha) = \boldsymbol{I}$ and

$$
J_3^\dagger(\alpha)\dot{J}_3(\alpha) = \begin{pmatrix} 0 & 1 & 0\\ -1 & 0 & 0\\ 0 & 0 & 0\end{pmatrix}
$$

(and likewise for J_1). Hence we have

$$
\begin{aligned}
\boldsymbol{p}^\dagger\dot{\boldsymbol{p}} &= \begin{pmatrix}\boldsymbol{p}_2^\dagger & 0\end{pmatrix}\left(-\dot{\alpha}J_1(-\beta)\begin{pmatrix} 0 & 1 & 0\\ -1 & 0 & 0\\ 0 & 0 & 0\end{pmatrix}J_1(\beta)\begin{pmatrix}\boldsymbol{p}_2\\0\end{pmatrix} - \dot{\beta}\begin{pmatrix} 0 & 0 & 0\\ 0 & 0 & 1\\ 0 & -1 & 0\end{pmatrix}\begin{pmatrix}\boldsymbol{p}_2\\0\end{pmatrix} + \begin{pmatrix}\dot{\boldsymbol{p}}_2\\0\end{pmatrix}\right) \\
&= -\dot{\alpha}\begin{pmatrix}\boldsymbol{p}_2^\dagger & 0\end{pmatrix}\begin{pmatrix} 0 & \cos(\beta) & -\sin(\beta)\\ -\cos(\beta) & 0 & 0\\ \sin(\beta) & 0 & 0\end{pmatrix}\begin{pmatrix}\boldsymbol{p}_2\\0\end{pmatrix} + \boldsymbol{p}_2^\dagger\dot{\boldsymbol{p}}_2
\end{aligned}
$$

$$= -\dot{\alpha}\cos(\beta)(p_{21}^* p_{22} - p_{21} p_{22}^*) + \boldsymbol{p}_2^\dagger \dot{\boldsymbol{p}}_2$$

The term $-(p_{21}^* p_{22} - p_{21} p_{22}^*)$ is shown to be equal to $\boldsymbol{j}\sin(2\chi)$ where χ is the excentricity of the ellipse described by $\boldsymbol{p}_2$.

The same kind of calculations allows to obtained

$$\begin{aligned}\boldsymbol{p}_0^\dagger \boldsymbol{p} &= \cos(\alpha-\alpha_0)p_{21}^*(0)p_{21} - \sin(\alpha-\alpha_0)\left(\cos(\beta_0)p_{22}^*(0)p_{21} - \cos(\beta)p_{21}^*(0)p_{22}\right) \\ &+ \left(\cos(\beta_0)\cos(\beta)\cos(\alpha-\alpha_0) + \sin(\beta_0)\sin(\beta)\right)p_{22}^*(0)p_{22}\end{aligned}$$

Thus if α and β are constant, we verify that $\boldsymbol{p}(t_0)^\dagger \boldsymbol{p}(t) = \boldsymbol{p}_2(t_0)^\dagger \boldsymbol{p}_2(t)$.

Appendix H Majorana stellar representation

A unit complex vector in dimension n as the normalized and symmetrized sum of the tensorial product of $n-1$ unit complex vector in dimension 2.

The geometry of Majorana's constellation for $n = 3$

Let $\boldsymbol{p} \in \mathbb{C}^3$ and $\boldsymbol{p}_1$ and $\boldsymbol{p}_2$ in $\mathbb{C}^2$. Note that $\boldsymbol{p}_i$ projects by π on the 2 dimensional sphere by a dot represented by euclidean vector $\boldsymbol{u}_i$ which satisfies $\pi(\boldsymbol{p}_i) = \boldsymbol{p}_i \boldsymbol{p}_i^\dagger = (\boldsymbol{I} + \boldsymbol{u}_i \cdot \sigma)/2$, where σ is the basis of Pauli matrices. Then

$$\boldsymbol{p} = \frac{1}{\sqrt{2A_2}}\tilde{\boldsymbol{p}} \quad \text{where} \quad \tilde{\boldsymbol{p}} = \boldsymbol{p}_1 \otimes \boldsymbol{p}_2 + \boldsymbol{p}_2 \otimes \boldsymbol{p}_1 \tag{H20}$$

where

$$A_2 = \boldsymbol{p}^\dagger \boldsymbol{p} = 1 + \frac{1 + \boldsymbol{u}_1 \cdot \boldsymbol{u}_2}{2} = \frac{3 + \boldsymbol{u}_1 \cdot \boldsymbol{u}_2}{2} \tag{H21}$$

This can be shown using Pauli representation and $|\boldsymbol{p}_1^\dagger \boldsymbol{p}_2|^2 = (1 + \boldsymbol{u}_1 \cdot \boldsymbol{u}_2)/2$.

Berry's connection is given by $\boldsymbol{j}A_{\boldsymbol{p}}(\dot{\boldsymbol{p}}) = \boldsymbol{j}\boldsymbol{p}^\dagger \dot{\boldsymbol{p}}$. We have

$$\begin{aligned}\dot{\boldsymbol{p}} &= -\frac{\dot{A}_2}{2\sqrt{2}A_2^{3/2}}\tilde{\boldsymbol{p}} + \frac{1}{\sqrt{2A_2}}\dot{\tilde{\boldsymbol{p}}} \\ &= -\frac{\dot{A}_2}{2\sqrt{2}A_2^{3/2}}\tilde{\boldsymbol{p}} + \frac{1}{\sqrt{2A_2}}\left(\dot{\boldsymbol{p}}_1 \otimes \boldsymbol{p}_2 + \dot{\boldsymbol{p}}_2 \otimes \boldsymbol{p}_1 + \boldsymbol{p}_1 \otimes \dot{\boldsymbol{p}}_2 + \boldsymbol{p}_2 \otimes \dot{\boldsymbol{p}}_1\right)\end{aligned}$$

leading to (recall $(\boldsymbol{p}_1 \otimes \boldsymbol{p}_2)^\dagger(\boldsymbol{y}_1 \otimes \boldsymbol{y}_2) = (\boldsymbol{p}_1^\dagger \boldsymbol{y}_1)(\boldsymbol{p}_2^\dagger \boldsymbol{y}_2)$

$$\begin{aligned}\boldsymbol{p}^\dagger \dot{\boldsymbol{p}} &= -\frac{\dot{A}_2}{4A_2} + \frac{1}{2A_2}\Big(\boldsymbol{p}_1^\dagger \dot{\boldsymbol{p}}_1 + (\boldsymbol{p}_1^\dagger \dot{\boldsymbol{p}}_2)(\boldsymbol{p}_2^\dagger \boldsymbol{p}_1) + \boldsymbol{p}_2^\dagger \dot{\boldsymbol{p}}_2 + (\boldsymbol{p}_1^\dagger \boldsymbol{p}_2)(\boldsymbol{p}_2^\dagger \dot{\boldsymbol{p}}_1) \\ &+ (\boldsymbol{p}_2^\dagger \dot{\boldsymbol{p}}_1)(\boldsymbol{p}_1^\dagger \boldsymbol{p}_2) + \boldsymbol{p}_2^\dagger \dot{\boldsymbol{p}}_2 + (\boldsymbol{p}_2^\dagger \boldsymbol{p}_1)(\boldsymbol{p}_1^\dagger \dot{\boldsymbol{p}}_2) + \boldsymbol{p}_1^\dagger \dot{\boldsymbol{p}}_1\Big) \\ &= -\frac{\dot{A}_2}{4A_2} + \frac{1}{2A_2}\Big(2\boldsymbol{p}_1^\dagger \dot{\boldsymbol{p}}_1 + 2\boldsymbol{p}_2^\dagger \dot{\boldsymbol{p}}_2 + 2(\boldsymbol{p}_1^\dagger \boldsymbol{p}_2)(\boldsymbol{p}_2^\dagger \dot{\boldsymbol{p}}_1) + 2(\boldsymbol{p}_2^\dagger \boldsymbol{p}_1)(\boldsymbol{p}_1^\dagger \dot{\boldsymbol{p}}_2)\Big)\end{aligned}$$

Now

$$\begin{aligned}
A_2 &= (\boldsymbol{p}_1 \otimes \boldsymbol{p}_2 + \boldsymbol{p}_2 \otimes \boldsymbol{p}_1)^\dagger \ (\boldsymbol{p}_1 \otimes \boldsymbol{p}_2 + \boldsymbol{p}_2 \otimes \boldsymbol{p}_1) \\
&= 2 + 2(\boldsymbol{p}_2^\dagger \boldsymbol{p}_1)(\boldsymbol{p}_1^\dagger \boldsymbol{p}_2) = 2 + 2|\boldsymbol{p}_2^\dagger \boldsymbol{p}_1|^2 \\
\dot{A}_2 &= 2\left(\boldsymbol{p}_2^\dagger \boldsymbol{p}_1(\dot{\boldsymbol{p}}_1^\dagger \boldsymbol{p}_2 + \boldsymbol{p}_1^\dagger \dot{\boldsymbol{p}}_2) + \boldsymbol{p}_1^\dagger \boldsymbol{p}_2(\dot{\boldsymbol{p}}_2^\dagger \boldsymbol{p}_1 + \boldsymbol{p}_2^\dagger \dot{\boldsymbol{p}}_1)\right)
\end{aligned}$$

Thus

$$\begin{aligned}
\boldsymbol{p}^\dagger \dot{\boldsymbol{p}} &= \frac{1}{A_2}\left(\boldsymbol{p}_1^\dagger \dot{\boldsymbol{p}}_1 + \boldsymbol{p}_2^\dagger \dot{\boldsymbol{p}}_2 + (\boldsymbol{p}_1^\dagger \boldsymbol{p}_2)(\boldsymbol{p}_2^\dagger \dot{\boldsymbol{p}}_1 - 1/2(\dot{\boldsymbol{p}}_2^\dagger \boldsymbol{p}_1 + \boldsymbol{p}_2^\dagger \dot{\boldsymbol{p}}_1)) + (\boldsymbol{p}_2^\dagger \boldsymbol{p}_1)(\boldsymbol{p}_1^\dagger \dot{\boldsymbol{p}}_2 - 1/2(\dot{\boldsymbol{p}}_1^\dagger \boldsymbol{p}_2 + \boldsymbol{p}_1^\dagger \dot{\boldsymbol{p}}_2)\right) \\
&= \frac{1}{A_2}\left(\boldsymbol{p}_1^\dagger \dot{\boldsymbol{p}}_1 + \boldsymbol{p}_2^\dagger \dot{\boldsymbol{p}}_2 + \frac{1}{2}(\boldsymbol{p}_1^\dagger \boldsymbol{p}_2)(\boldsymbol{p}_2^\dagger \dot{\boldsymbol{p}}_1 - \dot{\boldsymbol{p}}_2^\dagger \boldsymbol{p}_1) + \frac{1}{2}(\boldsymbol{p}_2^\dagger \boldsymbol{p}_1)(\boldsymbol{p}_1^\dagger \dot{\boldsymbol{p}}_2 - \dot{\boldsymbol{p}}_1^\dagger \boldsymbol{p}_2)\right)
\end{aligned}$$

Let us explicitely evaluate $(\boldsymbol{p}_1^\dagger \boldsymbol{p}_2)(\boldsymbol{p}_2^\dagger \dot{\boldsymbol{p}}_1 - \dot{\boldsymbol{p}}_2^\dagger \boldsymbol{p}_1)$ using Pauli matrices and real vectors $\boldsymbol{u}_i$. We have

$$\boldsymbol{p}_1^\dagger \boldsymbol{p}_2 \boldsymbol{p}_2^\dagger \dot{\boldsymbol{p}}_1 = \mathrm{Tr}\left(\boldsymbol{p}_2 \boldsymbol{p}_2^\dagger \dot{\boldsymbol{p}}_1 \boldsymbol{p}_1^\dagger\right)$$

But $\dot{\boldsymbol{p}}_1 \boldsymbol{p}_1^\dagger = \overbrace{\boldsymbol{p}_1 \boldsymbol{p}_1^\dagger}^{\cdot} - \boldsymbol{p}_1 \dot{\boldsymbol{p}}_1^\dagger$ so that $\dot{\boldsymbol{p}}_1 \boldsymbol{p}_1^\dagger(\boldsymbol{p}_1 \boldsymbol{p}_1^\dagger) = \overbrace{\boldsymbol{p}_1 \boldsymbol{p}_1^\dagger}^{\cdot}(\boldsymbol{p}_1 \boldsymbol{p}_1^\dagger) - \boldsymbol{p}_1 \dot{\boldsymbol{p}}_1^\dagger(\boldsymbol{p}_1 \boldsymbol{p}_1^\dagger)$ and this leads to $\dot{\boldsymbol{p}}_1 \boldsymbol{p}_1^\dagger = \overbrace{\boldsymbol{p}_1 \boldsymbol{p}_1^\dagger}^{\cdot}(\boldsymbol{p}_1 \boldsymbol{p}_1^\dagger) + \boldsymbol{p}_1^\dagger \dot{\boldsymbol{p}}_1(\boldsymbol{p}_1 \boldsymbol{p}_1^\dagger)$ because $\boldsymbol{p}_1$ is of unit norm.

Thus

$$\begin{aligned}
\boldsymbol{p}_1^\dagger \boldsymbol{p}_2 \boldsymbol{p}_2^\dagger \dot{\boldsymbol{p}}_1 &= \mathrm{Tr}\left(\boldsymbol{p}_2 \boldsymbol{p}_2^\dagger \dot{\boldsymbol{p}}_1 \boldsymbol{p}_1^\dagger\right) \\
&= \mathrm{Tr}\left(\boldsymbol{p}_2 \boldsymbol{p}_2^\dagger \overbrace{\boldsymbol{p}_1 \boldsymbol{p}_1^\dagger}^{\cdot}(\boldsymbol{p}_1 \boldsymbol{p}_1^\dagger)\right) + \boldsymbol{p}_1^\dagger \dot{\boldsymbol{p}}_1 \,\mathrm{Tr}\left(\boldsymbol{p}_2 \boldsymbol{p}_2^\dagger \boldsymbol{p}_1 \boldsymbol{p}_1^\dagger\right) \\
&= \frac{1}{8}\,\mathrm{Tr}\left((\boldsymbol{I} + \boldsymbol{u}_2 \cdot \sigma)(\dot{\boldsymbol{u}}_1 \cdot \sigma)(\boldsymbol{I} + \boldsymbol{u}_1 \cdot \sigma)\right) + \boldsymbol{p}_1^\dagger \dot{\boldsymbol{p}}_1 |\boldsymbol{p}_2^\dagger \boldsymbol{p}_1|^2
\end{aligned}$$

Recall that $(\boldsymbol{u}_1 \cdot \sigma)(\boldsymbol{u}_2 \cdot \sigma) = \boldsymbol{u}_1 \cdot \boldsymbol{u}_2 + \boldsymbol{j}(\boldsymbol{u}_1 \wedge \boldsymbol{u}_2) \cdot \sigma$ and that $\mathrm{Tr}\,\sigma_i = 0$. Then we have

$$\begin{aligned}
\boldsymbol{p}_1^\dagger \boldsymbol{p}_2 \boldsymbol{p}_2^\dagger \dot{\boldsymbol{p}}_1 &= \frac{1}{8}\,\mathrm{Tr}\left((\boldsymbol{I} + \boldsymbol{u}_2 \cdot \sigma)(\dot{\boldsymbol{u}}_1 \cdot \sigma + \dot{\boldsymbol{u}}_1 \cdot \boldsymbol{u}_1 \boldsymbol{I} + \boldsymbol{j}(\dot{\boldsymbol{u}}_1 \wedge \boldsymbol{u}_1) \cdot \sigma)\right) + \boldsymbol{p}_1^\dagger \dot{\boldsymbol{p}}_1 |\boldsymbol{p}_2^\dagger \boldsymbol{p}_1|^2 \\
&= \frac{1}{4}\left(\dot{\boldsymbol{u}}_1 \cdot \boldsymbol{u}_1 + \boldsymbol{u}_2 \cdot \dot{\boldsymbol{u}}_1 + \boldsymbol{j}\boldsymbol{u}_2 \cdot (\dot{\boldsymbol{u}}_1 \wedge \boldsymbol{u}_1)\right) + \boldsymbol{p}_1^\dagger \dot{\boldsymbol{p}}_1 |\boldsymbol{p}_2^\dagger \boldsymbol{p}_1|^2
\end{aligned}$$

Likewise we get for the other terms

$$\begin{aligned}
\boldsymbol{p}_1^\dagger \boldsymbol{p}_2 \dot{\boldsymbol{p}}_2^\dagger \boldsymbol{p}_1 &= \overline{\boldsymbol{p}_2^\dagger \boldsymbol{p}_1 \boldsymbol{p}_1^\dagger \dot{\boldsymbol{p}}_2} \\
&= \frac{1}{4}\left(\dot{\boldsymbol{u}}_2 \cdot \boldsymbol{u}_2 + \boldsymbol{u}_1 \cdot \dot{\boldsymbol{u}}_2 - \boldsymbol{j}\boldsymbol{u}_1 \cdot (\dot{\boldsymbol{u}}_2 \wedge \boldsymbol{u}_2)\right) + \overline{\boldsymbol{p}_2^\dagger \dot{\boldsymbol{p}}_2} |\boldsymbol{p}_2^\dagger \boldsymbol{p}_1|^2 \\
\boldsymbol{p}_2^\dagger \boldsymbol{p}_1 \boldsymbol{p}_1^\dagger \dot{\boldsymbol{p}}_2 &= \frac{1}{4}\left(\dot{\boldsymbol{u}}_2 \cdot \boldsymbol{u}_2 + \boldsymbol{u}_1 \cdot \dot{\boldsymbol{u}}_2 + \boldsymbol{j}\boldsymbol{u}_1 \cdot (\dot{\boldsymbol{u}}_2 \wedge \boldsymbol{u}_2)\right) + \boldsymbol{p}_2^\dagger \dot{\boldsymbol{p}}_2 |\boldsymbol{p}_2^\dagger \boldsymbol{p}_1|^2
\end{aligned}$$

$$
\begin{aligned}
\boldsymbol{p}_2^\dagger \boldsymbol{p}_1 \dot{\boldsymbol{p}}_1^\dagger \boldsymbol{p}_2 &= \overline{\boldsymbol{p}_1^\dagger \boldsymbol{p}_2 \boldsymbol{p}_2^\dagger \dot{\boldsymbol{p}}_1} \\
&= \frac{1}{4}\left(\dot{\boldsymbol{u}}_1 \cdot \boldsymbol{u}_1 + \boldsymbol{u}_2 \cdot \dot{\boldsymbol{u}}_1 - \boldsymbol{j}\boldsymbol{u}_2 \cdot (\dot{\boldsymbol{u}}_1 \wedge \boldsymbol{u}_1)\right) + \overline{\boldsymbol{p}_1^\dagger \dot{\boldsymbol{p}}_1} |\boldsymbol{p}_2^\dagger \boldsymbol{p}_1|^2
\end{aligned}
$$

$$
\begin{aligned}
\boldsymbol{p}^\dagger \dot{\boldsymbol{p}} &= \frac{1}{A_2}\left(\boldsymbol{p}_1^\dagger \dot{\boldsymbol{p}}_1 + \boldsymbol{p}_2^\dagger \dot{\boldsymbol{p}}_2 + \frac{1}{4}\boldsymbol{j}\boldsymbol{u}_2 \cdot (\dot{\boldsymbol{u}}_1 \wedge \boldsymbol{u}_1) + \frac{1}{4}\boldsymbol{j}\boldsymbol{u}_1 \cdot (\dot{\boldsymbol{u}}_2 \wedge \boldsymbol{u}_2) + |\boldsymbol{p}_2^\dagger \boldsymbol{p}_1|^2 \boldsymbol{j}\mathrm{Im}\left(\boldsymbol{p}_1^\dagger \dot{\boldsymbol{p}}_1 + \boldsymbol{p}_2^\dagger \dot{\boldsymbol{p}}_2\right)\right) \\
&= \frac{1}{A_2}\left(\boldsymbol{p}_1^\dagger \dot{\boldsymbol{p}}_1 + \boldsymbol{p}_2^\dagger \dot{\boldsymbol{p}}_2 + \frac{1}{4}\boldsymbol{j}(\dot{\boldsymbol{u}}_1 - \dot{\boldsymbol{u}}_2) \cdot (\boldsymbol{u}_1 \wedge \boldsymbol{u}_2)) + |\boldsymbol{p}_2^\dagger \boldsymbol{p}_1|^2 \boldsymbol{j}\mathrm{Im}\left(\boldsymbol{p}_1^\dagger \dot{\boldsymbol{p}}_1 + \boldsymbol{p}_2^\dagger \dot{\boldsymbol{p}}_2\right)\right)
\end{aligned}
$$

Writing $1/A_2 = 1 - (1 - A_2)/A_2$ and noting that $A_2 = |\boldsymbol{p}_2^\dagger \boldsymbol{p}_1|^2$ allows to write

$$
\boldsymbol{p}^\dagger \dot{\boldsymbol{p}} = \boldsymbol{p}_1^\dagger \dot{\boldsymbol{p}}_1 + \boldsymbol{p}_2^\dagger \dot{\boldsymbol{p}}_2 + \frac{1}{A_2}\left(\frac{1}{4}\boldsymbol{j}(\dot{\boldsymbol{u}}_1 - \dot{\boldsymbol{u}}_2) \cdot (\boldsymbol{u}_1 \wedge \boldsymbol{u}_2)) + |\boldsymbol{p}_2^\dagger \boldsymbol{p}_1|^2 \left(\boldsymbol{j}\mathrm{Im}\left(\boldsymbol{p}_1^\dagger \dot{\boldsymbol{p}}_1 + \boldsymbol{p}_2^\dagger \dot{\boldsymbol{p}}_2\right) - \boldsymbol{p}_1^\dagger \dot{\boldsymbol{p}}_1 + \boldsymbol{p}_2^\dagger \dot{\boldsymbol{p}}_2\right)\right)
$$

and finally

$$
\mathrm{Im}\left(\boldsymbol{p}^\dagger \dot{\boldsymbol{p}}\right) = \mathrm{Im}\left(\boldsymbol{p}_1^\dagger \dot{\boldsymbol{p}}_1\right) + \mathrm{Im}\left(\boldsymbol{p}_2^\dagger \dot{\boldsymbol{p}}_2\right) + \frac{1}{2}\frac{(\dot{\boldsymbol{u}}_1 - \dot{\boldsymbol{u}}_2) \cdot (\boldsymbol{u}_1 \wedge \boldsymbol{u}_2)}{3 + \boldsymbol{u}_1 \cdot \boldsymbol{u}_2}
$$

Let $\boldsymbol{R} = (\boldsymbol{u}_1 + \boldsymbol{u}_2)/2$ and $\boldsymbol{r} = \boldsymbol{u}_1 - \boldsymbol{u}_2$, and $\tilde{\boldsymbol{R}}$, $\tilde{\boldsymbol{r}}$ their normalized counterparts. Then $\boldsymbol{u}_1 = \boldsymbol{R} + \boldsymbol{r}/2$ and $\boldsymbol{u}_2 = \boldsymbol{R} - \boldsymbol{r}/2$.Then the time derivative of $\tilde{\boldsymbol{r}}$ can be written as

$$
\dot{\tilde{\boldsymbol{r}}} = \frac{\dot{\boldsymbol{r}}}{\|\boldsymbol{r}\|} - \tilde{\boldsymbol{r}}\frac{\dot{\boldsymbol{r}}^\top \boldsymbol{r}}{\boldsymbol{r}^\top \boldsymbol{r}}
$$

and thus

$$
\begin{aligned}
(\dot{\boldsymbol{u}}_1 - \dot{\boldsymbol{u}}_2) \cdot (\boldsymbol{u}_1 \wedge \boldsymbol{u}_2) &= (\boldsymbol{r} \wedge \boldsymbol{R}) \cdot \dot{\boldsymbol{r}} \\
&= \|\boldsymbol{r}\|\|\boldsymbol{R}\|(\tilde{\boldsymbol{r}} \wedge \tilde{\boldsymbol{R}}) \cdot (\|\boldsymbol{r}\|\dot{\tilde{\boldsymbol{r}}} + \gamma\tilde{\boldsymbol{r}}) \\
&= \|\boldsymbol{r}\|^2\|\boldsymbol{R}\|(\tilde{\boldsymbol{r}} \wedge \tilde{\boldsymbol{R}}) \cdot \dot{\tilde{\boldsymbol{r}}}
\end{aligned}
$$

Then, expliciting the different norms leads to

$$
\frac{1}{2}\frac{(\dot{\boldsymbol{u}}_1 - \dot{\boldsymbol{u}}_2) \cdot (\boldsymbol{u}_1 \wedge \boldsymbol{u}_2)}{3 + \boldsymbol{u}_1 \cdot \boldsymbol{u}_2} = f(\psi)\cos(\psi)\,(\tilde{\boldsymbol{r}} \wedge \tilde{\boldsymbol{R}}) \cdot \dot{\tilde{\boldsymbol{r}}}
$$

where $2\psi = \arccos(\boldsymbol{u}_1 \cdot \boldsymbol{u}_2)$ and $f(\psi) = \sin^2(\psi)/(1 + \cos^2(\psi))$.

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